SQL TIPS(2): 특정조건의 순서대로 번호를 부여하는 방법

Bulletin no : 10109

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* 특정조건의 순서대로 번호를 부여하는 방법

[Q1] 아래의 Record를 Deptno 순으로 Sequence를 부여하려면?

ENAME	JOB	DEPTNO
ALLEN	SALESMAN	20
WARD	SALESMAN	30
JONES	MANAGER	20
MARTINN	SALESMAN	30
BLAKE	MANAGER	30
CLARK	MANAGER	10
SCOTT	ANALYST	20
KING	PRESIDENT	10
TURNER	SALESMAN	30
ADAMS	CLERK	20
JAMES	CLERK	30
FORD	ANALYST	20
MILLER	CLERK	10

[A1]

1. Emp Table의 Deptno Column에 Index를 생성한다.

```
create index i_emp on emp(deptno);
```

2. Table에 Sequence Number Column을 추가한다.

```
alter table emp add (seq1  number(5));
```

3.
```
update emp
   set seq1 = rownum
   where deptno > '0';
```

ENAME	JOB	DEPTNO	SEQ1
CLARK	MANAGER	10	1
KING	PRESIDENT	10	2
MILLER	CLERK	10	3
ALLEN	SALESMAN	20	4
JONES	MANAGER	20	5
SCOTT	ANALYST	20	6
ADAMS	CLERK	20	7

```
FORD      ANALYST        20          8
WARD      SALESMAN       30          9
MARTINN   SALESMAN       30         10
BLAKE     MANAGER        30         11
TURNER    SALESMAN       30         12
JAMES     CLERK          30         13
```

[Q2] 각 부서별로 Sequence를 부여하려면?

[A2]

```
update emp A
set A.seq1 = (select A.seq1 - min(B.seq1)+1
              from emp B
              where B.deptno = A.deptno);
```

```
ENAME      JOB           DEPTNO   SEQ1
---------  ------------  ------   ----
CLARK      MANAGER       10         1
KING       PRESIDENT     10         2
MILLER     CLERK         10         3
ALLEN      SALESMAN      20         1
JONES      MANAGER       20         2
SCOTT      ANALYST       20         3
ADAMS      CLERK         20         4
FORD       ANALYST       20         5
WARD       SALESMAN      30         1
MARTINN    SALESMAN      30         2
BLAKE      MANAGER       30         3
TURNER     SALESMAN      30         4
JAMES      CLERK         30         5
```

SQL TRACE FACILITY
(TKPROF UTILITY 사용)

Bulletin no : 10134

SQL TRACE FACILITY 는 SQL 문 사용에대한 성능을 분석하기위해서 사용된다.
이러한 SQL TRACE FACILITY 를 이용하면 각 SQL 문에 대해서 다음과 같은 정보를 얻을수 있다.

. parse,execute,fetch count
. CPU 와 elapsed 시간
. physical reads 와 logical reads
. 처리된 row 의 수

SQL TRACE FACILITY 는 SESSION 혹은 INSTANCE 단위로 할 수 있고 TRACE 결과 화일은 tkprof UTILITY 에의해 사용자가 읽을수 있는 형태로 변환시킨다.

SQL TRACE SET UP 하여 사용하는방법
---

1. SQL TRACE enable 및 TRACE 화일 디렉토리 지정

< INSTANCE 단위 >

?/dbs/initSID.ora 화일에 다음 두개의 PARAMETER 를 추가하고 DATABASE 를 다시 STARTUP 시킨다.

```
sql_trace = true
timed_statistics = true
```

timed_statistics 은 시스템에 많은 LOAD 가 걸리므로 사용하지 않는 것이 좋다.

< SESSION 단위 >

SQL*PLUS

```
$ sqlplus scott/tiger
SQL > ALTER SESSION SET SQL_TRACE = TRUE;
```

SQLFORMS30, RUNFORM30 : -s OPTION 사용

```
$ runform30 -s frmfile scott/tiger -c vt220
```

PRO*C

```
EXEC SQL CONNECT :username;
EXEC SQL ALTER SESSION SET SQL_TRACE = TRUE;
```

TRACE FILE 디렉토리는 initSID.ora 화일에 다음과 같이 지정하고 지정하지 않으면 ?/rdbms/log 디렉토리에 생긴다.

```
user_dump_dest = /user/dump
```

2. SQL 문 실행

TRACE 화일은 SESSION 단위로 1 개씩 생김으로 한 SESSION 에서 사용된 모든 SQL 문의 TRACE 결과는 1개의 화일에 존재하게된다.

예. SQL> ALTER SESSION SET SQL_TRACE = TRUE;
SQL> SELECT COUNT(*) FROM EMP E,DEPT D

```
        WHERE E.DEPTNO = D.DEPTNO;
SQL> EXIT;
```

3. TRACE 화일 변환

2. 단계에서 SQL 문을 실행하면 user_dump_dest 에 지정된 디렉토리에 TRACE 화일이 생기고 tkprof 를이용하여 화일을 변환시킨다. TRACE 화일은 쉽게 찾을 수 있는 형태가 아니므로 SQL문을 실행하기 전에 dump 디렉토리에 있는 ora_xxxx.trc화일을 모두 삭제하거나 가장최근에 생긴 화일중에서 찾아야 한다.

```
예. $ cd $ORACLE_HOME/rdbms/log
    $ tkprof ora_1111 out sort=fchqry,fchcu explain=scott/tiger print=20
```

ora_1111 : TRACE 화일
out : OUTPUT 화일. 디렉토리에 out.prf 로 생긴다.
sort : 지정된 OPION(fchqry,fchcu) 에 ASCENDING 순으로 SQL 문을 SORTING 한다.
explain : SQL 문의 EXECUTION PLAN 을 발생시킨다.
print : 지정된 갯수의 SQL문에대해서만 TRACE 결과를 PRINT 한다.

4. SQL TRACE 결과 분석

```
********************************************************************************
count   = number of times OCI procedure was executed
cpu     = cpu time in seconds executing
elapsed = elapsed time in seconds executing
disk    = number of physical reads of buffers from disk
query   = number of buffers gotten for consistent read
current = number of buffers gotten in current mode (usually forupdate)
rows    = number of rows processed by the fetch or execute call
********************************************************************************
SELECT COUNT(*)
FROM EMP E,DEPT D WHERE E.DEPTNO=D.DEPTNO

call    count  cpu    elapsed   disk    query    current    rows
------------------------------------------------------------------
Parse     2     0.00    0.00       0       0        0         0
Execute   2     0.00    0.00       0       0        0         0
Fetch     2     0.00    0.00       0      33        2         1

Misses in library cache during parse: 1
Parsing user id: 8  (SCOTT)
Rows     Execution Plan
-------  ---------------------------------------------------
      0  SELECT STATEMENT
      0    SORT (AGGREGATE)
     16      NESTED LOOPS
     16        TABLE ACCESS (FULL) OF 'EMP'
```

```
    16          INDEX (UNIQUE SCAN) OF 'DEPT_PRIMARY_KEY' (UNIQUE)

********************************************************************************
```

분석 예)

a. cpu, elapsed 정보가없는 경우는 initSID.ora 에 timed_stattistic = false 로 되어 있기 때문이다.

b. Execute count 와 Fectch count 가 동일하게 크다고하면 ARRAY FETCH 사용을 고려함.

c. fetch 된 rows 수 : query + current = 1 : 4 이하이면 SQL 문은 적절히 사용된 경우이고, row 수에 비하여 query + count 가 상당히 크면 부적하게 사용된 SQL 문이므로 (count,sum,distinct 등 Group function 을 사용하는경우는 예외) 다음 내용들에대해서 재검토가 필요하다.
   .INDEX 사용,구성여부
   .ROWID 사용
   .COST_BASED OPTIMIZER 사용(ORACLE7)
   .ARRAY FETCH 사용
   .SORTING 을 피할수 있는 SQL 문 구사

d. Parse count,Execute count 가 비슷한경우 RELEASE_CURSOR, HOLE_CURSOR OPTION 사용하여 Parse count 를 줄임.

LONG COLUMN을 가진 TABLE의 COPY

Bulletin no : 10147

문제설명

==========

LONG COLUMN 을 가진 테이블을 COPY 를 해야 하는경우 LONG COLUMN은 SUB-QUERY 를 사용 할수 없기 때문에 다음과 같은 현상이 발생한다.

1. "CREATE TABLE .. AS SELECT .." 명령이나
   "INSERT INTO .. SELECT .." 를 사용하는 경우 ORA-00997 에러가 발생한다.

2. SQL*Plus 의 COPY 명령을 사용 하는 경우 CPY0005 에러가 발생한다.

해결방법

==========

가장 쉬운 방법은 PL/SQL에서 VARCHAR2 혹은 LONG DATA TYPE을 이용하여 COPY한다. PL/SQL에서 VARCHAR2 및 LONG 변수는 32K 까지 사용할 수 있기 때문에 32K 이하 데이타라면 가능하고 그 이상의 데이타는 3GL을 이용한다.

< TABLE>

```
SQL> CREATE TABLE LONGTAB(NO NUMBER(2), TEXT LONG);
```

< PL/SQL Script >

```
DECLARE
  string varchar2(32000);
  no     number(2);
BEGIN
  SELECT no,longcolumn INTO no, string FROM longtable;
  INSERT INTO longtab VALUES(no, string);
END;
/
```

SQL TIPS(3): DATE의 기간을 계산하는 방법

Bulletin no : 10158

```
$ sqlplus scott/tiger

SQL>  SELECT
        trunc(((DATE2-DATE1)*60*60)/3600)||' DAYS '||
        floor(mod(((DATE2-DATE1)*24*60*60)/3600,24))||' HOURS '||
        floor(((date1-date2)*24*60*60)
        - floor(((date1-date2)*24*60*60)/3600)*3600)/60)|| ' MINUTES ' ||
        round((((date1-date2)*24*60*60)
        - floor(((date1-date2)*24*60*60)/3600)*3600
        - (floor((((date1-date2)*24*60*60)
        - floor(((date1-date2)*24*60*60)/3600)*3600)/60)*60))||' SECS '
      TIMEDIFF
      FROM  dual
/
```

SQL TIPS(4): 중복된 RECORD 삭제 방법

Bulletin no : 10185

중복된 RECORD 를 삭제 하는 방법은 2가지가 있다.

방법1.

```
SQL> CREATE TABLE emp2 AS SELECT distinct * FROM emp;
SQL> DROP TABLE emp;
SQL> RENAME emp2 TO emp;
```

방법2.

```
SQL> DELETE FROM emp A
           WHERE rowid > (SELECT min(rowid) FROM emp B
                          WHERE B.empno = A.empno);
```

외부에서 값을 받아서 SQLPLUS의 SCRIPT 실행하는 방법

Bulletin no : 10211

---

1. PARAMETER 를 받을수 있도록 SQL SCRIPT 를 작성한다.

```
< SQL_SCR1.sql >
 select count(*) from emp where deptno = &1;

< SQL_SCR2.sql >
 select count(*) from emp where deptno = &1;
```

2. SHELL 프로그램을 작성 한다.

```
< SQL_SCR >
 read dno
 sqlplus -s scott/tiger @SQL_SCR1 $dno;
 sqlplus -s scott/tiger @SQL_SCR2 $dno;
```

3. SHELL 프로그램 기동

```
 $ ./SQL_SCR
```

SQL TIPS(5): DECODE이용사례 (1) : 월별로 집계하는 방법

Bulletin no : 10219

```
sql> select nvl(sum(decode(to_char(hiredate,'mm'),'01',sal)),0) jan,
            nvl(sum(decode(to_char(hiredate,'mm'),'02',sal)),0) feb,
            nvl(sum(decode(to_char(hiredate,'mm'),'03',sal)),0) mar,
            nvl(sum(decode(to_char(hiredate,'mm'),'04',sal)),0) apr
     from emp
     where hiredate between to_date('810101','yymmdd') and
                            to_date('811231','yymmdd')+0.99999
sql>/
       JAN        FEB        MAR        APR
---------- ---------- ---------- ----------
         0       2850          0       2975
```

SQL TIPS(6): 해당 COLUMN에 걸려 있는 CONSTRAINT확인

Bulletin no : 10331

SQL TIPS(6):

```
/*
**  Name       :  Show_Constraints.Sql
**
**  Usage      :  @Show_Constraints   Table_Name   Column_Name
**  Description  :  해당 Table의 Column에 걸려 있는 Constraint를 보여준다.
**
*/

SET VERIFY OFF
CLEAR BREAK
BREAK ON TABLES ON COL_NAME
SELECT  SUBSTR(TABLE_NAME,1,15)    TABLES,
          SUBSTR(COLUMN_NAME,1,15)  COL_NAME,
          SUBSTR(CONSTRAINT_NAME,1,25)  CONSTRAINT_NAME
FROM    USER_CONS_COLUMNS
WHERE  TABLE_NAME = UPPER('&1')
AND     COLUMN_NAME = UPPER('&2');
```

< 실행 예 >

```
SQL>  @SHOW_CONSTRAINTS  WIDGETS  LENGTH

TABLES           COL_NAME        CONSTRAINT_NAME
---------------  --------------- --------------------
WIDGETS          LENGTH          CK_WIDGETS_LENGTH
                                 CK_WIDGETS_PRODUCTION
```

SQL TIPS(7): 한 CONSTRAINT에서 REFERENCE하는 COLUMN확인

Bulletin no : 10332

SQL TIPS(7):

```
/*
**  Name        :  Show_Columns.Sql
**
**  Usage       :  @Show_Columns   Constraint_Name
**  Description  :  Shows The Columns Bound By A Constraint
**
*/

SET VERIFY OFF
CLEAR BREAK
BREAK ON CONSTRAINT_NAME ON TABLES
SELECT  SUBSTR(CONSTRAINT_NAME,1,25)  CONSTRAINT_NAME,
          SUBSTR(TABLE_NAME,1,15)    TABLES,
          SUBSTR(COLUMN_NAME,1,15)  COL_NAME,

FROM    ALL_CONS_COLUMNS
WHERE  CONSTRAINT_NAME = UPPER('&1');
```

< 실행 예 >

```
SQL>  @SHOW_COLUMNS  CK_WIDGETS_PRODUCTION

CONSTRAINT_NAME           TABLES           COL_NAME
------------------------- ---------------  ----------
CK_WIDGETS_PRODUCTION     WIDGETS          LENGTH
                                           WIDTH
                                           HEIGHT
```

SQL TIPS(8): PARENT-CHILD관계를 갖는 TABLE의 PRIMARY,FOREIGN KEY의 COLUMN명과 POSITION확인

Bulletin no : 10333

SQL TIPS(8):

```
/*
**  Name        :  Show_Position.Sql
**
**  Usage       :  @Show_Positions  Parent_Table  Child_Table
**  Description :  Shows Primary And Foreign Key Positions
**
**  WARNING   :  이 문장은 해당 Table의 Constraint생성시 Naming
**                  Convention을 따른 경우에 적용되도록 되어 있다.
**
*/

SET VERIFY OFF
CLEAR BREAK
BREAK ON CONSTRAINT_NAME ON TABLES

SELECT  SUBSTR(CONSTRAINT_NAME,1,27)  CONSTRAINT_NAME,
          SUBSTR(TABLE_NAME,1,15)         TABLES,
          SUBSTR(COLUMN_NAME,1,15)      COL_NAME,
          SUBSTR(POSITION,1,3)              POSITION,
          SUBSTR(OWNER,1,7)                OWNER
FROM    USER_CONS_COLUMNS
WHERE  TABLE_NAME = UPPER('&1')
AND     CONSTRAINT_NAME LIKE  'PK%'
UNION
SELECT  SUBSTR(CONSTRAINT_NAME,1,27)  CONSTRAINT_NAME,
          SUBSTR(TABLE_NAME,1,15)         TABLES,
          SUBSTR(COLUMN_NAME,1,25)      COL_NAME,
          SUBSTR(POSITION,1,3)              POSITION,
          SUBSTR(OWNER,1,7)                OWNER
FROM    USER_CONS_COLUMNS
WHERE  TABLE_NAME = UPPER('&2')
AND     CONSTRAINT_NAME LIKE  'FK%'
ORDER BY  1 DESC , 4  ASC;
```

< 실행 예 >

```
SQL> @SHOW_POSITIONS  EMP_SERVICE  EMP_SERVICE_LOG

CONSTRAINT_NAME          TABLES             COL_NAME   POS
-----------------------  -----------------  --------  -----
PK_EMP_SERVICE           EMP_SERVICE        EMP_ID        1
                                            CUST_ID       2
FK_EMP_SERVICE_LOG_EC    EMP_SERVICE_LOG    EMP_ID        1
                                            CUST_ID       2
```

SQL TIPS(9): 해당 TABLE에 걸려있는 CONSTRAINT의 확인

Bulletin no : 10334

SQL TIPS(9):

```
/*
**  Name        :  Show_Table_Constrinats.Sql
**
**  Usage       :  @Show_Table_Constraints  Table_Name
**  Description  :  Returns The Constraint Names And Types
**
*/

SET VERIFY OFF
CLEAR BREAK
BREAK ON COLUMN_NAME ON CONSTRAINT_TYPE

SELECT  SUBSTR(A.COLUMN_NAME,1,15)    COLUMN_NAME,
          DECODE(B.CONSTRAINT_TYPE,  'P','PRIMARY KEY',
                                          'U','UNIQUE KEY',
                                          'C','CHECK OR NOT NULL',
                                          'R','FOREIGN KEY')  CONSTRAINT_TYPE,
          A.CONSTRAINT_NAME    CONSTRAINT_NAME
FROM    USER_CONS_COLUMNS  A,  USER_CONSTRAINTS  B
WHERE  A.TABLE_NAME = UPPER('&1')
AND     A.TABLE_NAME = B.TABLE_NAME
AND     CONSTRAINT_NAME = B.CONSTRAINT_NAME
ORDER BY 1,2;
```

< 실행 예 >

```
SQL>  @SHOW_TABLE_CONSTRAINTS  EMP_SERVICE

COLUMN_NAME          CONSTRAINT_TYPE      CONSTRAINT_NAME
-------------- -------------- --------------------
CUST_ID                 PRIMARY KEY            PK_EMP_SERVICE
                          FOREIGN KEY            FK_EMP_SERVICE_CUST_ID
EMP_ID                  PRIMARY KEY            PK_EMP_SERVICE
                          FOREIGN KEY            FK_EMP_SERVICE_EMP_ID
LENGTH                CHECK OR NOT NULL    SYS_C0014089
```

주의 : SYS로 시작하는 CONSTRAINT명은 USER가 CONSTRAINT NAME을 지정하지 않아 SYSTEM에서 DEFAULT로 생성한 경우를 보여준다.

SQL TIPS(10): CHECK CONSTRAINT의 조건 확인

Bulletin no : 10335

SQL TIPS(10):

```
/*
**  Name        :  Show_Search_Condition.Sql
**
**  Usage       :  @Show_Search_Condition  Constraint_Name
**  Description  :  Shows The Rule For A CHECK Constraint
**
*/

SET VERIFY OFF
SET ARRAYSIZE 1
SET LONG 75

SELECT  SEARCH_CONDITION
FROM    USER_CONSTRAINTS
WHERE   CONSTRAINT_NAME = UPPER('&1');
```

< 실행 예 >

```
SQL>  @SHOW_SEARCH_CONDITION   CK_WIDGETS_PRODUCTION

SEARCH_CONDITION
--------------------------------------------------
(HEIGHT >= 0)
    AND (LENGTH <= HEIGHT + 10)
    AND (WIDTH <= LENGTH + 10)
```

SQL TIPS(11) :TABLE에 대하여 EXTENT 발생량 산정방법

Bulletin no : 10351

```
/* 모든 TABLE에 대하여 각각의 EXTENT 발생량을 산정한다 */

set linesize 80
set pages 66
set echo off
set termout off
spool report/extent.lst

column owner no print new_value owner
break on owner page
ttitle cen 'Extent Definition Owner : ' owner skip 1 -
       cen '----------------------------------------' skip 1;

col f1 format a40 heading 'OBJECT NAME'
col f2 format a10 heading 'TYPE'
col f3 format 99,999,999 heading 'TOTAL EXTS'
select a.owner owner, a.segment_name f1,
   segment_type f2,
   extents f3
from dba_segments a
where a.owner not in ('SYS','SYSTEM','SCOTT')
and extents > 1
/
spool off
exit
```

SQL TIPS(12): PRINTING A HIERARCHICAL LIST OF REFERRING AND REFERENCED

Bulletin no : 10364

This SQL script generates a hierarchical list of tables with a level number (tables that are linked with foreign key constraints), from any Oracle schema. The list is useful in identifying the 'order of tables', while loading data, dropping tables or deleting rows from tables.

The script collects referential integrity constraints data from the data dictionary tables, creates and populates temporary tables and prints the hierarchical table list.

PURPOSE: Print an Hierarchical list of referring and referenced tables in any Oracle Schema.

```
    ------------------------------------------------------------------------
    FILE   : table_levels.sql
    OUTPUT : level_tree.lis
    USAGE  : sqlplus userid/password @table_levels.sql

/
--
--  Create temporary tables, collecting data from Dictionary tables
--  ------------------------------------------------------------------

set termout off
drop table parent_child_tab
/
drop table parent_child_temp_tab
/
drop table level_tab
/
drop table parent_child_level
/
create table parent_child_level (p_level number,c_level number)
/
insert into parent_child_level values (1,1)
/
commit
/
```

```
create table parent_child_tab as
  (select x.p_level,
          p.table_name parent_table_name,
          p.owner p_owner,
          x.c_level,
          c.table_name child_table_name,
          c.constraint_name c_constraint_name,
          c.owner c_owner
     from user_constraints p, user_constraints c,
          parent_child_level x
    where p.constraint_type IN ('P','U') AND
          c.constraint_type = 'R' AND
          p.constraint_name = c.r_constraint_name  )
/
-- ------------------------------------------------------------
--  Collect data if the referenced tables are in another schema.
-- ------------------------------------------------------------
create table parent_child_temp_tab as
  (select x.p_level,
          a.table_name parent_table_name,
          c.r_owner p_owner,
          x.c_level,
          c.table_name child_table_name,
          c.constraint_name c_constraint_name,
          c.owner c_owner
     from user_constraints c, all_constraints a,
          parent_child_level x
    where c.owner <> c.r_owner and
          c.r_owner = a.owner and
          c.r_constraint_name = a.constraint_name )
/
insert into parent_child_tab
               ( p_level, c_constraint_name, parent_table_name, p_owner,
                 c_level, child_table_name , c_owner)
          (select p_level, c_constraint_name, parent_table_name, p_owner,
                  c_level, child_table_name , c_owner
             from parent_child_temp_tab )
/

create table level_tab
  ( table_name     varchar2(30),
    level_number   number )
/
-- ------------------------------------------------------------------
-- << Create performance indexes on the tables - rw1 >>
-- ------------------------------------------------------------------
```

```
create index child_level_owner_ix
    on PARENT_CHILD_TAB(child_table_name,c_level,c_owner)
/
create index parent_owner_ix
    on PARENT_CHILD_TAB(parent_table_name,c_owner)
/
create index level_table_name_ix
    on LEVEL_TAB(table_name)
/
create index level_level_table_ix
    on LEVEL_TAB(level_number,table_name)
/
-- ----------------------------------------------------------------
--  The following pl/sql block will populate the 'level_tab' table
--  with 'table name' and 'level number'.
--  At the end it updates 'parent_child_tab' with level numbers.
-- ----------------------------------------------------------------
declare
   dummy_val            char(1);
   level_num            number;
   total_levels         number;
   tab_name             varchar2(30);
   child_tab_name       varchar2(30);
--
   cursor get_child_tab_name(pk1 number) is
     select distinct child_table_name
       from parent_child_tab
      where parent_table_name in (select table_name
                                    from level_tab
                                   where level_number = pk1 ) ;
--
   cursor delete_duplicates(pk2 number) is
     select table_name
       from level_tab
      where level_number = pk2 ;
--
begin
-- ---------------------------------------------------------
--  Insert all Master table names with out any foreign key
--  definition, into 'level_tab' table.
-- ---------------------------------------------------------
  insert into level_tab
    (select distinct parent_table_name, 1
       from parent_child_tab
       where parent_table_name not in
        (select child_table_name from parent_child_tab) ) ;
   commit;
```

```
-- ------------------------------------------------------------------
-- << Update self-referencing FK's by appending an '@' onto the end of
--           the parent_table_name - rwl >>
-- ------------------------------------------------------------------
update parent_child_tab
   set parent_table_name = parent_table_name || '@'
 where parent_table_name = child_table_name
   and p_owner = c_owner;
commit;
-- ------------------------------------------------------------------
-- << Update bi-directional FK's by appending an '!' onto the end of
--           the parent_table_name - rwl >>
-- ------------------------------------------------------------------
update parent_child_tab
   set parent_table_name = parent_table_name ||'!'
 where p_owner||'.'||parent_table_name||':'||
       c_owner||'.'||child_table_name in
       (select c_owner||'.'||child_table_name||':'||
               p_owner||'.'||parent_table_name
          from parent_child_tab);
commit;
-- ------------------------------------------------------------
--  Identify all tables with foreign key definition, calculate
--  level number and insert 'table name', 'level number'
--  into 'level_tab' table.
-- ------------------------------------------------------------
  level_num := 1;
  <>
  loop
     open get_child_tab_name(level_num);
     fetch get_child_tab_name into child_tab_name;  /* first record */
     if get_child_tab_name%notfound then
       total_levels := level_num;
       --  'total_levels' variable gives highest depth value in the Schema
       close get_child_tab_name;
       exit outer_loop;
     else
       level_num := level_num + 1;
       insert into level_tab values(child_tab_name, level_num);
     end if;
     <>
     loop
        fetch get_child_tab_name into child_tab_name;  /* other records */
        exit  inner_loop when get_child_tab_name%notfound ;
        insert into level_tab values(child_tab_name, level_num);
     end loop inner_loop;
     close get_child_tab_name;
  end loop outer_loop;
```

```
  commit;
-- ------------------------------------------------------
--  Keep only the highest level number row for a table name
--  and delete rest of the duplicate rows from 'level_tab'.
-- ------------------------------------------------------
  ◇
  for loop_var in reverse 1..total_levels loop
     open delete_duplicates(loop_var);
     ◇
     loop
        fetch delete_duplicates into tab_name;
        exit  for_inner_loop when delete_duplicates%notfound ;
        delete from level_tab
         where level_number < loop_var and table_name = tab_name;
     end loop for_inner_loop;
     close delete_duplicates;
  end loop for_loop;
  commit;
-- ------------------------------------------------
--  update level numbers into parent_child_tab table
--  by fetching the value from 'level_tab' table.
-- ------------------------------------------------
  update parent_child_tab
     set p_level = (select level_number
                      from level_tab
                     where table_name = parent_table_name );
  commit;
--
  update parent_child_tab
     set c_level = (select level_number
                      from level_tab
                     where table_name = child_table_name );
  commit;
end ;
/
-- ------------------------------------------------------
--  prepare spoolfile 'level_tree.lis' to capture the output
--  generated by dbms.output utility package
-- ------------------------------------------------------
--
set serveroutput on size 1000000
-- set linesize 132
spool level_tree.lis
--
-- ------------------------------
--  Format and print the output
-- ------------------------------
```

```
declare
   level_num             number;
   text_line             varchar2(132);
--
   cursor get_parent_child_tab is
     select distinct child_table_name, c_level, c_owner
       from parent_child_tab
      order by 2 desc, 1 ;
--
   cursor get_level_one_tab is
     select distinct parent_table_name, p_owner from parent_child_tab
       where p_level = 1
      order by 1 ;
--
   cursor print_tree(key1  varchar2) is
     select lpad(' ',2*(level)) lpad_value,
            parent_table_name t_name, p_owner, c_owner, c_constraint_name
       from parent_child_tab
       start with child_table_name = key1
       connect by prior parent_table_name = child_table_name;
--
   cursor get_level_numbers(key2 varchar2) is
     select level_number from level_tab
       where table_name = key2;
--
begin
   for xrec in get_parent_child_tab loop
     dbms_output.put_line(
        xrec.c_owner||'.'||xrec.child_table_name||'('||xrec.c_level||')');
       for yrec in print_tree(xrec.child_table_name) loop
         open get_level_numbers(yrec.t_name);
         fetch get_level_numbers into level_num;
         close get_level_numbers;
         text_line := null;
         if level_num = 1 then
            text_line := '-'||yrec.lpad_value||yrec.p_owner||'.'
                  ||yrec.t_name||'('||level_num||')';
         else
            text_line := '-'||yrec.lpad_value||yrec.c_owner||'.'
                  ||yrec.t_name||'('||level_num||')';
         end if;
```

```
-- ------------------------------------------------------------------------
-- << if you don't want the constraint name listed, comment-out
--    the following line of code.
--    SET LINESIZE is located at the SPOOL statement some 40+ lines above.
--    For padding out to fill an 80 column page, rpad to 47.
--    For padding out to fill a 132 column page, rpad to 99  - rwl  >>
-- ------------------------------------------------------------------------
   text_line  := rpad(text_line,47,' .')
              ||'['||yrec.c_constraint_name||']';
-- ------------------------------------------------------------------------
         dbms_output.put_line(text_line);
       end loop;
       dbms_output.put_line('-----------------------');
   end loop;
   for xrec in get_level_one_tab loop
     dbms_output.put_line(
         xrec.p_owner||'.'||xrec.parent_table_name||'(1)');
   end loop;
end ;
/
-- ------------------------------------------------------------------------
spool off

exit
-------------------------------------------------------------
/* ************************************************************************
   The following scripts are included for operating against the
   PARENT_CHILD_TAB.  The first two provide an 'immediate neighbour' picture
   one level above or below this table:

rem
rem   FKNAME.SQL  - tables whose key is referenced by a field in this table
rem                 (immediate 'neighbours' - 1 level outward) - rwl Sep/96
rem

column ref heading "Has an FK named ..." format a35
column tab heading "Pointing to This Table's PK" format a35
ttitle center 'This Table: &&1.'  SKIP 2

set feedback off
set verify off
set pagesize 60

select c_constraint_name ref, parent_table_name tab
-- or if you want 'pure' names translate(parent_table_name,'@!','  ') tab
  from parent_child_tab
 where child_table_name = upper('&&1.')  -- 'normal' name
    or child_table_name = upper('&&1.@') -- self-referencing name
    or child_table_name = upper('&&1.!') -- bi-directional name
```

```
 order by 2,1
/

rem
rem   FKREF.SQL   - tables which reference this table's key
rem                 (immediate 'neighbours' - 1 level only)  - rwl Sep/96
rem
rem   NOTE: if you run this and specify the table name with an @ or !,
rem         you will get a listing of all self_referencing relationships (@)
rem         or all tables which refer to each other (! bi-directional)
rem

column tab heading "... in This Table" format a35
column ref heading "Is Referenced by This FKey" format a35
ttitle center 'The PK of This Table: &&1.'  SKIP 2

set feedback off
set verify off
set pagesize 60

select c_constraint_name ref, child_table_name tab
-- or if you want 'pure' names translate( child_table_name,'@!','  ') tab
  from parent_child_tab
 where parent_table_name = upper('&&1.')  -- 'normal' name
    or parent_table_name = upper('&&1.@') -- self-referencing name
    or parent_table_name = upper('&&1.!') -- bi-directional name
 order by 2,1
/

   A third script provides a subset of the entire tree starting with
   the 'seed' table. It's like a recursive FKNAME.SQL listing:

rem
rem   FKTREE.SQL  - a subset of the entire application's FK tree
rem
rem NOTE: it may be necessary to run this for each of
rem       &&1, &&1.!!@ , and &&1.!!! to get the entire subset picture
rem       depending on the complexity of your application's relationships
rem                                                       - rwl Sep/96

column ref heading "Is Referenced BY These Tables" format a48
column fk  heading "... via This Foreign Key"      format a30
column parent_table_name NEW_VALUE a noprint format a1 trunc
break on parent_table_name skip 0 on report
ttitle center 'This Table''s Primary Key: &&1.'  SKIP 2
```

```
set pagesize 60
set linesize 79
set newpage 0
set verify off

select parent_table_name,
       lpad('>',2*(LEVEL-1))||child_table_name ref,
       c_constraint_name FK
  from parent_child_tab
start with parent_table_name = UPPER('&&1.')
connect by parent_table_name = PRIOR child_table_name
/
```

SQL TIPS(13): DROPPING DATABASE OBJECTS FOR A USER

Bulletin no : 10365

```
--_____________________________________________________________________
--
--MODULE NAME:
--     drop_obj.sql
--
--SYNOPSIS:
--     sqlplus / @drop_obj.sql
--
--DESCRIPTION:
--     This script will create another SQL script which will
--     drop all of the objects owned by a user.
--
--_____________________________________________________________________

set pages 5000
set echo off
set heading off
set feedback off

spool drop.sql

select 'spool drop.lis' from dual
/

SELECT 'DROP ' || OBJECT_TYPE || ' '
               || OBJECT_NAME ||
                  DECODE(OBJECT_TYPE,'TABLE',' CASCADE CONSTRAINTS',NULL)
               || ';'
  FROM USER_OBJECTS
 ORDER BY DECODE(OBJECT_TYPE,'INDEX','A_INDEX',
                             'TRIGGER','A_TRIGGER',
                             OBJECT_TYPE),
          OBJECT_NAME
/

select 'spool off' from dual
/
```

```
spool off

set echo on
set heading on
set feedback on

rem
rem  end of script
rem
```

SQL TIPS(14): PROVIDING USERS WITH SESSION INFORMATION

Bulletin no : 10366

```
rem
rem orasessn.sql
rem
 ----------------------------------------------------------------------------
 --
rem
rem This script is used to provide users with information regarding their oracle
rem sessions.  The USER_SESSION view provides information related only to the
rem current session for a user; while the ALL_SESSIONS relates to all sessions
rem within the database for a user.  The DBA_SESSION can only be viewed by the
rem Oracle sys id (and anyone granted select on this view).  The DBA_SESSION
rem lists all oracle sessions excluding the Oracle generic processes (dbwr,
rem pmon, smon, etc.).  The SID and SERIAL# can be used with killing sessions.
rem
rem This script must be run from the Oracle sys id.
rem
rem
 ----------------------------------------------------------------------------
 --
rem

set echo on;
spool orasessn;

rem
 ----------------------------------------------------------------------------
 --

drop public synonym ALL_SESSIONS;

create or replace view ALL_SESSIONS
      (USERNAME, OS_USER, SID, SERIAL#, STATUS, SERVICE, DATABASE,
       CURSORS, MACHINE, TERMINAL, PROGRAM)
as
select substr(v1.username,1,20), substr(v1.osuser,1,12),
       to_char(v1.sid,'999'), to_char(v1.serial#,'999999'),
       v1.status, v1.server, substr(v3.value,1,10), v2.value,
```

```
       substr(v1.machine,1,15), v1.terminal, v1.program
  from v$session v1, v$sesstat v2, v$parameter v3
 where v1.username   = user
   and v2.sid        =  v1.sid
   and v2.statistic# =  3
   and v3.name       =  'db_name' ;

grant  select on      ALL_SESSIONS to  PUBLIC;
create public synonym ALL_SESSIONS for ALL_SESSIONS;

rem
 ----------------------------------------------------------------------

drop public synonym USER_SESSIONS;

create or replace view USER_SESSIONS
      (USERNAME, OS_USER, SID, SERIAL#, STATUS, SERVICE, DATABASE,
       CURSORS, MACHINE, TERMINAL, PROGRAM)
as
select substr(v1.username,1,20), substr(v1.osuser,1,12),
       to_char(v1.sid,'999'), to_char(v1.serial#,'999999'),
       v1.status, v1.server, substr(v3.value,1,10), v2.value,
       substr(v1.machine,1,15), v1.terminal, v1.program
  from v$session v1, v$sesstat v2, v$parameter v3
 where v1.audsid     = userenv('SESSIONID')
   and v2.sid        =  v1.sid
   and v2.statistic# =  3
   and v3.name       =  'db_name' ;

grant  select on      USER_SESSIONS to  PUBLIC;
create public synonym USER_SESSIONS for USER_SESSIONS;

rem
 ----------------------------------------------------------------------

create or replace view DBA_SESSIONS
      (USERNAME, OS_USER, SID, SERIAL#, STATUS, SERVICE, DATABASE,
       CURSORS, MACHINE, TERMINAL, PROGRAM)
as
select substr(v1.username,1,20), substr(v1.osuser,1,12),
       to_char(v1.sid,'999'), to_char(v1.serial#,'999999'),
       v1.status, v1.server, substr(v3.value,1,10), v2.value,
       substr(v1.machine,1,15), v1.terminal, v1.program
  from v$session v1, v$sesstat v2, v$parameter v3
 where v1.username   is not null
   and v2.sid        =  v1.sid
```

```
   and v2.statistic# =  3
   and v3.name       =  'db_name' ;

rem ================================================================

spool off;
```

SQL TIPS(15): FORECASTING ON NEXT EXTENTS

Bulletin no : 10367

A DBA needs to know when a table or index is reaching its maximum number of extents, and you can use theSQL*PLUS script XEXTENTS.SQLto report this information. The output lists tables/indexes that are within "X" extents of reaching their maximum. XEXTENTS.SQL is operating-system independent.

```
/*
  Constructions:
  [1] Find out the Oracle data block size by querying V$PARAMETER;
  [2] Translate it into a maximum number of extents. Because the size of the Oracle data block
      is operating-system dependent, the following approach:
      select decode(value, 2048, 121, 4096, 240) from v$parameter is not appropriate.
  [3] Without the lose of generality, a weighted average method is used for constructing
      a stored function named expected_extents. This function is OS Independent and it returns
      max_ext when called.
*/

/*
   XEXTENTS.SQL
*/

REM
REM Creating stored function expected_extents.
REM

DROP FUNCTION expected_extents;

CREATE OR REPLACE FUNCTION expected_extents(vkey IN varchar2)
RETURN varchar2 IS
        e_value varchar2(20);
BEGIN
        e_value:= ROUND((1/2)*((121/2048)+(240/4096))*vkey, 0);
        RETURN e_value;
END expected_extents;
/

REM
```

```
REM Generating a forecasting report.
REM

set linesize 78;

ttitle
   center 'Report of Next Extent Within ' &X ' of maximum extents' skip -
   center 'Date: ' datevar              skip -
   right  '(T/I - Table/Index)'         skip 2

column today    new_value datevar       format A1 noprint;
column owner    new_value owner         format A9;
column object   new_value object        format A15;
column ts       new_value tablespace    format A25;

select  sysdate                         "today",
        t.owner                         "owner",
        t.table_name                    "object",
        t.tablespace_name               "ts",
        'T/I'                           "T/I",
        t.max_extents                   "max_extents",
        s.extents                       "extents"
from    sys.dba_tables                  t,
        sys.dba_segments                s,
        v$parameter                     p
where   table_name = segment_name
and     (t.max_extents < s.extents + &X or
        expected_extents(p.value) < s.extents + &X)
and     p.name = 'db_block_size'
UNION
select  sysdate,
        i.owner,
        i.index_name,
        i.tablespace_name,
        'I'                     indicator,
        i.max_extents max_extents,
        s.extents current_extent
from    sys.dba_indexes                 i,
        sys.dba_segments                s,
        v$parameter                     p
where   index_name = segment_name
and     (i.max_extents < s.extents + &X or
        expected_extents(p.value) < s.extents + &X)
and     p.name = 'db_block_size';

EXIT;
```

SQL TIPS(16): CREATING SCRIPTS TO RECREATE A TABLE STRUCTURE

Bulletin no : 10368

The script creates scripts that can be used to recreate a table structure. For example, this script can be used when a table has become fragmented or to get a defintion that can be run on another database.

```
/*

   CREATES SCRIPT TO RECREATE A TABLE-STRUCTURE
   INCL. STORAGE, CONSTRAINTS, TRIGGERS ETC.
   -----------------------------------------

   This script creates scripts to recreate a table structure.
   Use the script to reorganise a table that has become fragmented,
   to get a definition that can be run on another database/schema or as
   a basis for altering the table structure (eg. drop a column!).
```

IMPORTANT: Running the script is safe as it only creates two new scripts and does not do anything to your database! To get anything done you have to run the scripts created.

The created scripts does the following:

1. save the content of the table
2. drop any foreign key constraints referencing the table
3. drop the table
4. creates the table with an Initial storage parameter that will accomodate the entire content of the table. The Next parameter is 25% of the initial.
   The storage parameters are picked from the following list:
   64K, 128K, 256K, 512K, multiples of 1M.
5. create table and column comments
6. fill the table with the original content
7. create all the indexes incl storage parameters as above.
8. add primary, unique key and check constraints.
9. add foreign key constraints for the table and for referencing tables.
10. Create the table's triggers.
11. Compile any depending objects (cascading).
12. Grant table and column privileges.
13. Create synonyms.

```
This script must be run as the owner of the table.
If your table contains a LONG-column, use the COPY command in SQL*Plus to store/restore
the data.

USAGE
------

from SQL*Plus:

start reorgtb

This will create the scripts REORGS1.SQL and REORGS2.SQL

REORGS1.SQL contains code to save the current content of the table.
REORGS2.SQL contains code to rebuild the table structure.

---------------------------------------------------------------
---------------------------------------------------------------
*/

undef tab;
set echo off
column a1 new_val stor
column b1 new_val nxt

select
decode(sign(1024-sum(bytes)/1024),-1,to_char((round(sum(bytes)/(1024*1
024))+1))||'M', /* > 1M new rounded up to nearest Megabyte */
decode(sign(512-sum(bytes)/1024), -1,'1M',
decode(sign(256-sum(bytes)/1024), -1,'512K',
decode(sign(128-sum(bytes)/1024), -1,'256K',
decode(sign(64-sum(bytes)/1024) , -1,'128K',
                      '64K'
)))))
 a1,
decode(sign(1024-sum(bytes)/4096),-1,to_char((round(sum(bytes)/(4096*1
024))+1))||'M', /* > 1M new rounded up to nearest Megabyte */
decode(sign(512-sum(bytes)/4096), -1,'1M',
decode(sign(256-sum(bytes)/4096), -1,'512K',
decode(sign(128-sum(bytes)/4096), -1,'256K',
decode(sign(64-sum(bytes)/4096) , -1,'128K',
                      '64K'
)))))
 b1
 from user_extents
where segment_name=upper('&1');
```

```
set pages 0 feed off verify off lines 150
col c1 format a80


spool reorgs1.sql

PROMPT drop table bk_&1
prompt /
PROMPT create table bk_&1 storage (initial &stor) as select * from &1
prompt /

spool off
spool reorgs2.sql

PROMPT spool reorgs2

select 'alter table '||table_name||' drop constraint
'||constraint_name||';'
from user_constraints where r_constraint_name
in (select constraint_name from user_constraints where
table_name=upper('&1')
and constraint_type in ('P','U'));

PROMPT drop table &1
prompt /

prompt create table &1
select decode(column_id,1,'(',',')
     ||rpad(column_name,40)
     ||decode(data_type,'DATE'    ,'DATE           '
                       ,'LONG'    ,'LONG           '
                       ,'LONG RAW','LONG RAW       '
                       ,'RAW'     ,'RAW      '
                       ,'CHAR'    ,'CHAR     '
                       ,'VARCHAR' ,'VARCHAR  '
                       ,'VARCHAR2','VARCHAR2 '
                       ,'NUMBER'  ,'NUMBER   '
                       ,'unknown')
     ||rpad(
       decode(data_type,'DATE'    ,null
                       ,'LONG'    ,null
                       ,'LONG RAW',null
                       ,'RAW'     ,decode(data_length,null,null
                                         ,'('||data_length||')')
                       ,'CHAR'    ,decode(data_length,null,null
                                         ,'('||data_length||')')
```

```
                    ,'VARCHAR' ,decode(data_length,null,null
                                       ,'('||data_length||')')
                    ,'VARCHAR2',decode(data_length,null,null
                                       ,'('||data_length||')')
                    ,'NUMBER'  ,decode(data_precision,null,'   '
                                      ,'('||data_precision||

decode(data_scale,null,null

,',','||data_scale)||')')
                    ,'unknown'),8,' ')
     ||decode(nullable,'Y','NULL','NOT NULL') c1
from user_tab_columns
where table_name = upper('&1')
order by column_id
/

prompt )

select 'pctfree        '||t.pct_free    c1
      ,'pctused        '||t.pct_used    c1
      ,'initrans       '||t.ini_trans    c1
      ,'maxtrans       '||t.max_trans    c1
      ,'tablespace     '||s.tablespace_name    c1
      ,'storage (initial     '||'&stor'    c1
      ,'         next        '||'&stor'    c1
      ,'         minextents  '||t.min_extents    c1
      ,'         maxextents  '||t.max_extents    c1
      ,'         pctincrease '||t.pct_increase||')'    c1
from  user_Segments s, user_tables t
where s.segment_name = upper('&1') and
      t.table_name = upper('&1')
 and  s.segment_type = 'TABLE'
/

prompt /

select 'comment on table &1 is '''||comments||''';' c1 from
user_tab_comments
where table_name=upper('&1');

select 'comment on column &1..'||column_name||
' is '''||comments||''';' c1 from user_col_comments
where table_name=upper('&1');

prompt insert into &1 select * from bk_&1
```

```
prompt /

set serveroutput on

declare
cursor c1 is select index_name,decode(uniqueness,'UNIQUE','UNIQUE')
unq
from user_indexes where
table_name = upper('&1');
indname varchar2(50);
cursor c2 is select
decode(column_position,1,'(',',')||rpad(column_name,40) c1
from user_ind_columns where table_name = upper('&1') and
 index_name = indname
order by column_position;
l1 varchar2(100);
l2 varchar2(100);
l3 varchar2(100);
l4 varchar2(100);
l5 varchar2(100);
l6 varchar2(100);
l7 varchar2(100);
l8 varchar2(100);
l9 varchar2(100);
begin
dbms_output.enable(100000);
for c in c1 loop
 dbms_output.put_line('create '||c.unq||' index '||c.index_name||' on
&1');
 indname := c.index_name;
 for q in c2 loop
  dbms_output.put_line(q.c1);
 end loop;
  dbms_output.put_line(')');
select 'pctfree        '||i.pct_free    ,
      'initrans       '||i.ini_trans    ,
      'maxtrans       '||i.max_trans    ,
      'tablespace     '||i.tablespace_name    ,
      'storage (initial     '||
decode(sign(1024-sum(e.bytes)/1024),-1,
to_char((round(sum(e.bytes)/(1024*1024))+1))||'M',
decode(sign(512-sum(e.bytes)/1024), -1,'1M',
decode(sign(256-sum(e.bytes)/1024), -1,'512K',
decode(sign(128-sum(e.bytes)/1024), -1,'256K',
decode(sign(64-sum(e.bytes)/1024) , -1,'128K',
                      '64K'))))))    ,
```

```
      '         next        '||
decode(sign(1024-sum(e.bytes)/4096),-1,
to_char((round(sum(e.bytes)/(4096*1024))+1))||'M',
decode(sign(512-sum(e.bytes)/4096), -1,'1M',
decode(sign(256-sum(e.bytes)/4096), -1,'512K',
decode(sign(128-sum(e.bytes)/4096), -1,'256K',
decode(sign(64-sum(e.bytes)/4096) , -1,'128K',
                      '64K'))))) ,
      '         minextents  '||s.min_extents    ,
      '         maxextents  '||s.max_extents    ,
      '         pctincrease '||s.pct_increase||')'
into l1,l2,l3,l4,l5,l6,l7,l8,l9
from  user_extents e,user_segments s, user_indexes i
where s.segment_name = c.index_name
 and  s.segment_type = 'INDEX'
 and  i.index_name = c.index_name
 and  e.segment_name=s.segment_name
group by s.min_extents,s.max_extents,s.pct_increase,
i.pct_free,i.ini_trans,i.max_trans,i.tablespace_name ;
  dbms_output.put_line(l1);
  dbms_output.put_line(l2);
  dbms_output.put_line(l3);
  dbms_output.put_line(l4);
  dbms_output.put_line(l5);
  dbms_output.put_line(l6);
  dbms_output.put_line(l7);
  dbms_output.put_line(l8);
  dbms_output.put_line(l9);
  dbms_output.put_line('/');
end loop;
end;
/

declare
cursor c1 is
select constraint_name, decode(constraint_type,'U',' UNIQUE',' PRIMARY
KEY') typ,
decode(status,'DISABLED','DISABLE',' ') status from user_constraints
where table_name = upper('&1')
and   constraint_type in ('U','P');
cname varchar2(100);
cursor c2 is
select decode(position,1,'(',',')||rpad(column_name,40) coln
from user_cons_columns
where table_name = upper('&1')
and   constraint_name = cname
```

```
order by position;
begin
for q1 in c1 loop
 cname := q1.constraint_name;
 dbms_output.put_line('alter table &1');
 dbms_output.put_line('add constraint '||cname||q1.typ);
 for q2 in c2 loop
  dbms_output.put_line(q2.coln);
 end loop;
  dbms_output.put_line(')' ||q1.status);
  dbms_output.put_line('/');
end loop;
end;
/

declare
cursor c1 is
select c.constraint_name,c.r_constraint_name cname2,
       c.table_name table1, r.table_name table2,
       decode(c.status,'DISABLED','DISABLE',' ') status,
       decode(c.delete_rule,'CASCADE',' on delete cascade ',' ')
delete_rule
from   user_constraints c,
       user_constraints r
where c.constraint_type='R' and
      c.r_constraint_name = r.constraint_name and
      c.table_name = upper('&1')
union
select c.constraint_name,c.r_constraint_name cname2,
       c.table_name table1, r.table_name table2,
       decode(c.status,'DISABLED','DISABLE',' ') status,
       decode(c.delete_rule,'CASCADE',' on delete cascade ',' ')
delete_rule
from   user_constraints c,
       user_constraints r
where c.constraint_type='R' and
      c.r_constraint_name = r.constraint_name and
      r.table_name = upper('&1');
cname varchar2(50);
cname2 varchar2(50);
cursor c2 is
select decode(position,1,'(',',')||rpad(column_name,40) colname
from user_cons_columns
where   constraint_name = cname
order by position;
cursor c3 is
```

```
select decode(position,1,'(',',')||rpad(column_name,40) refcol
from user_cons_columns
where constraint_name = cname2
order by position;
begin
dbms_output.enable(100000);
for q1 in c1 loop
 cname := q1.constraint_name;
 cname2 := q1.cname2;
 dbms_output.put_line('alter table '||q1.table1||' add constraint ');
 dbms_output.put_line(cname||' foreign key');
 for q2 in c2 loop
   dbms_output.put_line(q2.colname);
 end loop;
 dbms_output.put_line(') references '||q1.table2);
 for q3 in c3 loop
   dbms_output.put_line(q3.refcol);
 end loop;
 dbms_output.put_line(') '||q1.delete_rule||q1.status);
 dbms_output.put_line('/');
end loop;
end;
/


col c1 format a79 word_wrap
set long 32000
set arraysize 1

select 'create or replace trigger ' c1,
       description c1,
      'WHEN ('||when_clause||')' c1,
      trigger_body ,
      '/'         c1
from user_triggers
where table_name = upper('&1') and when_clause is not null
/
select 'create or replace trigger ' c1,
       description c1,
      trigger_body ,
      '/'         c1
from user_triggers
where table_name = upper('&1') and when_clause is null
/

select 'alter trigger '||trigger_name||decode(status,'DISABLED','
```

```
DISABLE',' ENABLE')
from user_Triggers where table_name='&1';

set serveroutput on

declare
cursor c1 is
select 'alter table
'||'&1'||decode(substr(constraint_name,1,4),'SYS_',' ',
      ' add constraint ') a1,
       decode(substr(constraint_name,1,4),'SYS_','
',constraint_name)||' check (' a2,
      search_condition a3,
      ') '||decode(status,'DISABLED','DISABLE','') a4,
      '/'         a5
from user_constraints
where table_name = upper('&1') and
constraint_type='C';
b1 varchar2(100);
b2 varchar2(100);
b3 varchar2(32000);
b4 varchar2(100);
b5 varchar2(100);
f1 number;
begin
open c1;
loop
 fetch c1 into b1,b2,b3,b4,b5;
 exit when c1%NOTFOUND;
 select count(*) into f1 from user_tab_columns where table_name =
upper('&1') and
        upper(column_name)||' IS NOT NULL' = upper(b3);
 if f1 = 0 then
   dbms_output.put_line(b1);
   dbms_output.put_line(b2);
   dbms_output.put_line(b3);
   dbms_output.put_line(b4);
   dbms_output.put_line(b5);
 end if;
end loop;
end;
/


create or replace procedure dumzxcvreorg_dep(nam varchar2,typ
varchar2) as
```

```
cursor cur is
select type,decode(type,'PACKAGE BODY','PACKAGE',type) type1,
name from  user_dependencies
where referenced_name=upper(nam) and referenced_type=upper(typ);
begin
dbms_output.enable(500000);
for c in cur loop
 dbms_output.put_line('alter '||c.type1||' '||c.name||' compile;');
 dumzxcvreorg_dep(c.name,c.type);
end loop;
end;
/
exec dumzxcvreorg_dep('&1','TABLE');

drop procedure dumzxcvreorg_Dep;

select 'grant '||privilege||' on '||table_name||' to '||grantee||
decode(grantable,'YES',' with grant option;',';') from
user_tab_privs where table_name = upper('&1');

select 'grant '||privilege||' ('||column_name||') on &1 to
'||grantee||
decode(grantable,'YES',' with grant option;',';')
from user_col_privs where grantor=user and
table_name=upper('&1')
order by grantee, privilege;

select 'create synonym '||synonym_name||' for
'||table_owner||'.'||table_name||';'
from user_synonyms where table_name=upper('&1');

PROMPT REM
PROMPT REM YOU MAY HAVE TO LOG ON AS SYSTEM TO BE
PROMPT REM ABLE TO CREATE ANY OF THE PUBLIC SYNONYMS!
PROMPT REM

select 'create public synonym '||synonym_name||' for
'||table_owner||'.'||table_name||';'
from all_synonyms where owner='PUBLIC' and table_name=upper('&1') and
table_owner=user;

prompt spool off

spool off
set echo on feed on verify on
```

```
/*
-----------------------------------------------------------
     The scripts REORGS1.SQL and REORGS2.SQL have been
     created. Alter these script as necesarry.
     To recreate the table-structure, first run REORGS1.SQL.
     This script saves the content of your table in a table
     called bk_.
     If this script runs successfully run REORGS2.SQL.
     The result is spooled to REORGTB.LST.
     Check this file before dropping the bk_ table.
  -----------------------------------------------------------
*/
```

SQL TIPS(17): AVAILABLE SPACE, FRAGMENTATION, AND EXTENTS

Bulletin no : 10369

The script below lists available space, fragmentation, and potential extent problems.

```
/*
   This script displays information about space in tablespaces, fragmentation and
   potential extent-problems. It is divided in 3:

   1. Space available per tablespace.
      Shows total size in bytes, total free space in bytes, percentage free space
      to total space, the size of the largest contigous free segment and the number of
      free segments.

   2. Lists tables and indexes with more than 20 extents.

   3. Lists tables/indexes whose next extent will not fit into their tablespace.
*/

set pagesize 300

column sumb format 999,999,999
column extents forma 9999
column bytes format 999,999,999
column largest format 999,999,999
column Tot_Size format 999,999,999
column Tot_Free format 999,999,999
column Pct_Free format 999,999,999
column Chunks_Free format 999,999,999
column Max_Free format 999,999,999
set echo off
spool space

PROMPT  SPACE AVAILABLE IN TABLESPACES

select a.tablespace_name,sum(a.tots) Tot_Size,
sum(a.sumb) Tot_Free,
sum(a.sumb)*100/sum(a.tots) Pct_Free,
sum(a.largest) Max_Free,sum(a.chunks) Chunks_Free
from
```

```
(
select tablespace_name,0 tots,sum(bytes) sumb,
max(bytes) largest,count(*) chunks
from dba_free_space a
group by tablespace_name
union
select tablespace_name,sum(bytes) tots,0,0,0 from
 dba_data_files
group by tablespace_name) a
group by a.tablespace_name;

column owner format a15
column segment_name format a30


PROMPT   SEGMENTS WITH MORE THAN 20 EXTENTS

select owner,segment_name,extents,bytes ,
max_extents,next_extent
from  dba_segments
where segment_type in ('TABLE','INDEX') and extents>20
order by owner,segment_name;


PROMPT SEGMENTS WHERE THERE'S NOT ENOUGH ROOM FOR THE NEXT EXTENT

select  a.owner, a.segment_name, b.tablespace_name,
        decode(ext.extents,1,b.next_extent,
        a.bytes*(1+b.pct_increase/100)) nextext,
        freesp.largest
from    dba_extents a,
        dba_segments b,
        (select owner, segment_name, max(extent_id) extent_id,
          count(*) extents
          from dba_extents
          group by owner, segment_name
        ) ext,
        (select tablespace_name, max(bytes) largest
         from dba_free_space
         group by tablespace_name
        ) freesp
where   a.owner=b.owner and
        a.segment_name=b.segment_name and
        a.owner=ext.owner and
        a.segment_name=ext.segment_name and
        a.extent_id=ext.extent_id and
```

```
        b.tablespace_name = freesp.tablespace_name and
        decode(ext.extents,1,b.next_extent,
                a.bytes*(1+b.pct_increase/100)) > freesp.largest
/

spool off
```

SQL TIPS(18): DBMS_OUTPUT을 이용하여 SQL*PLUS에서 REPORT 작성하기

Bulletin no : 10370

SQL*PLUS에서 원하는 값을 화면에 DISPLAY하여 확인하거나 REPORT로 출력해 보려면 어떻게 해야 할까?

I. HINT
1. DBMS_OUTPUT PACKAGE를 사용한다.
1 SQL*PLUS상에서 SET SERVEROUTPUT ON을 하거나
1 DBMS_OUTPUT.ENABLE을 사용한다.
2. 필요하다면 PL/SQL BUILD-IN FUNCTION을 사용한다.
3. CHAR DATATYPE을 출력할 때는 RPAD를 사용한다.
4. NUMBER나 INTEGER DATATYPE을 출력할 때는 LPAD를 사용한다.

II. EXAMPLE : TABLESPACE의 사용 정도에 대한 REPORT작성

```
SET SERVEROUTPUT ON;

DECLARE
    VAR_BYTES   NUMBER(9);
    CURSOR FIND_INFO IS
    SELECT SUBSTR(D.TABLESPACE_NAME,1,22)       TSPACE,
           D.FILE_ID                                FILE_ID,
           ROUND(D.BYTES/1024/1024,4)             TOT_MB,
           D.BYTES                                  ORA_BLKS,
           SUM(E.BLOCKS)                          TOT_USED,
           ROUND(SUM(E.BLOCKS)/D.BYTES,4)*100     PCT_USED
      FROM   SYS.DBA_DATA_FILES D,
             SYS.DBA_EXTENTS E
     WHERE D.FILE_ID = E.FILE_ID (+)
     GROUP BY D.TABLESPACE_NAME,
                      D.FILE_ID,
                      D.BYTES/1024/1024,
                      D.BYTES;
BEGIN
    DBMS_OUTPUT.PUT_LINE(
            RPAD('TSPACE',20,' ')||
            LPAD('FILE_ID',10,' ')||
            LPAD('TOT_MB',10,' ')||
            LPAD('ORA_BLKS',10,' ')||
            LPAD('TOT_USED',10,' ')||
```

```
        LPAD('PCT_USED',10,' '));
    DBMS_OUTPUT.PUT_LINE(
        RPAD('--------------------',20,' ')||
        LPAD('-------',10,' ')||
        LPAD('------',10,' ')||
        LPAD('--------',10,' ')||
        LPAD('--------',10,' ')||
        LPAD('--------',10,' '));

    FOR SEARCH IN FIND_INFO
    LOOP
        SELECT DISTINCT BYTES/BLOCKS
          INTO VAR_BYTES
          FROM SYS.DBA_SEGMENTS;

        DBMS_OUTPUT.PUT_LINE(
            RPAD(SEARCH.TSPACE,20,' ')||
            LPAD(SEARCH.FILE_ID,10,' ')||
            LPAD(SEARCH.TOT_MB,10,' ')||
            LPAD(SEARCH.ORA_BLKS/VAR_BYTES,10,' ')||
            LPAD(SEARCH.TOT_USED,10,' ')||
            LPAD(SEARCH.PCT_USED*VAR_BYTES,10,' '));
    END LOOP;
END;
/
```

SQL TIPS(19): MAKE A MAP OF YOUR ORACLE FILES

Bulletin no : 10372

The script below generates a report on all the Oracle files (database, redo log, and control) in an Oracle UNIX system.
It's like a map of your Oracle files. Moreover, it also makes available other information (size and physical I/Os, if applicable) related to the files.

THE SCRIPT

```
rem
rem  File: filemap.sql
rem
rem  This script provides information on the location/path, file name,
rem  size and physical I/Os (if applicable) of all the Oracle files
rem  (database, control and redo log) in an Oracle/UNIX platform.
rem
rem
set pages 999
col path  format a20     heading 'Path'
col fname format a15     heading 'File Name'
col fsize format 999b    heading 'M bytes'
col pr    format 999999b heading 'Phy. Reads'
col pw    format 999999b heading 'Phy. Writes'
break on path skip 1
spool zzfilemap.rep
select substr(name,1,instr(name, '/', -1)-1 ) path,
       substr(name,instr(name, '/', -1)+1 )  fname,
       bytes/1048576  fsize,
       phyrds pr,
       phywrts pw
  from v$datafile df, v$filestat fs
 where df.file# = fs.file#
UNION
select substr(name,1,instr(name, '/', -1)-1 ) path,
       substr(name,instr(name, '/', -1)+1 ) fname,
       0 fsize,
       0 pr,
       0 pw
  from v$controlfile
```

```
UNION
select substr(lgf.member,1,instr(lgf.member,'/', -1)-1) path,
       substr(lgf.member,instr(lgf.member, '/', -1)+1 ) fname,
       lg.bytes/1048576 fsize,
       0 pr,
       0 pw
  from v$logfile lgf, v$log lg
 where lgf.group# = lg.group#
order by 1,2
/
spool off
exit
```

```
-------------------------Sample Output---------------------------------

Path                 File Name      M bytes Phy. Reads Phy. Writes
-------------------- -------------- ------------ ---------- -----------

/u01/oradata/PDB7    control3.ctl
                     rdmxf1.dbf            50        694        2781

                     redo1b.log            10

                     redo2b.log            10

                     redo3b.log            10

                     tempf1.dbf            30



/u02/oradata/PDB7    control1.ctl
                     rbsf1.dbf             40          6        2813

                     redo1a.log            10

                     redo2a.log            10

                     redo3a.log            10
```

Path	File Name	M bytes	Phy. Reads	Phy. Writes
/u03/oradata/PDB7	control2.ctl			
	genf1.dbf	50	1551	396
	rdmf1.dbf	100	4403	2335
	systemf1.dbf	50	5905	1116
	toolsf1.dbf	15	2979	

16 rows selected.

SQL TIPS(20): HIERARCHICAL DATA의 SORT와 ORDERING

Bulletin no : 10373

ORACLE6에서의 HIERARCHICAL QUERY에서는 SORT를 하기 위한 ORDER BY 절을 사용할 수 없었다. 그러나 ORACLE 7.1이상 VERSION에서는 USER-DEFINED STORED PROCEDURE를 이용하여 HIERARCHY순서로 출력되면서 ORDERING할 수 있게 되었다.

세계의 지역에 관한 자료를 예로 보자.

```
CREATE TABLE UNIVERSE(
 PARENT VARCHAR2(30) REFERENCES UNIVERSE,
 NAME  VARCHAR2(30) PRIMARY KEY
);

REM SOME TEST DATA
INSERT INTO UNIVERSE VALUES ( NULL, 'WORLD' ) ;
INSERT INTO UNIVERSE VALUES ( 'WORLD', 'EUROPE' ) ;
INSERT INTO UNIVERSE VALUES ( 'EUROPE', 'ENGLAND' ) ;
INSERT INTO UNIVERSE VALUES ( 'EUROPE', 'THE NETHERLANDS' ) ;
INSERT INTO UNIVERSE VALUES ( 'EUROPE', 'GERMANY' ) ;
INSERT INTO UNIVERSE VALUES ( 'WORLD', 'ASIA' ) ;
INSERT INTO UNIVERSE VALUES ( 'ASIA', 'JAPAN' ) ;
INSERT INTO UNIVERSE VALUES ( 'ASIA', 'CHINA' ) ;
INSERT INTO UNIVERSE VALUES ( 'WORLD', 'AMERICA' ) ;
INSERT INTO UNIVERSE VALUES ( 'AMERICA', 'UNITED STATES' ) ;
INSERT INTO UNIVERSE VALUES ( 'AMERICA', 'MEXICO' ) ;
INSERT INTO UNIVERSE VALUES ( 'WORLD', 'AFRICA' ) ;
INSERT INTO UNIVERSE VALUES ( 'AFRICA', 'EGYPT' ) ;
INSERT INTO UNIVERSE VALUES ( 'AFRICA', 'MOROCCO' ) ;
```

위의 자료를 다음과 같이 보고자 하는 경우

```
WORLD
        AFRICA
                EGYPT
                MOROCCO
        AMERICA
                MEXICO
                UNITED STATES
        ASIA
                CHINA
                JAPAN
```

```
        EUROPE
                ENGLAND
                GERMANY
                THE NETHERLANDS
```

만약,ORDER BY절이 없이 QUERY하면

```
     SELECT RPAD( ' ', LEVEL * 5 ) || NAME FROM UNIVERSE
      CONNECT BY PRIOR NAME = PARENT START WITH PARENT IS NULL;
```

다음과 같은 결과를 얻게 된다.

```
    WORLD
        EUROPE
                ENGLAND
                GERMANY
                THE NETHERLANDS
        ASIA
                JAPAN
                CHINA
        AMERICA
                UNITED STATES
                MEXICO
        AFRICA
                EGYPT
                MOROCCO
```

만약, 위 문장에 ORDER BY절을 사용하면

```
     SELECT RPAD( ' ', LEVEL * 5 ) || NAME FROM UNIVERSE
      CONNECT BY PRIOR NAME = PARENT START WITH PARENT IS NULL
      ORDER BY NAME;
```

다음과 같은 원치 않는 결과를 얻게 된다.

```
        AFRICA
        AMERICA
        ASIA
                CHINA
                EGYPT
                ENGLAND
        EUROPE
                GERMANY
                JAPAN
                 MEXICO
                MOROCCO
                THE NETHERLANDS
                UNITED STATES
     WORLD
```

7. 1이상 VERSION에서는 다음과 같이 USER DEFINED FUNCTION을 이용하여 원하는 자료를 얻을 수 있다.

```
CREATE OR REPLACE FUNCTION UNIVERSESORTORDER( PKEY UNIVERSE.NAME%TYPE )
     RETURN VARCHAR2 IS
        PATH VARCHAR2(2000);
     BEGIN
        PATH := PKEY;
        -- INSERT ALL PREVIOUS PARENT RECORDS LIKE A DIRECTORY STRUCTURE
        -- E.G. WORLD/EUROPE/...
        FOR CREC IN ( SELECT PARENT FROM UNIVERSE
                      CONNECT BY PRIOR PARENT = NAME
                      START WITH NAME = PKEY ) LOOP
           PATH := CREC.PARENT || '/' || PATH;
        END LOOP;
        RETURN PATH;
     END;
     /

SELECT SUBSTR( RPAD( ' ', LEVEL * 5) || NAME, 1, 40) "THE UNIVERSE"
      FROM UNIVERSE
      CONNECT BY PRIOR NAME = PARENT
      START WITH PARENT IS NULL
      ORDER BY UNIVERSESORTORDER( NAME )
;
```

SQL TIPS(21): DBA가 가지고 있어야 할 10개의 SQL

Bulletin no : 10374

```
============================================
 NO 1: Display the Current Archivelog Status
============================================
```

The command below, when entered via SQLDBA, will display the current archivelog status of the database. The script
will display the active log_archive_dest location, as well as the current and oldest online redo log sequence numbers.

```
ARCHIVE LOG LIST;
```

```
============================================
 NO 2: Creating a Control File Trace File
============================================
```

The command below, when entered via SQLDBA, will create a trace file in your trace file destination directory. The
trace file will contain the complete CREATE CONTROLFILE command for the current database, saving you vast
quantities of time if you ever need to generate that command.

```
ALTER DATABASE BACKUP CONTROLFILE TO TRACE;
```

```
============================================
 NO 3: Tablespace Free Extents and Free Space
============================================
```

The query below will ONLY work in Oracle7.2 and higher. In previous versions, it was difficult to query
DBA_DATA_FILES and DBA_FREE_SPACE in one query, since both could have multiple rows for each tablespace.
Using 7.2's FROM clause subquery feature, you can group the two queries separately, then join them on the
Tablespace_Name value. The result shows the largest free extent in the tablespace, the number of free extents in the
tablespace, the total free space in the tablespace, and the percentage of the tablespace's available space that is free.

```
column Tablespace_Name format A20
column Pct_Free format 999.99
```

```
select Tablespace_Name,
Max_Blocks,
Count_Blocks,
Sum_Free_Blocks,
100*Sum_Free_Blocks/Sum_Alloc_Blocks AS Pct_Free
from
(select Tablespace_Name, SUM(Blocks) Sum_Alloc_Blocks
from DBA_DATA_FILES
group by Tablespace_Name),
   (select Tablespace_Name FS_TS_NAME,
   MAX(Blocks)  AS Max_Blocks,
   COUNT(Blocks)  AS Count_Blocks,
   SUM(Blocks) AS Sum_Free_Blocks
   from DBA_FREE_SPACE
   group by Tablespace_Name)
where Tablespace_Name = FS_TS_NAME;
```

```
================================================
 No. 4: Display Allocated Space & Quota by User
================================================
```

DBA_TS_QUOTAS displays the allocated space, by user, in each tablespace, along with the quota for each of those
users. If a user has unlimited quota in a tablespace, the quota will be displayed as 0 or negative. Running this script is a
great quick way to see which users own objects in which tablespaces, and how close they are to their quotas. You'll
need to SET LINESIZE 132 before running this script.

```
select * from DBA_TS_QUOTAS
order by Tablespace_Name, Username;
```

```
================================================
 No  5: Show Allocated Storage for All Objects
================================================
```

DBA_SEGMENTS shows the allocated storage for all objects in the database. Valid values for the &segment_type
variable include:

TEMPORARY--for currently used temporary segments. Watch them grow!
ROLLBACK--for rollback segments
INDEX--for indexes
TABLE--for tables

There is one record in DBA_SEGMENTS for each segment. There is a one-to-many relationship between
DBA_SEGMENTS and DBA_EXTENTS: DBA_SEGMENTS shows you the total space allocated to a segment:

DBA_EXTENTS shows the space allocated to each extent within the segment.

```
column Segment_Name format A40

 select Segment_Name, Extents, Blocks
 from DBA_SEGMENTS
 where Segment_Type = '&segment_type'
 order by Segment_Name;
```

```
============================================
 No. 6: Map a Tablespace's Used and Free Space
============================================
```

Since DBA_EXTENTS shows the starting block of each extent (the combination of File_ID and Block_ID), plus the
length of each extent (the Blocks column), you can map out the allocated space within a given tablespace as a series of
consecutive extents. DBA_FREE_SPACE shows the free extents, so the combination of DBA_EXTENTS and
DBA_FREE_SPACE produces a map of the used and unused free space within a tablespace. You can use this map of
the tablespace's space to determine whether an object has been properly sized or if the tablespace needs to be
defragmented.

```
select Segment_Name, File_Id, Block_Id, Blocks
 from DBA_EXTENTS
 where Tablespace_Name = '&&tablespacename'
 UNION
select 'Free Space', File_Id, Block_Id, Blocks
 from DBA_FREE_SPACE
 where Tablespace_Name = '&&tablespacename'
 order by 2,3;
```

```
======================================
 No. 7: Blocks Used by Data in a Table
======================================
```

When you replace "sometable" with the name of a table in your database, the query below will return the number of
blocks used by the data in that table. The query examines the Block_ID (columns 1-8) and the File_ID (columns 15-18)
of the ROWID. The query then returns the number of distinct blocks used by the records within the table. The
ANALYZE command can also provide this information, but this query is faster.

```
select COUNT(DISTINCT(SUBSTR(ROWID,1,8)||SUBSTR(ROWID,15,4)))
      Blocks_Used
```

```
 from sometable;
```

```
================================
 No. 8: Reset a User's Password
================================
```

The script below generates an ALTER USER command that will reset a user's password (the user whose username is
passed to the script as &&1). If you spool the output of this command, you will be able to reset the user's password
without ever having to know what the password was. You can change the user's password to anything you like, use the
account for testing purposes, and then reset the password when your testing is complete. For a detailed discussion, see
Chapter 9 of the Oracle DBA Handbook.

```
select 'ALTER USER &&1 IDENTIFIED BY VALUES '
 ||''''||Password||''''||';'
  from DBA_USERS where Username = UPPER('&&1');
```

```
=========================
 No. 9: Query V$PARAMETER
=========================
```

Don't try to guess what parameters are in effect. You can never tell for sure which INIT.ORA file was used to start an
instance, or if the INIT.ORA file has been modified since the database was started. The only sure way to tell the setting
of an initialization parameter is to query V$PARAMETER.

```
column Name format A50
column Value format A28

select Name, Value from V$PARAMETER;
```

```
====================================
 No. 10: Show Statement Execution Path
====================================
```

This query selects the execution path from the PLAN_TABLE, ordered according to the hierarchy of the steps
involved. The query shown in the listing specifically queries for the execution path of a statement using the
Statement_ID of 'TEST'. Prior to running this query, you should use the EXPLAIN PLAN command to populate
PLAN_TABLE.

```
select
 LPAD(' ',2*LEVEL)||Operation||' '||Options
```

```
                ||' '||Object_Name   Q_PLAN
from PLAN_TABLE
where Statement_ID = 'TEST'
connect by prior ID = Parent_ID and Statement_ID = 'TEST'
start with ID=1;
```

SQL TIPS(22): DETAILED LOOK AT DATABASE INDEX

Bulletin no : 10376

This script prompts the user for an index name and owner, and provides a detailed look at any index in the database by
using the little known index_stats view.

```
=====================================================set verify off
set pagesize 23
set linesize 80
set pause on
set pause 'Hit enter to continue'
set feedback off
set showmode off
set echo off

set space 0
set heading off

set termout off pause off
column blsk new_value BLOCK_SIZE_K
select value / 1024 blsk
from v$parameter
where name = 'db_block_size';
set termout on pause on

PROMPT
ACCEPT USER_INPUT1 CHAR PROMPT 'Please enter an index to analyze:>'
ACCEPT USER_INPUT2 CHAR PROMPT 'Please enter the owner of the index:>'
PROMPT
PROMPT Working, Please wait.....
PROMPT

analyze index &USER_INPUT2..&USER_INPUT1 validate structure;

col ROWS_PER_KEY          format 999.99     heading '      '
col BLKS_GETS_PER_ACCESS  format 99,999.99  heading '         '
```

```
select
'********************************************************************************',
'                                  Name of the index = ' || NAME,
'                                 Height of the b-tree = ' || HEIGHT,
'                         Blocks allocated to the index = ' || BLOCKS,
'            Number of leaf rows (values in the index) = ' || LF_ROWS,
'                  Number of leaf blocks in the b-tree = ' || LF_BLKS,
'               Sum of the lengths of all the leaf rows = ' || LF_ROWS_LEN,
'                         Useable space in a leaf block = ' || LF_BLK_LEN,
'                                 Number of branch rows = ' || BR_ROWS,
'                 Number of branch blocks in the b-tree = ' || BR_BLKS,
'Sum of lengths of all the branch blocks in the b-tree = ' || BR_ROWS_LEN,
'                       Useable space in a branch block = ' || BR_BLK_LEN,
'             Number of deleted leaf rows in the index = ' || DEL_LF_ROWS,
'        Total length of all deleted rows in the index = ' || DEL_LF_ROWS_LEN,
'                  Number of distinct keys in the index = ' || DISTINCT_KEYS,
'     How many times the most repeated key is repeated = ' ||
MOST_REPEATED_KEY,
'        Total space currently allocated in the b-tree = ' || BTREE_SPACE,
'Totl space that is currently being used in the b-tree = ' || USED_SPACE,
'% of space allocated in the b-tree that is being used = ' || PCT_USED,
'              Average number of rows per distinct key = ', ROWS_PER_KEY,
'Expected number of consistent mode block gets per row = ',
BLKS_GETS_PER_ACCESS, ' '
from index_stats;

col NAME                 format a30       fold_after
col dummy_col_0                           fold_after
col dummy_col_1                           fold_after
col dummy_col_2                           fold_after
col LF_ROWS              format 9,999,990           heading '         '
col DEL_LF_ROWS          format 9,999,990           heading '         '
col BR_ROWS              format 9,999,990           heading '         '
col DISTINCT_KEYS        format 9,999,990           heading '         '
col MOST_REPEATED_KEY    format 9,999,990           heading '         '
col ROWS_PER_KEY         format 9,999,990           heading '         '
col HEIGHT               format 0         fold_after heading '          '
col BLKS_GETS_PER_ACCESS format 99,999.99 fold_after heading '         '
col LF_BLK_LEN           format 9,990     fold_after heading '     '
col lf_row_size          format 990       fold_after heading '      '
col lf_row_per_blk       format 990       fold_after heading '      '
col BR_BLK_LEN           format 9,990     fold_after heading '     '
col br_row_size          format 990       fold_after heading '      '
col br_row_per_blk       format 990       fold_after heading '      '

col meg                  format 999.90    fold_after heading '      '
```

```
col lf_meg              format 999.90              heading '      '
col lf_meg_pct          format 99.90     fold_after heading '      '
col br_meg              format 999.90              heading '      '
col br_meg_pct          format 99.90     fold_after heading '      '
col uu_meg              format 999.90              heading '      '
col uu_meg_pct          format 99.90     fold_after heading '      '
col bt_meg              format 999.90              heading '      '
col bt_meg_pct          format 99.90     fold_after heading '      '
col btu_meg             format 999.90              heading '      '
col btu_meg_pct         format 99.90     fold_after heading '      '
col btuu_meg            format 999.90              heading '      '
col btuu_meg_pct        format 99.90     fold_after heading '      '
col btuub_meg           format 999.90              heading '      '
col btuub_meg_pct       format 99.90     fold_after heading '      '

select
 '********************************************************************************'
dummy_col_0,
 'Index Name.........', NAME, ' ' dummy_col_1,
 'Leaf Rows..........', LF_ROWS, '      ',
 'Leaf Block Size........', LF_BLK_LEN,
 'Deleted Leaf Rows..', DEL_LF_ROWS, '      ',
 'Leaf Row Size..........', LF_ROWS_LEN / LF_ROWS lf_row_size,
 'Branch Rows........', BR_ROWS, '      ',
 'Leaf Rows Per Block....', LF_BLK_LEN / (LF_ROWS_LEN / LF_ROWS) lf_row_per_blk,
 'Distinct Keys......', DISTINCT_KEYS, '      ',
 'Branch Block Size......', BR_BLK_LEN,
 'Max Common Key.....', MOST_REPEATED_KEY, '      ',
 'Branch Row Size........', BR_ROWS_LEN / (BR_ROWS + 1) br_row_size,
 'Avg Common Key.....', ROWS_PER_KEY, '      ',
 'Branch Rows Per Block..', BR_BLK_LEN / ((BR_ROWS_LEN / (BR_ROWS + 1)) + 1)
br_row_per_blk,
 'Height Of B-Tree...', HEIGHT,
 'Reads Per Access...', BLKS_GETS_PER_ACCESS,
 ' ' dummy_col_2,
 'Index Meg.................', (BLOCKS * &BLOCK_SIZE_K) / 1024 meg,
 'Leaf Meg/Pct.............', (LF_BLKS * &BLOCK_SIZE_K) / 1024 lf_meg, ' /',
                            (LF_BLKS / BLOCKS) * 100 lf_meg_pct,
 'Branch Meg/Pct...........', (BR_BLKS * &BLOCK_SIZE_K) / 1024 br_meg, ' /',
                            (BR_BLKS / BLOCKS) * 100 br_meg_pct,
 'Unused Meg/Pct...........', ((BLOCKS - (LF_BLKS + BR_BLKS)) *, &BLOCK_SIZE_K) / 1024 uu_meg,
' /',
                            ((BLOCKS - (LF_BLKS + BR_BLKS)) / BLOCKS) * 100 uu_meg_pct,
 'B-Tree Meg/Pct...........', ((LF_BLKS + BR_BLKS) * &BLOCK_SIZE_K) / 1024 bt_meg, ' /',
                            ((LF_BLKS + BR_BLKS) / BLOCKS) * 100 bt_meg_pct,
 'B-Tree Used Meg/Pct......', (USED_SPACE / 1024) / 1024 btu_meg, ' /',
```

```
                              PCT_USED btu_meg_pct,
 'B-Tree UnUsed Meg/Pct....', ((BTREE_SPACE - USED_SPACE) / 1024) / 1024 btuu_meg, ' /',
                              100 - PCT_USED btuu_meg_pct,
 'B-Tree UnUsable Meg/Pct..', (DEL_LF_ROWS_LEN / 1024) / 1024 btuub_meg, ' /',
                              (((DEL_LF_ROWS_LEN / 1024) / 1024) /
                              (((LF_BLKS + BR_BLKS) * &BLOCK_SIZE_K) / 1024))
                              * 100 btuub_meg_pct from index_stats;

exit;
=======================================================
```

SQL TIPS(23): EXPORTING TABLES BASED ON TABLESPACE NAME

Bulletin no : 10377

This script allows the user to enter the tablespace name and creates a list of tables in exp format. Additional choices
allow the export of over-extended tables or tables over a certain size.

```
*****************************************************************
rem list.sql
rem Give list of tables depending on parameters

set termout on linesize 75 pagesize 0 verify off feedback off
clear buffer
clear columns
prompt
prompt This script will create list.out, which can be used
prompt to give export tables parameter file
prompt

accept Segment_Name Prompt 'Enter Seg Name              : '
accept owner        Prompt 'Enter Seg Owner             : '
accept tsname       prompt 'Enter TableSpace Name       : '
accept min_extents  Prompt 'Enter Minimum num of exts   : '
accept min_size     Prompt 'Enter Minimum seg size(M)   : '
prompt
accept expchoice    Prompt 'Is this list for exports ?  : '

col srt                         noprint
SPOOL list.out

-- start bit
select 0 srt, decode(upper('&expchoice'),'Y','tables=(','')
from   dual
union
-- data bit without end segment
select 1 srt, segment_name||decode(upper('&expchoice'),'Y',',','')
from   dba_segments
where  segment_type    = 'TABLE'
and    segment_name    like upper('&&Segment_Name%')
```

```
and    owner           like upper('&&owner%')
and    tablespace_name like upper('&&tsname%')
and    extents         >= (&&min_extents + 0)
and    bytes           >= (&&min_size + 0) * (1024*1024)
minus
-- remove last one (with ',' after the name)
select 1 srt, max(segment_name)||decode(upper('&expchoice'),'Y',',','')
from   dba_segments
where  segment_type    = 'TABLE'
and    segment_name    like upper('&&Segment_Name%')
and    owner           like upper('&&owner%')
and    tablespace_name like upper('&&tsname%')
and    extents         >= (&&min_extents + 0)
and    bytes           >= (&&min_size + 0) * (1024*1024)
union
-- data bit with end segment
select 3 srt, max(segment_name)||decode(upper('&expchoice'),'Y',')','')
from   dba_segments
where  segment_type    = 'TABLE'
and    segment_name    like upper('&&Segment_Name%')
and    owner           like upper('&&owner%')
and    tablespace_name like upper('&&tsname%')
and    extents         >= (&&min_extents + 0)
and    bytes           >= (&&min_size + 0) * (1024*1024)
order by 1,2
/

spool off

set heading on
**********************************************************************
```

SQL TIPS(24): CALCULATING HIT RATIO

Bulletin no : 10378

Calculating hit ratios is sometimes a drudgery. This little script helps and requires no parameters. The script uses
PL/SQL.

```
=====================================================
rem
rem  le hit ratio
rem
col global_name  for A20
variable d number
variable c number
variable r number
begin
select value into :c from v$sysstat
where name = 'consistent gets';

select value into :d from v$sysstat
where name = 'db block gets';
select (1- value/( :c + :d)) into :r   from v$sysstat where name in
   ( 'physical reads');

end;
/
select  global_name, :r "Hit Ratio by ChMe" from global_name ;
=====================================================
```

SQL TIPS(25): ANALYZING TABLESPACE USAGE WITH AN OS-INDEPENDENT SQL SCRIPT

Bulletin no : 10379

When tables, indexes, and rollback segments are created, they are pre-assigned a storage allocation (extent) that is
reserved and cannot be used by any other object. Although these objects may not use all of the space allocated at
creation, the amount of available free space diminishes over time as more information is placed into the database. The
following query helps you instantly find application problems resulting from space shortage. Run this query at regular
intervals to obtain the best information. Note that this script assumes that you have Oracle data blocks of any size m*K,
where 'm' is an positive integer and 'K' equals 1,024 bytes.

The query uses a PL/SQL variable called VAR_BYTES to eliminate the operating system-dependent feature of the size
of the Oracle data blocks. You will not have to modify the query; this SQL script is operating system-independent.

```
REM
REM Drop TABLESPACETEMP before creation.
REM

DROP TABLE TABLESPACETEMP;

REM
REM Create TABLESPACETEMP.
REM

CREATE TABLE TABLESPACETEMP
                          (TSPACE             CHAR(22),
                            FILE_ID           NUMBER(5),
                            TOT_MB              NUMBER(10,2),
                            ORA_BLKS          NUMBER(10,2),
                            TOT_USED          NUMBER(10,2),
                            PCT_USED          NUMBER(10,2));
```

```
COMMIT;

DECLARE
        VAR_BYTES   NUMBER(9);

        CURSOR FIND_INFO IS
        SELECT SUBSTR(D.TABLESPACE_NAME,1,22)      TSPACE,
            LTRIM(TO_CHAR(D.FILE_ID,'99990'))                          FILE_ID,

        LTRIM(TO_CHAR(D.BYTES/1024/1024,'99999990.99'))

                  TOT_MB,
            LTRIM(TO_CHAR(D.YTES,'99999990.99'))                 ORA_BLKS,
            LTRIM(TO_CHAR(SUM(E.BLOCKS),'99999990.99'))  TOT_USED,
            LTRIM(TO_CHAR(ROUND(SUM(E.BLOCKS)
                /(D.BYTES),4)*100,'99999990.99'))                  PCT_USED
           FROM         SYS.DBA_DATA_FILES D,
                        SYS.DBA_EXTENTS E
        WHERE   D.FILE_ID = E.FILE_ID (+)
         GROUP BY       D.TABLESPACE_NAME,
                        D.FILE_ID,
                        D.BYTES/1024/1024,
                        D.BYTES;

BEGIN
        FOR SEARCH IN FIND_INFO
        LOOP
                    SELECT  DISTINCT BYTES/BLOCKS
                        INTO        VAR_BYTES
                       FROM         SYS.DBA_SEGMENTS;

                    INSERT INTO TABLESPACETEMP VALUES
                       (SEARCH.TSPACE,
                                SEARCH.FILE_ID,
                                SEARCH.TOT_MB,
                    SEARCH.ORA_BLKS/VAR_BYTES,
                                SEARCH.TOT_USED,
                    SEARCH.PCT_USED*VAR_BYTES);
        END LOOP;
END;
/

REM
REM Now, you can select information from TABLESPACETEMP without
REM modifying anything.
REM
```

```
SELECT * FROM TABLESPACETEMP;

REM
REM You may drop TABLESPACETEMP!
REM

DROP TABLE TABLESPACETEMP;
```

SQL TIPS(26): CREATING A DATABASE LAYOUT

Bulletin no : 10388

The script below is one I used quite frequently when I was with Oracle Consulting. It is handy for walking into a
company and getting a quick layout of their database. I call it dbmap.sql.

```
spool dbmap.txt;
prompt
prompt ======================================
prompt   Tablespace/Datafile Listing
prompt ======================================
prompt
prompt
column  "Location"              format A30;
column  "Tablespace Name"       format A15;
column  "Size(M)"               format 999,990;

break on "Tablespace Name" skip 1 nodup;
compute sum of "Size(M)" on "Tablespace Name";

SELECT  tablespace_name "Tablespace Name",
        file_name "Location",
        bytes/1048576 "Size(M)"
FROM sys.dba_data_files
order by tablespace_name;

prompt
prompt ======================================
prompt   Redo Log Listing
prompt ======================================
prompt
prompt
column  "Group"         format 999;
column  "File Location" format A40;
column  "Bytes (K)"     format 99,990;

break on "Group" skip 1 nodup;
```

```
select  a.group# "Group",
        b.member "File Location",
        a.bytes/1024 "Bytes (K)"
from    v$log a,
        v$logfile b
where a.group# = b.group#
order by 1,2;


prompt
prompt ======================================
prompt   Rollback Listing
prompt ======================================
prompt
prompt
column  "Segment Name"  format A15;
column  "Tablespace"    format A15;
Column  "Initial (K)"   Format 99,990;
Column  "Next (K)"      Format 99,990;
column  "Min Ext."      FORMAT 990;
column  "Max Ext."      FORMAT 990;
column  "Status"        Format A7;

select  segment_name "Segment Name",
        tablespace_name "Tablespace",
        initial_extent/1024 "Initial (K)",
        next_extent/1024 "Next (K)",
        min_extents "Min Ext.",
        max_extents "Max Ext.",
        status "Status"
from    sys.dba_rollback_segs
order by tablespace_name,
        segment_name;
spool off;
```

SQL TIPS(27): DECODE 이용사례 (2)

Bulletin no : 10396

다음과 같은 자료를 갖는 TABLE을 가정해보자.

```
Employee Name     Employee Address   City         State     ZIP
-------------     ----------------   -----------  -----    ---------
Donald Duck       101 Cool Pond      Bedford      OR       12345
Bugs Bunny        540 Carrot Way     Atlanta      GA       29292
Elmer Fudd
Taz Devil         999 Wildwoods      Nashville    TN       99999
```

REPORT를 생성할 때는 종종 CITY와 STATE를 COMMA(,)로 연결하여 출력하곤한다.
이때 다음과 같이 COLUMN간에 CONCATENATING를 이용하면 가능하다.

```
select employee_name,
       employee_address,
       city || ', ' || state || ' ' || zip  city_state_zip
  from decode_sample;
```

위의 문장은 다음과 같은 결과를 같는다.

```
EMPLOYEE_NAME        EMPLOYEE_ADDRESS     CITY_STATE_ZIP
-------------------- -------------------- ----------------------
Donald Duck          101 Cool Pond        Bedford, OR 12345
Bugs Bunny           540 Carrot Way       Atlanta, GA 29292
Elmer Fudd                                ,
Taz Devil            999 Wildwoods        Nashville, TN 99999
```

위와 같은 경우에서 주소의 자료가 없는 경우 필요없는 COMMA만 출력되게 된다.
이런경우에 DECODE 을 이용하여 다음과 같이 처리하면 깔끔한 OUTPUT을 받을 수 있게 된다.

```
select employee_name,
       employee_address,
       city || decode(', ' || state,', ','',', ' || state) || ' '
            || zip  city_state_zip
  from decode_sample;

EMPLOYEE_NAME        EMPLOYEE_ADDRESS     CITY_STATE_ZIP
-------------------- -------------------- ----------------------
Donald Duck          101 Cool Pond        Bedford, OR 12345
Bugs Bunny           540 Carrot Way       Atlanta, GA 29292
Elmer Fudd
Taz Devil            999 Wildwoods        Nashville, TN 99999
```

SQL TIPS(28): DECODE 이용사례 (3)

Bulletin no : 10397

예 : salary가 1100보다 큰 경우엔 (+)를
같은 경우엔 (=)를
작은 경우엔 (-)를 표시하려면?

```
select  DECODE(sign(sal-1100),0,'(=)',1,'(+)','(-)'),SAL from emp;

DEC        SAL
--- ----------
(-)        800
(+)       1600
(+)       1250
(+)       2975
(+)       1250
(+)       2850
(+)       2450
(+)       3000
(+)       5000
(+)       1500
(=)       1100
(-)        950
(+)       3000
(+)       1300
```

(V1.1) REPORTWRITER 에서 LOCAL PRINTER로 출력하는 방법

Bulletin no : 10001

ReportWriter에서 단말기에 연결된 printer로 출력하는 방법

---

forms30에서 host문을 이용하여 ReportWriter를 호출할때 printer bypass on/off에 해당하는 escape sequence를 parameter로 전송한다.
ReportWriter에서 file로 출력한 후 cat을 이용하여 printer로 전송한다.

on : ^[[5i : fast단말기 기준
off : ^[[4i

주의; emulator를 사용할 경우 해당 emulator에서 나타내는 bypass on/off 의 값을 확인하여야 한다.

ex> ReportWriter : . menu bar -> parameter 에서 P_ON, P_OFF parameter를 add한다.(type :char)
. menu bar -> parameter 에서 destination을 file로 지정한다.
screen으로 지정할 경우, 출력시 holding되는 현상이 생기는 경우 발생하기 때문
. menu bar -> text에서 page header,tailor에 각각 &P_ON,&P_OFF를 등록한다.

forms30 : trigger에setting

```
begin
   select chr(27)||chr(91)||chr(53)||chr(105),
          chr(27)||chr(91)||chr(52)||chr(105)
   into :global.P_ON,:global.P_OFF from dual;
   host('runrep k1 userid=scott/tiger term=vt100
         paramform=no P_ON=' ||:global.P_ON||'
         P_OFF='||:global.P_OFF||' ',no_screen);
   host('cat k1.lis',no_screen);
   redisplay;
end;
```

(V1.1) 출력물을 현 단말기에 연결된 PRINTER로 출력하는 방법

Bulletin no : 10114

---

* Forms30에서 Host문을 이용하여 ReportWriter를 호출할 때 Printer bypass on/off에 해당하는 Escape Sequence를 Parameter로 전송한다.

* ReportWriter에서 File로 출력한 후 Cat을 이용하여 Printer로 전송한다.
  on : [[5i : fast단말기 기준
  off : [[4i

[주의] Emulator를 사용할 경우 해당 Emulator에서 나타내는 bypass on/off값을 확인하여야 한다.

예) * ReportWriter
- Menu Bar : Parameter에서 P_ON, P_OFF parameter를 Add한다.(type :char)
- Menu Bar : Parameter에서 Destination을 FIle로 지정한다. Screen으로 지정할 경우, 출력시 Holding되는 현상이 생기는 경우가 발생하기 때문.
- Menu Bar Text에서 Page Header, Tailor에 각각 &P_ON, &P_OFF를 등록.

```
* Forms30 - Trigger 에 Setting
begin
   select chr(27)||chr(91)||chr(53)||chr(105),
          chr(27)||chr(91)||chr(52)||chr(105)
   into :global, P_ON, :global.P_OFF from dual;
    host ('runrep k1 userid=scott/tiger term=vt100 paramform=no P_ON=
   '||:global.P_ON||' P_OFF='|| :global.P_OFF||';catk1.lis',no_screen);
    redisplay;
end;
```

(V1.1) PAGE HEIGHT가 66보다 작은 SPECIAL FORM지에 출력하는 방법

Bulletin no : 10115

---

* Height를 33으로 가정할 때 : 예를 들어 연속지로용지 출력시

1. 우선 Report Source에서 Page Height를 33으로 Setting한다.
   Menu Bar * Report * Height조정

2. $ORACLE_HOME/sqlreport/admin/printer directory에서 Page Height = 33인 Printer File을 생성한다. (Printerdef Utility 이용)

   * SQL*Report에서 Report의 Height를 조정하고 Printer File의 Page Height를 조정하여도 실제로 출력해 보면 66Line을 기준으로 Form Feed가 일어나게 된다.

   * 이것은 SQL*Report에서는 33Line에 한번씩 Form Feed를 보내주지만 실제 Printer(H/W)에서는 한 Page의 Length를 66으로 인식하고 있기 때문이다.

3. 위의 현상을 방지하기 위해서는 Printer에서의 Page Length Setting을 33 Line으로 Setting하여야 한다. 만약 Priinter에서 Page Setting하는 기능이 없다면 다음의 방법을 참조한다.

   * 실제로, SQL*Report에서 출력을 File로 받아보면 해당 Page의 끝에^(Control L :form Feed)이 삽입되어 있다. 이것은 SQL*Report Tool에서 붙여주는 Character 로써 User가 임의대로 처리할 수가 없다.
   * User가 위에서의 질문처럼 임의대로 Page Length를 조정하려면 3가지 정도의 방법이 있다.

[방법1] 생성된 File에서 Form Feed Character(^L)의 기능을 없애는 방법.
- 출력하고자 하는PrinterDescription($ORACLE_HOME/sqlreport/admin/printer 에 위치)에서 ff값을 다른 것으로 교체합니다. 이것은 SQL*Report에서 날아오는 ff값이 기능을 상실되므로 빈 한줄이 넘어가는 결과를 갖게 됩니다.

[방법2] Unix Command에서 Form Feed Character의 기능을 없애는 방법.
- SQL*Report에서는 정상적으로 출력하고, 내부에서 실행되는 Spoolcmd (SQL*Report 에서 Printer 출력시 Unix System에서 실제로 출력하는 부분임.)내부에서 lp나 lpr command를 사용하기 전에 Unix의 Sed Command나 C Program을 이용하여 ff(^L)값이 들어가 있는 Line을 삭제하거나 ff값을 삭제합니다.

[방법3] Report에서 매 Page에서 Printer의 Page Length를 33으로 Setting.
- Page마다 Page Length를 33으로 Setting on/off esc key 값을 Settiing.
  Report의 맨 앞에 맨 뒤에 Page Length Setting on/off 값을 Setting.
  (SQL*Report에서 Esc Key 값 Setting하는 참고자료 참조 할 것.)

(V2.5) ASCII DRIVER로 WIDE FORMAT 출력

Bulletin no : 10126

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1. Oracle Reports ASCII dirver를 install한다.

(1) orawin\reports25\printers\asciidrv.txt를 참조.
(2) orawin\reports25\printers directory하에 사용하실 수 있는 printer destination format file 들이 존재한다.
- default : dflt.prt 사용, 80column용
- 80 column이상을 출력할 경우 : wide.prt, wide180.prt를 사용 (general)
- 일반적인 PC Editor를 사용하여 필요에 따라 새로운 .prt file을 생성하여도 된다.

2. Reports2.5 Runtime Module을 사용 할 경우

(1) Mode = Character
(2) Desformat = Wide or Wide180

3. Reports2.5 Designer Module를 사용 할 경우

(1) Tools * Properties * Character Mode : Page Width 를 조정
(2) Tools * Tools Option * Runtime Parameters : Mode = Character로 변경
(3) File * Print Setup * Oracle Reports ASCII Driver 선택: 출력방향이나 용지선택은 할 필요가 없다.
(4) Default Parameter Form * Desformat 을 첨가

(V1.1) SQL*REPORTWRITER 1.1에서 프로그램의 BACK-UP 및 RESTORE 방법

Bulletin no : 10142

SQL*ReportWriter에서는 사용자가 DESIGN MODULE인 SQLREP를 통하여 프로그램을 작성한 후 컴파일 시 또는, GENREP를 수행할 경우에도 실행 모드인 *.REP 화일(예, TEST1.REP)만을 생성하고 SOURCE 화일을 만들지 않는다.

따라서, 작성한 SQL*ReportWriter 프로그램의 SOURCE는 DUMPREP를 이용하여 생성하여야 한다. DUMPREP는 데이타베이스로부터 SQL*ReportWriter의 REPORT DEFINITION을 ASCII 화일(*.REX 확장자를 가짐)인 백업 화일을 만든다.

이러한 ASCII 화일인 BACK-UP 화일을 ORACLE 데이타베이스로 로드하기 위해서는 LOADREP를 이용한다.

1) DUMPREP 사용 방법

```
$ dumprep report=report_name userid=scott/tiger file=backup_file
    report  : SQL*ReportWriter 프로그램의 이름(*.rep 화일의 이름)
    userid  : ORACLE userid/password
    file    : 로드 화일의 이름
```

2) LOADREP 사용 방법

SQL*ReportWriter 프로그램의 백업 화일을 ORACLE 데이타베이스에 로드시키는 명령이다.

```
$ loadrep file=backup_file userid=scott/tiger
    file     : 로드할 백업 화일(*.REX)의 이름
    userid   : ORACLE userid/password
```

(V1.1) SQL*REPORTWRITER 1.1에서의 TERMINAL FUNCTION KEY 수정방법

Bulletin no : 10143

여기서는 SQL*ReportWriter의 SQLREP, RUNREP를 수행할때 사용되는 TERM타입을 설정하는 방법을 살펴보기로 한다.

SQL*R/W는 TERMINAL과 KEYMAPPING을 식별하기 위하여 TERMINAL/KEY FILE이라는 엔코드된 화일을 사용하며 이러한 TERMINAL DEFINITION은 TERMDEF 유틸리티를 사용하여 생성하거나 수정한다.

SQL*R/W는 $ORACLE_HOME/sqlreport/admin/terminal 디렉토리 아래에 기본적인 TERMINAL DEFINITION(termdef.dat)및 KEY DEFINITION(srw_100.kdf) 화일을 제공하며, 이 화일을 이용하거나 특정한 TERMINAL DEFINITION을 생성할 수 있다.

KEY DEFINITION 화일에는 SQL*R/W의 FUNCTION NAME, FUNCTION에 대한 KEY ASSIGNMENT 그리고 KEY에 대한 CHARACTER SEQUENCE를 가지고 있다. 예를 들어, srw_100.kdf 화일은 다음과 같이 구성되어 있다.

```
------------------------------------------------------------------------
   ; Function            ASCII Code     Key Assignment
   ; --------            ----------     --------------
   right:                 \E[C  : Right
   left:                  \E[D  : Left
   accept                ^XA           : ^X A
   delete backward:       \010         : BS
   delete character:       ^D   : ^D
   next word:            ^W^N   : ^W ^N
   previous word:        ^W^P   : ^W ^P
   delete word:          ^W^D   : ^W ^D
   undelete word:        ^W^U   : ^W ^U
------------------------------------------------------------------------
```

(표-1 SRW_100.KDF 화일의 예)

TERMINAL DEFINITION 화일에는 TERMDEF 유틸리티에서 참조하는 TERMINAL NAME과 CLEAR SCREEN, TERMINAL에 대한 INITIALIZATION STRING,DISPALY 되는 컬럼의 수, 필드에 대한 HIGHLIGHT CODE 등의 TERMINAL CAPABILITY CODE를 가지고 있다.

```
------------------------------------------------------------------------
# VT100.
d0¦vt100¦vt-100¦pt100¦pt-100¦dec vt100:\
        :is=\017\E(B\E)0\E[?3l\E[20l\E[4l\E[m\E[?6h:\
        :cl=\E[;H\E[2J:ce=\E[K:\
        :co#80:li#24:bs:cm=\E[%i%d;%dH:nd=\E[C:up=\E[A:do=J:do=\E[B:\
        :ks=\E=:ke=\E>:\
        :gl=\244j\234k\204l\214m\250n\236q\215t\245u\247v\237w\212x:
```

(표-2 TERMDEF.DAT 화일의 예)

다음은 실제 FUNCTION KEY 수정방법을 살펴보기로 한다.
- SRW_100.KDF 화일에서 [ACCEPT] FUNCTION KEY를 [F3] KEY로 수정하는 방법을 살펴보자.

(1) $ORACLE_HOME/sqlreport/admin/terminal 디렉토리 아래 SRW_100.KDF 화일을 에디터로 오픈한다.
- 예를 들어 VT220 타입을 수정시는 SRW_220.KDF 화일을 이용한다.

(2) 표-1의 SRW_100.KDF 화일에서 [ACCEPT] FUNCTION에 대한 ASCII CODE와 KEY ASSIGNMENT를 변경한다.

- [ACCEPT]의 KEY ASSIGNMENT를 "F3"으로 변경
- [ACCEPT]의 ASCII CODE를 실제로 [F3] KEY에서 발생되는 CHARACTER SEQUENCE인 " ^[OR " 로 변경
- 수정할 FUNCTION KEY를 위의 방법과 동일하게 KEY ASSIGNMENT와 ASCII CODE를 변경

(3) SRW_100.KDF의 KEY DEFINITION 화일을 수정한 다음 TERMDEF 유틸리티를 사용하여 새로운 TERM 타입을 생성한다.

```
$ termdef terminal=vt100 outfile=new_100 infile=srw_100.kdf
  userid=system/manager tdfile=termdef.dat
```

- terminal : TERMDEF.DAT 화일에서 지정된 TERMINAL NAME으로 두 번째 인수가 해당된다.
  outfile : 생성될 새로운 TERM 타입
  infile : 수정한 KEY DEFINITION 화일 (예, SRW_100.KDF 화일)
  userid : ORACLE USER ID
  tdfile : TERMINAL DEFINITION 화일 (예, TERMDEF.DAT 화일)

(4) TERMDEF을 사용하여 생성된 TERM 타입을 SQLREP, RUNREP 시에 지정한다

```
$ sqlrep term=new_100 userid=scott/tiger
```

(V1.1) SQL*REPORTWRITER 1.1에서 PRINTER DRIVER DEFINITION 수정방법

Bulletin no : 10144

SQL*ReportWriter의 SQLREP, RUNREP에서 프로그램 출력시 제어되는 PRINTER DRIVER DEFINITION을 수정하는 방법을 살펴보자. 새로운 타입의 DRIVER를 생성하기 위해서는 PRINTER DEFINITION FILE과 이 화일을 참조하는 PRINTDEF 명령을 수행하여야 한다.

PRINTER DEFINITION 화일에는 SQL*R/W 프로그램을 프린터로 출력시 사용되는 한 LINE의 컬럼 수, CARRIAGE RETURN PRINTER, PRINTER 초기화 값 등의 제어 코드를 포함하고 있다. 예를 들어, PRINTER DEFINITION인 printdef.dat($ORACLE_HOME/sqlreport/admin/printer 디렉토리에 존재)은 다음과 같이 구성되어 있다.

```
------------------------------------------------------------------------
# DEC -- DEC printers in portrait mode
DC¦DEC¦DEC printers:\                  <------------------------ (A)

      :ff=^L:cr=^M:nl=^J:is=^L:so=\E[1m:se=\E[0m:li#66:co#80:\ <-- (B)
      :ap:us=\E[4m:ue=\E[0m:

# DEC180 -- DEC printers, landscape wide mode (/FORM=LANDWIDE)
D1¦DEC180¦DEC printers:\
       :ff=^L:cr=^M:nl=^J:so=\E[1m:se=\E[0m:li#66:co#180:\
       :ap:us=\E[4m:ue=\E[0m:
       :is=\E[!p\E[?21 J\E[15m:\
       :fs=\E[!p:

# DFLT -- Printer with no special codes -- also used for files
DF¦DFLT¦default logical printer:\
        :ap:li#66:co#80:
#
------------------------------------------------------------------------
```

(표 - 1 PRINTDEF.DAT 화일의 예)

표-1에서 (A)는 PRINTER DEFINTION 화일의 LABEL 규칙이며, (B)는 FORMAT CODE의 규칙을 나타낸다.

```
------------------------------------------------------------------------
(A) LABEL SYNTAX
    PRINTER ID ¦ PRINTER NAME ¦ COMMENT :\
```

- PRINTER ID : 내부적으로 사용되는 일반적인 이름,
  2 문자로 구성, PRINTER DEFINITION 화일내에서는 유일
  PRINTER NAME : PRINTDEF 내의 DRIVER 특성을 지정
  PRINTER DEFINTION 화일내의 유일한 PRINTER 이름
  COMMENT : DRIVER 기능 설명

(표 - 2 PRINTER DEFINTION 화일의 LABEL)

(B) FORMAT CODE SYNTAX
FORMAT CODE [=STRING ¦ #NUMBER]

- FORMAT CODE : PRINTER FORMAT CODE 로써 DEFAULT 형태로 제공되거나
  사용자가 정의할 수 있음.
  STRING : PRINTER 제어 코드에 해당되는 CONTROL CHARACTER 또는
  ESCAPE SEQUENCE
  #NUMBER : 정의된 PRINTER FORMAT CODE의 NUMBER 타입의 값
  POUND 기호를 지정

(표 - 3 PRINTER DEFINITION 화일의 FORMAT CODE)

표-1의 PRINTER DEFINITION 화일은 SQL*ReportWriter에서 제공하는 DEFAULT DRIVER(DFTL, DEC, DEC180)이며, 각 DRIVER DEFINITION은 아래 형태의 PRINTER FORMAT CODE들로 구성되어 있다.

NAME	TYPE	DESCRIPTION
am	BOOLEAN	여백을 조정
ap	BOOLEAN	LINE FEED
co	NUMBER	한 라인의 컬럼 수
cr	STRING	CARRIAGE RETURN(DEFAULT=^M)
ff	STRING	PRINTER PAGE EJECT (DEFAULT=^L)
fs	STRING	PRINTER TERMINATION STRING
is	STRING	PRINTER INITIALIZATION STRING
li	NUMBER	한 페이지의 라인 수
nl	STRING	NEWLINE STRING
nnn	STRING	사용자가 정의하는 PRINTER 제어 코드. 'nnn'은 숫자로 명시.
os	STRING	OVER STRIKE
se	STRING	HIGHLIGHT MODE의 종료
so	STRING	HIGHLIGHT MODE의 시작
ue	STRING	UNDERSCORE MODE의 종료
uo	STRING	UNDERSCORE MODE의 시작

(표 - 4 PRINTER FORMAT CODE의 예)

PRINTER DEFINITION 일의 PRINTER FORMAT CODE를 수정하거나 생성할 경우에는 다음의 규칙에 따라야 한다.

- COLON(:)은 CODE LIST의 시작.
- COLON은 CODE LIST의 종료.
- COLON은 각 CODE DEFINITION의 분리자.
- FORMAT CODE의 라인 끝에 SLASH(/) 기호를 사용하여 다음 라인에 연속적으로 FORMAT CODE를 지정.

다음은 실제 PRINTER DRIVER DEFINITION을 수정하는 방법을 살펴보기로 한다.

- "DFLT" PRINTER DRIVER를 이용하여 한 라인의 컬럼을 132 컬럼으로 출력할 수 있도록 변경하고자 한다.
  즉, :co#80(한 라인의 컬럼 수 = 80컬럼)을 132 컬럼으로 변경하는 방법

(1) $ORACLE_HOME/sqlreport/admin/printer 디렉토리 아래의 PRINTDEF.DAT 화일을 에티터로 오픈한다.
(2) PRINTER FORMAT CODE 작성 규칙에 따라 PRINTER DEFINTION 파일 (예, 표-1의 PRINTDEF.DAT)의 "DFLT" PRINTER DRIVER에서 "한 라인 컬럼 수"를 나타내는 :co#80을 :co#132로 변경한다.
(3) 수정된 PRINTDEF.DAT 화일을 저장한다.
(4) SQL*ReportWriter의 PRINTDEF 유틸리티를 사용하여 새로운 PRINTER DRIVER DEFINTION을 생성한다.
    PRINTDEF 유틸리티의 사용 방법은 다음과 같다.
    - $ printdef printer=DFLT outfile=NEW_DFLT pdfile=printdef.dat
    - printer : PRINTER의 이름.
      PRINTER DEFINITION 화일의 LABEL에 명시된 두번째 인수를 참조
    - outfile : 새로운 PRINTER DRIVER DEFINITION 이름
    - pdfile : PRINTER DESCRIPTION FILE (예, PRINDEF.DAT)
(5) SQL*ReportWriter 프로그램 출력시 [PARAMETR VLAUE FORM]의 [PRINTER DESCRIPTION] 옵션에서 수정된 NEW_DFLT를 지정한다.

(V1.1) 고정된 위치에 합계를 PRINT 하는 방법

Bulletin no : 10179

합계를 프린트하는 행의 윗부분에 인쇄되는 데이타의 행이 가변적일 때 고정된 (ex.세금계산서양식) 위치에 합계를 프린트하기 위해서는

1> 두개의 query생성

- 개별 data인쇄 query
- 합계 data인쇄 query

2> text option에서 합계데이타를 인쇄하는 group의 body부분에서

- relative position : margin
- lines before : 20
- space before : 10

3> 항시 합계는 (20,10)에서 인쇄된다.

(V1.1) PRINTER로 찍을때 확대축소하려면?

Bulletin no : 10181

---

1. PRINTER MANUAL에서 확대값을 찾는다.
   EX) 행정전산망용 : ^[y1 ^[y0

2. cd $ORACLE_HOME/sqlreport/admin/printer

3. vi printdef.dat : printer device에 값을 입력한다.
   HP¦HPLB¦HP LaserJet¦\
      :ff=^L:cr=^M:nl=^J:co#80:l#63:\
      :005=^[y1:006=^[y0:

4. printdef hplb newhp

5. sqlrep scott/tiger
      Text Screen : &5 확대할 문자 &6
      Runform parameter:
            Printer Description File : newhp

6. 결과를 확인한다.
      제대로 안 나오면 먼저 output을 file로 보낸 후 그 file내에
      ^[y1,^[y0 으로 되어 있는지 확인한다.

(V2.5) ORACLE REPORT SERVER에 관하여

Bulletin no : 10251

Report Server에 관하여

정의와 목적

Oracle Reports Server는 Oracle Forms나 Oracle Graphics로 부터 호출되어 큐에 쌓여있는 Report들을 쉽고도 강력하게 관리하여 준다. 이것은 쌓여있는 실행될 Report들을 나타내주고, Report들의 상태를 볼수 있게하며, 큐안에서 Report를 위,아래로 움직일 수도 있게하며,큐에서 삭제 할 수도 있다.

Oracle Report Server가 초기화될때, DLL들은 먼저 Load되어 있다. 이것은 Oracle Report,Oracle Forms,Oracle Graphics의 Start시간을 줄여주는 효과를 준다. Executable이 이미 Memory에 Load되어 있고 Report Server는 이미 Database에 Connect되어 있기때문에 Report의 실행시간 역시 빠르게 해준다. 처음 Report Server가 호출되면 이것은 특별한 Userid로 Database에 Connect하게된다. 다른 Client에서 다음 Report Server가 호출되면 자동적으로 다른 Logon String으로 Connect하게 된다.

Server는 다음과 같은 경우에 자동적으로 기동된다.
. Oracle Forms나 Oracle Graphics에서 RUN_PRODUCT을 이용하여 Report를 호출할 때
. 다른 Report에서 SRW.RUN_PRODUCT를 이용하여 Report를 호출 할 때

RUN_PRODUCT를 사용하면, Report Server Icon이 Window화면밑에 위치하게된다. Reports Server는 User가 임의로 Close하기전까지 Open된 상태로 남아있게 된다.

Reports Server Action Menu

---

1. Cancelling/Removing Reports

Reports Server 큐에 쌓여있는 Print Job을 삭제하려면
방법1 : Highlight/Select된 Report가 Cancel된다.
방법2: ACTION Menu에서 CANCEL JOB이나 REMOVE JOB를 선택한다.

CANCEL JOB은 Report List에서 현재 실행중인 Report를 종료시킨다. 이 Menu는 현재 실행중인 Report가 선택되었을 때만 Displayed/Enabled된다.
REMOVE JOB은 Report List에서 현재 선택된 Report를 삭제한다. 이 Menu는 현재 실행중인 Report가 아닌것을 선택하였을때만 Displayed/Enabled된다.

2. SHOW/HIDE LOG

이 Menu는 각 Report의 Logging을 별도의 다른 Pane으로 나타내주거나 없애 주거나 한다.
Report Server를 종료시키면 이 Log의 내용들은 현재작업 Directory내에 Reports.Log로 저장된다.

3. Exiting The Reports Server

ACTION Menu에서 QUIT를 선택하면 현재작업 중인 Report를 포함한 모든 Report들이 종료되고, Server Log들은 Reports.Log에 저장된다.

4. Report List

Report Server내의 Report List는 현재 실행중인 Report와 큐에 쌓인 Report들에 대한 우선순위와 상태를 나타내준다.

RANK는 큐내에서의 현재의 우선순의를 의미한다. RANK 1인 것은 현재 실행 중인 Report이고, RANK 2는 다음에 실행될 것이다. Report를 선택하여 위아래로 당겨 새로운 위치에 가져다 놓음으로써 Report의 우선순위를 바꿀 수도 있다. 단 이것은 큐에 쌓여있는 Report에 한해서만 가능하여 현재 실행중인 Report에 대해서는 할 수가 없다.

STATUS는 현재 실행중인 Report의 상태를 나타내준다.

. Formatting Page X : Report의 Page X가 작업중이다.
. In Parameter Form : Runtime Parameter Form에 화면에 나타나있다.
User가 Parameter Form에서 "Run Reports"를 선택하 기전 까지 Report 는 실행되지 않는다.
. In Reviewer : Report는 현재 Oracle Reports Previewer상태로 나타나있다.
다음 Report는 Previewer가 종료되기 전까지 실행되지 않는다.
. Initalizing Report : Report 실행이 시작중이다.
. Waiting : Report가 큐에 쌓여 실행되기 기다리고 있다.

(V2.5) ORACLE REPORTS PERFORMANCE TIPS

Bulletin no : 10257

Oracle Reports Performance Tips

1. 일반적으로

1) Formula나 Summary Field를 이용하는 것보다는 Query생성시에 계산을 직접하는 것이 빠르다.
2) Group Filter나 Format Trigger에서 Record를 선택해 내는 것보다 WHERE절에서 처리하는 것이 빠르다.
3) Oracle Reports에서의 Truncate기능을 사용하는 것보다 SUBSTR Function에 사용하는 것이 빠르다.

2. SRW.DO SQL

SRW.DO_SQL은 Report상에서 DDL이나 DML처리를 할 수 있게한다.
이 기능을 매우 유용하지만, 잘못사용하면 많은 시간을 낭비하게 되므로 꼭 필요한 경우에만 사용하도록 한다. SRW.DO_SQL문장이 Parsing되면,Database에 Cursor가 Open되고 문장이 실행된다. Query에 와 달리, SRW.DO_SQL은 해당 Group에서 Data가 Fetch 될 때마다 매번 그렇게 처리된다. 예를 들어, 10건의 Record를 Fetch하게될 Group에 SRW.DO_SQL문이 포함되어 있다면 이 문장은 10번 Parsing되고, 10번 Cursor가 Open되고,10번 실행되게 된다. SRW.DO_SQL을 사용하는 것보다는 Query나 Group의 PL/SQL내에서 계산을 수행해 주는 것이 좋다.

3. CDE MM.GET REF

CDE_MM.GET_REF Package역시 꼭 필요한 경우에만 사용하는 것이 좋다.
이것은 Oracle Reports에서 사용할 Temporary Space의 양을 감소시키는 결과를 가져오기 때문이다.
Temporary Space에 대한 요구가 줄어드는 동안, Column의 값을 Database로 부터 가져와야 하기 때문에 Performance가 낮아지게 된다.

4. Multi-Query Data Model을 사용할 때

가능한데로 Report내에 Query의 수를 적게 가져가도록 한다. Query의 수가 적을 수록 실행속도는 빨라지게된다. Multi-Query Data Model이 이해하기는 쉬우나,Single Query가 실행속도는 빠르다.

다음과 같은 경우라면 Multi-Query를 사용한다.
1) Parent로 부터 많은 자료를 Fetch하여 적은 양의 Child값을 가져올 때
2) SELECT문으로부터 직접적으로 자료를 가져올 수 없을 경우
3) 복잡한 View를 가진 경우 (분산된 Query들이나, Group By Query사용시)
4) View가 필요하지만 View를 가지고 있지 않거나, View를 사용하고 싶지 않을 때

5. Index에 관하여

WHERE절의 Key Column이나, Detail Query에 걸린 Table에 대해 Index를 생성한다.

Master Query에 걸린 Index는 대부분 한번 Database를 찾기 때문에 큰 영향을 주지 않으나, Master/Detail형식의 Report의 Performance향상에는 중요한 역할을 하게된다. Two-Query의 Master에 대한 Detail Record의 비율이 적어질 수록,Detail Query에 걸린 Index는 더욱 중요하게 된다.

Oracle Report에서는 Detail Query를 생성하면 내부적으로 Parent/Child관계를 갖는 WHERE절을 생성하기때문에 Datail Query에 연결된 Table에 대한 Index 생성을 권장한다.

6. Query문 수정

Oracle Reports는 User의 Query들을 다음과 같은 경우에 변경한다.

1) Link를 생성하면 Oracle Reports는 Child Query에 해당 Link에 대한 다음과 같은 절을 추가한다.

예를 들어;

```
SELECT   Deptno,Ename,Sal
   FROM  Emp
   WHERE Sal > 1000
```

만약, 위의 Query에 대해 Child Column으로 DEPTNO를 사용하는 Query에 Link를 생성하면 User의 Query는 다음과 같이 수정된다.

```
SELECT   Deptno,Ename,Sal
   FROM  Emp
   WHERE (Sal>1000) And (Deptno = :Deptno)
```

단, Multi-Query Matrix에서는 위의 경우와 다르다.

2) Oracle Reports는 Break Order Set을 갖는 Database Column에 대해서 Query에 ORDER BY절을 추가한다.

예를 들어;

```
SELECT  Deptno,Ename,Sal
   FROM Emp
   ORDER BY Sal
```

만약, 위의 Query에 대해 DEPTNO를 Break Column을 하는 Break Group을 생성하면 Query는 다음과 같이 변경된다.

```
SELECT  Deptno,Ename,Sal
   FROM Emp
   ORDER BY 1,Sal
```

이러한 SQL문장은 Databse로 보내져서,SQL Optimizer가 Database로 부터 Data를 가져오는 데 적당한 방법을 결정하여 찾은 Data를 Oracle Reports로 보내오게 된다.

Optimizer는 Index를 사용할 것인지 아닌지, 어느 것을 "Driving" Table로 사용할 것인지 등을 결정하게 된다.

7. Break Column

Break Group을 생성한다면, 가능한데로 Group내의 Column을 적게 가져가도록 한다. Break Group과 Break Column간에 1:1비율이 되는 것이 좋다. 또한 Break Column이 가능하면 작도록 한다. Break Column이 긴것보다는 짧은 것이 일반적으로 좋은 Performance를 가져온다. 만약 긴 Column에 대해 Break Column을 가져가야 한다면,SUBSTR Function을 이용하여 Length를 줄이는 것도 Performance에 도움이 된다.

각각의 Break Group에 대해,Oracle Reports는 각각의 Break Column에 대해 ORDER BY절을 가져간다. (한가지의 예외가 있다면 Break Column이 Formula Coulmn일때) Break Group의 Break Column의 수를 적게가져갈 수록, 추가되는 ORDER BY절이 적어지게 된다. ORDER BY절에 추가되는 Column이 적을수록 Query가 수행될 때수행되는 Processing이 적어지게 된다. 이것은 결국 Oracle Report가 내부적으로 생성하는 Index Key의 Size가 작아지게 되고, 결국 좋은 Performance를 가져오게 되는 것이다.

8. Maximum Rows And Group Filters

Query Property화면에서 Maximun Rows Property를 지정하는 것은 Report의 Query 에서 Retrieve하는 Record의 수를 줄여준다. 많은 양의 Data를 Access하나, Retrieve 하는 Data의 양에 제한 주면 Test하는 동안 Report는 좀더 빠르게 수행될 것이다.

Query Property화면에서의 Maximum Rows를 지정하면 Query에 의해 Fetch 되는 Record의 수를 제한 할 수 있다. Group Filter는 어떤 Record를 포함할 것인가,제외 할 것인가를 결정해준다. Maximum Rows를 지정하는 것이 실제로 Retrieve되는 Data의 양에 제한을 주었기때문에,대부분의 경우에 있어서 Group Filter보다 빠르다.

만약 Group Filter를 마지막이나 조건적으로 지정하였다면,Oracle Reports는 Group내에서 Filtering을 하 전에 모든 Record를 Retrieve해야한다. Maximun Rows나 First Filter Setting이 더 빠르게 된다. 일반적으로 Maximun Rows의 Setting이 First Filter보다 빠른데,그 이유는 Record의 수가 결정하게 된다.

Performance의 차이는 User가 지정하는 ARRAYSIZE에 따라 달라 질수 있다.

9. Unused Data Model Objects

Report에서 실제로 사용하지 않는 Data Model Object가 있는지 확인하고 있다. 면 Remove하도록 한다. 특정한 경우에만 사용하는 경우가 있다면 SRW.SET_MAXROW Packaged Procedure를 이용하여 조건적으로 Suppress하도록 한다. SRW.SET_MAXROW(Query_Name,0)로 Setting하면 Query는 한건의 Data도 Fetch하지 않게 된다.

10. Unused Frames

Layout에서 필요없는 Frame들은 지우도록 한다. Oracle Reports는 Default Layout을 생성 할 때 모든것에 대해 가상의 Frame을 생성한다. 이것은 Output에서 다른 Object에 의해 Frame내의 Object들이 Overwritten 되는 것을 막기위한 것이다. 만약 User가 Frame내의 Object가 Over Written될 가능성이 없다고 판단된다면, Report의 Output에 영향을 주지 않는 Frame은 제거해도 된다.

Layout상에 Object수가 적을 수록, Oracle Report가 Runtime시에 Format해야 하는 Object수가 적어지게 되어 결국 Performance가 향상되게 된다.

11. Total Number Of Pages

Field Source의 Total Number Of Page의 값을 가능한 쓰지 않는 것이 좋다.
Field Source에서 Total Number Of Pages를 사용하면,Oracle Reports는 Total Page의 수를 결정하기 위해 모든 Page를 Temporary Space에 저장해야만 하기 때문이다.
이것은 결정적으로 Report가 사용하는 Temporary Disk Space의 양을 증가시키게 되고, File에 Write하기 때문에 속도가 느려지게 된다.

12. Format Triggers

Object에 빈도수가 적은 Object에 Format Trigger을 걸도록 한다. 일반적으로 Field보다 Frame에 지정한 Format Trigger의 PL/SQL이 빠르다.

Format Trigger내의 PL/SQL은 그 Object가 나타날 때 마다 실행된다. Object의 출현빈도수가 적을수록, PL/SQL이 실행되는 빈도수가 적어지므로 속도가 빨라지게 되는 것이다.

13. Oracle Graphics Integration

Oracle Report의 자료를 일부 혹은 전부를 이용하는 Oracle Graphics Display를 Report에서 사용한다면, Report에서 Display로 Data를 전송하는 것이 좋다.
Report와 Display가 같은 Data를 사용한다면,Data를 Passing하는 것이 Fetch되는 Data의 양을 줄여주기 때문이다. 만약 Report에서 Display로 Data를 전송하지 않는다면, Data는 실제로 한번은 Report에서 한번은 Display에서 Fetch되어 두번 Fetch된다.

(V1.1) REPORT SOURCE LISTING GENERATION

Bulletin no : 10259

Oracle에서는 SQL*ReportWriter Base Table에 저장되어있는 Report들의 List를 생성하는 Program을 제공하고있다.

이것은 SRW80.rex, SRW132.rex, SRW175.rex의 3가지 Report로써 80,132,175은 출력되는 Report의 Column Width를 의마한다. 즉,SRW80 Report를 실행하면 결과가 80 Column용지에 맞게 출력된다.

* Location

1. VMS - ORA_ROOT:[SQLREPORTWRITER.SRW_DEMO]
2. UNIX - $ORACLE_HOME/sqlreport/demo

* 생성방법

1. .rex File을 Load한다.

   loadrep file=srw80 userid=your_user_id

2. Generate한다.

   genrep file=srw80 userid=your_user_id

3. srw80을 Execute한다.
   Listing을 하고싶은 Report의 이름과 출력방법(Screen/File/...)을 선택하는
   Dialog Box가 나타난다.

(V1.1) SQL*REPORTWRITER에서 SOURCE를 자동으로 BACKUP받는 방법

Bulletin no : 10260

Report가 Drop되거나,Source가 들어있는 SRW Table에 손상이 생겨 예전의 File로부터 복구가 필요할 때가 있다.
이때 Source를 이전에 Dump받아놓은 .rex File이 있어야만 복구가 가능하다.

Version 2.x에서는 .rex File이 Report를 생성할 때 자동으로 생성될 수 있지만,1.1에서는 기본적으로 제공하지 않기 때문에 User는 주기적으로 Report Source를Dump받아 .rex File로 보관하여야만 한다.

이 자료에서는 매일매일 이 작업을 손쉽게 처리하도록 해준다.

내용 : System Date를 받아 SRW Table로 부터 해당 일자에 변경된 Report들을 Dump받는다.
Background Job으로 처리 할 수도 있다.

```
+-------------------------------------------------------------------------+

# Autometic REX generation of modified reports

# Set some local variables
#

set REX_DIR=/usr/user_home/srw/rex    # dir to hold .rex files
set SQL_DIR=/usr/user_home/plus         # dir to hold .sql files
set USERNAME=scott/tiger
set BEFORE_DATE=`date '+%y%m%d%H%M%S'`

#
# Startup SRW
#

  $ORACLE_HOME/bin/sqlrep $USERNAME

#
# Create list on all reports that changed
#

set  Reports=`sqlplus -s scott/tiger @$SQL_DIR/get_mod_srw "$BEFORE_DATE"`
```

```
#
# Loop for all the reports found to be modified
#

foreach f ($Reports)
    echo Dumping $REX_DIR/$f
    dumprep reports=$f  file=$REX_DIR/$f userid=scott/tiger >! /dev/null
end
```

```
+-------------------------------------------------------------------------+
```

아래의 get_mod_srw SQL script는 위에서 호출한 SQL문이다.

```
+-------------------------------------------------------------------------+
set  heading  off
set  feed  0
set  pagesize   999
set ver  off

select  report_name
   from srw_report
   where  to_char(modified_date,'YYMMDDHH24MISS') > &1;

exit;
+-------------------------------------------------------------------------+
```

(V2.5) REPORT에서 자주 사용되는 SRW PACKAGED PROCEDURE

Bulletin no : 10264

Introduction

==========

Oracle Reports는 User가 Library나 Report에서 Reference하여 이용할 수 있는 많은 Function이나 Procedure, Exception들을 제공하는 PL/SQL Program을 제공한다.
이 Oracle Reports의 Package의 이름이 SRW이다. User는 언제고 Reference하여 사용할 수 있으며 사용할때는 예를 들어 SRW.DO_SQL처럼 반드시 SRW를 앞에 붙여서 사용하여 한다.

Oracle Reports에서 제공하는 SRW Package내에는 다음과 같은 Function 혹은 Procedure들이 있다.

- SRW.BREAK
- SRW.CONTEXT_FAILURE
- SRW.DO_SQL
- SRW.DO_SQL_FAILURE
- SRW.GETERR_NUM
- SRW.GET_PAGE_NUM
- SRW.INTEGER_ERROR
- SRW.MAXROW_INERR
- SRW.MAXROW_UNSET
- SRW.MESSAGE
- SRW.NULL_ARGUMENTS
- SRW.PROGRAM_ABORT
- SRW.REFERENCE
- SRW.RUN_REPORT
- SRW.SET_ATTR
- SRW.RUN_REPORT_FAILURE
- SRW.RUN_REPORT_BATCHNO
- SRW.SET_FIELD_CHAR
- SRW.SET_FIELD_DATE
- SRW.SET_FIELD_NUM
- SRW.SET_MAXROW
- SRW.TRACE_ADD_OPTION
- SRW.TRACE_END
- SRW.TRACE_REM_OPTION
- SRW.TRACE_START
- SRW.TRUNCATED_VALUE
- SRW.UNKNOWN_QUERY
- SRW.UNKNOWN_USER_EXIT
- SRW.USER_EXIT
- SRW.USER_EXIT20
- SRW.USER_EXIT_FAILURE

자주 사용하는 SRW Package와 Sample들

==========

1.SRW.DO_SQL은 특별히 지정하는 SQL문장을 실행한다. 이 Procedure는 어떤 DDL이나 DML문장을 실행하지만 일반적으로 SRW.DO_SQL을 통해 실행한 것보다 PL/SQL에서 실행한 DML문장이 빠르다. PL/SQL에서는 DDL문장의 사용을 할 수 없기때문에 이 Packaged Procedure를 User Exit을 사용하는 것보다는 유용하다고 본다.

Reports에서 사용하는 DDL 문장에서 Table을 Modify하는 것은 가능한 피하는 것이 좋다. Report를 실행하기전에 Snapshot이 생성되어 Report가 살행되는 동안 계속 남아있기때문이다.

Syntax/Explanation

```
SRW.DO_SQL(Sql_Statement CHAR); -- 지정한 SQL문을 실행한다.
SRW.DO_SQL_FAILURE;          -- SRW.DO_SQL에서 실패한 경우 Report의 실행을 중지한다.
SRW.MESSAGE(Msg_Number NUMBER,Msg_Text CHAR);
                                --지정된Message와Message번호를 화면에 뿌린다.
SRW.PROGRAM_ABORT;         -- Rreport의 실행을 중지한다.
```

Example

Runtime Parameter Form이 뜨기전에 TEMP라는 Temporary Table을 생성하는 예를 본다.

```
BEGIN
   SRW.DO_SQL(CREATE TABLE Temp
                              ( Empno NUMBER NOT NULL  PRIMARY KEY,
                                Sal      NUMBER(10,2))
                               PCTFREE 5  PCTUSED 75);
   RETURN TRUE;
EXCEPTION
   WHEN SRW.DO_SQL_FAILURE THEN
         SRW.MESSAGE(100,'ERROR WHILE CREATING CHECK TABLE');
         SRW.MESSAGE(50,'REPORT WAS STOPPED BEFORE THE PARM FORM');
         RAISE SRW.PROGRAM_ABORT;
END;
```

2.

Syntax/Explanation

```
SRW.RUN_REPORT(Command_Line CHAR); -- R25RUN Command를 실행.
SRW.RUN_REPORT_FAILURE;   -- SRW.RUN_REPORT에서 오류가 발생하면
                              -- Report의 실행을 정지한다.
```

Example

Report Button으로부터 화면으로 출력하는 Drill Down Report를 호출한다.

```
FUNCTION  Foo  RETURN BOOLEAN IS
BEGIN
   SRW.RUN_REPORT('Report=REP_A P_Parm1=20');
EXCEPTION
   WHEN SRW.RUN_REPORT_FAILURE THEN
            SRW.MESSAGE(100,'ERROR CALLING REPORT.');
            RAISE SRW.PROGRAM_ABORT;
   RETURN TRUE;
END;
```

3.

Syntax/Explanation

SRW.SET_ATTR는 Layout Object에 대한 Font Size혹은 Color와 같은 Attribute의 Setting을 처리한다.

이 Procedure는 Current Frame,Repeating Frame,Field, Boilerplate Object Attribute를 처리할 수 있게한다.

```
SRW.SET_ATTR(Object_Id NUMBER, Attr SRW.GENERIC_ATTR);
                        Object_Id : 항상 ZERO 이다.
                                    Object는 항상 자신의 Attribute를 지정해야만 한다.
                        Attr      : SRW.ATTR이다.
                                    예를 들어 바꾸고자하는 Attribute와 그 값.
```

Example

Salary가 2000이면 다음의 Attribute를 Setting한다.

```
            Border Width=1
            Foreground Border Color = Red
            Background Border Color = Blue
            Border Pattern = Checker
            Foreground Fill Color = Yellow
            Background Fill Color = Green
            Fill Pattern = Crisscross

  IF :Sal = 2000 THEN
    SRW.ATTR.MASK :=  SRW.BORDERWIDTH_ATTR +
                                SRW.FBCOLOR_ATTR        +
                                SRW.BBCOLOR_ATTR        +
                                SRW.BORDPATT_ATTR      +
                                SRW.FFCOLOR_ATTR        +
                                SRW.BFCOLOR_ATTR        +
                                SRW.FILLPATT_ATTR;
   SRW.ATTR.BORDERWIDTH    := 1;
   SRW.ATTR.FBCOLOR          := 'Red';
   SRW.ATTR.BBCOLOR          := 'Blue';
   SRW.ATTR.BORDPATT        := 'Checker';
   SRW.ATTR.FFCOLOR          := 'Yellow';
   SRW.ATTR.BFCOLOR          := 'Green';
   SRW.ATTR.FILLPATT          := 'Crisscross';

   SRW.SET_ATTR(0,  SRW.ATTR);
  END IF;
  RETURN TRUE;
```

4.

Syntax/Explanation

```
SRW.SET_FIELD_CHAR(Object_Id,  Text  CHAR);
SRW.SET_FIELD_DATE(Object_Id,  Date  DATE);
SRW.SET_FIELD_NUM(Object_Id,    Number  NUMBER);
```

이 SRW.SET_FIELD Procedure는 Format Trigger에서 매우 유용하게 쓰인다.
이것은 Report에서는 Field의 Value를 직접 Assign할 수 없기 때문에 특별한 조건인 경우에만 나타나게 하려는 경우에 자주 사용된다.

Example

Salary가 2000보다 적은 직원들의 경우에 'LOW'라는 문자를 출력하려면 다음과 같은 Format Trigger를 Salary_Range_Field에 걸어야 한다.

```
FUNCTION  Chgfield RETURN BOOLEAN IS
BEGIN
   IF :Sal < 2000 THEN
      SRW.SET_FIELD_CHAR(0,'LOW');
   ELSE
   END IF;
   RETURN TRUE;
END;
```

(V2.5.X) 조건에 따라 HIGHLIGHTING, COLORING 처리하는 방법

Bulletin no : 10266

DATA MODEL에서

==========

다음의 간단한 Query를 생성한다.

```
Q_Highlinght      :    SELECT Empno,Ename,Sal
                       FROM  Emp
                       ORDER BY Empno
```

LAYOUT에서

==========

1> F_ENAME Field의 Property화면 * General Layout * Format Trigger

```
+-------------------------------------------------------------------+
 FUNCTION F_Enameformattrigger
 RETURN  BOOLEAN IS
 BEGIN
    IF :Sal > 2000 THEN
      SRW.ATTR.MASK := SRW.WEIGHT_ATTR + SRW.FACE_ATTR;
      SRW.ATTR.WEIGHT := SRW.BOLD_WEIGHT;
      SRW.ATTR.FACE := *Helvetica*;
      SRW.SET_ATTR(0,SRW.ATTR);
    END IF;
    RETURN TRUE;
 END;
+-------------------------------------------------------------------+
```

2> F_SAL Field의 Property화면 * General Layout * Format Trigger

```
+-------------------------------------------------------------------+
 FUNCTION F_Salformattrigger
 RETURN  BOOLEAN IS
 BEGIN
    IF :Sal > 2000 THEN
      SRW.ATTR.MASK := SRW.WEIGHT_ATTR +
                       SRW.FACE_ATTR   +
                       SRW.BORDERWIDTH_ATTR +
                       SRW.FBCOLOR_ATTR +
```

```
                          SRW/BORDPATT_ATTR;
      SRW.ATTR.WEIGHT := SRW.BOLD_WEIGHT;
      SRW.ATTR.FACE := *Helvetica*;
      SRW.ATTR.FBCOLOR  := *BLACK*;
      SRW.ATTR.BORDPATT  := *SOLID*;
      SRW.SET_ATTR(0,SRW.ATTR);
    END IF;
    RETURN TRUE;
 END;
+-------------------------------------------------------------------+
```

위의 Trigger들은 Sal이 2000이상이 되면 SRW.ATTRIBUTES의 값을 새롭게 적용하게된다.

3> F_SAL의 Format Mask를 -NNN.00로 Setting한다.
4> Report를 Save하고 Run시킨다.

Highlighting대신에 Color를 Setting하는 조건Trigger
==========

F_ENAME에 걸려있는 Trigger를 다음의 Color를 Setting하는 Format Trigger로 대치한다.

```
+-------------------------------------------------------------------+
 FUNCTION F_Enameformattrigger
 RETURN  BOOLEAN IS
 BEGIN
    IF :Sal > 2000 THEN
      SRW.ATTR.MASK := SRW.WEIGHT_ATTR + SRW.FACE_ATTR +
                          SRW.FILLPATT_ATTR + SRW.FFCOLOR_ATTR ;
      SRW.ATTR.WEIGHT := SRW.BOLD_WEIGHT;
      SRW.ATTR.GCOLOR := *RED*;
      SRW.ATTR.FILLPATT := *GRAY10*;
      SRW.ATTR.FFCOLOR := *BLUE*;
 SRW.SET_ATTR(0,SRW.ATTR);
    END IF;
    RETURN TRUE;
 END;
+-------------------------------------------------------------------+
```

F_SAL에 걸려있는 Trigger를 다음의 Color를 Setting하는 Format Tirgger로 대치한다.

```
+-------------------------------------------------------------------+
 FUNCTION F_Salformattrigger
 RETURN  BOOLEAN IS
 BEGIN
    IF :Sal > 2000 THEN
      SRW.ATTR.MASK := SRW.WEIGHT_ATTR + SRW.FACE_ATTR +
                          SRW.FILLPATT_ATTR + SRW.FFCOLOR_ATTR +
                          SRW.BORDEFWIDTH_ATTR + SRW.FBCOLOR_ATTR +
```

```
                          SRW.BORDPATT_ATTR;
      SRW.ATTR.WEIGHT := SRW.BOLD_WEIGHT;
      SRW.ATTR.GCOLOR := *RED*;
      SRW.ATTR.FILLPATT := *GRAY10*;
      SRW.ATTR.FFCOLOR := *BLUE*;
      SRW.ATTR.BORDERWIDTH := 0;
      SRW.ATTR.FBCOLOR := *BLUE*;
      SRW.ATTR.BORDPATT := *SOLID*;
      SRW.SET_ATTR(0,SRW.ATTR);
    END IF;
    RETURN TRUE;
 END;
+-----------------------------------------------------------------------+
```

위에서 사용한 SRW Attributes의 예

```
=====================================

ATTRIBUTE                     DEFINITION
-------------------------------------------------------------
SRW.ATTR.MASK                 User가 Create하는 Mask
SRW.WEIGHT_ATTR TEXT WEIGHT
SRW.FACE_ATTR           TEXT FONT
SRW.GCOLOR_ATTR TEXT COLOR
SRW.FILLPATT_ATTR       FIELD를 채우는 PATTERN
SRW.FFCOLOR_ATTR        FOREGROUND FILL COLOR
SRW.ATTR.FBCOLOR        BACKGROUND FILL COLOR
SRW.ATTR.BORDPATT       BORDER PATTERN
```

(V2.5) ORACLE REPORTS에서의 FORMAT TRIGGER사용에 관하여

Bulletin no : 10270

Format Tigger는 Report를 Fomat하는데 아주 Powerful하게 사용된다. 이것은 어떤 Object에 대한 Highlighting이나suppressing 혹은 Column의 값이나 Heaing을 변경할 수도 있게한다. Format Trigger는 Pl/Sql Function 으로서, True와 False의 Boolean값을 Return한다. Trigger내에서 Oracle Reports의 Packaged Function이나 Procedure, Exception을 사용할 수도 있다. Format Trigger는 Oracle Reports가 해당 Layout Object를 Format할 때마다 매번 실행되게 된다.

조건에 따라 해당 Column의 값을 바꾸어 나타내거나, 값을 나타나지 않도록 막으려한다면 해당 Layout Object에 Format Trigger를 생성한다. 여기에서 만약 Report내의 다른 Object의 을 사용해야 한다면, Frequency가 맞는지에 주의하여야 한다.

==========
Format Trigger에 대한 기본 Rule
==========

1. 동일한 Frequency를 갖는 Column이나 Parameter의 값을 읽어들이는 것은 가능하지만, 값을 직접 주는 것은 안된다. (예: :Count = 10)

2. Format Trigger가 생성된 Object와 다른 Frequency를 작는 Column이나 값은 Reference할 수 없다. (예: Master/Detail Report에서, Master의 Format Trigger에서는 Detail Item의 값을 Reference할 수 없다.)

3. Format Trigger에서 False를 Return하여 Object가 Format되니 않는다면 이것 때문에 다른 Object들이 Print되지 않을 수도 있다. (예: Group Frmae에 Format Trigger에서 False가 Return되었다면 , 그속에 속한 모든 Object들 역시 Format되지 않게 된다.)

4. Repeating Frame에 Format Trigger가 생성되어있다면, Return되는 Record의 수만큼 Format Trigger가 실행된다.

5. 같은 Object에 대해 Format Trigger는 일관된 결과를 Return한다. (예: 한 Object가 두Page에 걸쳐있다면 Trigger는 한 Page에 한번씩 실행된다.)

6. Matrix형 Report의 Repeating Frame에 생성된 Format Trigger에서 False를 Return하게된다면,Matrix내의 모든 Column이나 Row의 값도 나타나지 않게 된다.

7. Matrix의 Cell의 조건에 따라 처리하고 싶다면, Cell내부의 Field를 감싸는 Frame을 생성하여 그 Frame에 Format Trigger를 생성하면 된다 (여기에서 정확한 Level위에 Frame을 위치시켜야만 한다.)

========
Sample
========

Case 1. 값이 Null인 Column에 대해서는 Zero를 표시해 주려면?

```
Function  My_Formatting  Return Boolean Is
Begin
If :Comm  Is Null Then
    Srw.Set_Field_Num(0,'0');
 End If;
 Return True;
End;
```

Case 2. 값을 른 값으로 대치하여 표시하려면?

```
Function  My_Formatting  Return Boolean Is
  Tmp Number;
Begin
If :Comm  Is Not  Null Then
    Tmp := :Comm * 1.5;
    Srw.Set_Field_Num(0,Tmp);
 End If;
 Return True;
End;
```

Tips : 위의 예에서 Field의 Type이 Date 혹은 Char라면 각각 Srw.Set_Field_Date, Srw.Set_Field_Char를 사용하면 된다.

Case 3. Master/Detail Report에서 Detail자료가 없는 Master Record가 있나고 가정하자. Master는 그대로 출력되지만 Detail에 관한 Header부분은 생략되는 것이 좋을 것이다.

Step 1 : Detail Group의 Record수를 Count하는 Summary Column을 생성한다.
Master Group에서 Reset하도록 Setting한다.

Step 2 : Detail Group의 M_Grper Object에 Format Trigger를 생성한다.

```
Function My_Formatting  Return Boolean Is
Begin
If :Count_Col = 0 Then
   Return(False);
Else
   Return(True);
End If;
End;
```

Case 4. Report를 생성하였지만 조건에 맞는 자료만 출력하고싶은 경우를 가상한다. 일반적으로는 Report를 생성할 ㅛ ㅐ Query에 조건을 주어 원하지 않는 Report는 아예 Retrieve를 하지 않는다.. 그러나 Record들은 모두 Retrieve하지만 출력만 하지 않으려면 (예를 들어 Retrieve된 Record의 Summary는 구하고, 단지 출력만 하지 않으려는 경우) Repeating Frame에 다음과 같은 Format Trigger를 생성한다.

Commision이 3000보다 큰 사원 만 출력하려는 경우

```
Function My_Formatting  Return Boolean Is
Begin
If :Comm > 3000  Then
   Return(False);
Else
   Return(True);
End If;
End;
```

Case 5. User가 Input한 값에 따라서 Record/Column을 생략하기도 하고, Column의 값이나 Column Heading, Report Title/Header를 바꾸기도 하는 경우를 생각해보자. 우선 User의 Input을 받을 Parameter를 생성한 뒤 해당 Object에 다음과 같은 Format Trigger를 생성한다.

```
Function My_Formatting  Return Boolean Is
Begin
If :P1 = *Condition1*  Then
   Srw.Set_Field_Char(0,'This Is My New Value.');
Elsif  :P1 = *Condition2* Then
   Srw.Set_Field_Char(0,'This Is My New Value 2.');
End If;
Return(True);
End;
```

(V2.5) TABLE에 IMAGE FILE에 대한 PATH만 가지고 있는 경우의 처리

Bulletin no : 10321

Image가 Table에 직접 들어있는 경우는 Forms에서나 Report에서 Default Format으로 간단히 처리할 수 있다. 그러나 Table의 Column에 실제의 Image가 아닌 Image File에 대한 Path와 명칭을 가지고 있는 경우에는 다음과 같이 처리하여야 한다.

1. Forms4.5에서는
   작업하고자 하는 Block에 Image Item을 생성한다.
   적당한 Trigger에서 Read_Image_File Procedure를 이용하여 Image를 화면으로 읽어들인다.
   -> Forms Reference Manual참조

2. Report2.5에서는
   1) Image명칭을 가진 Column을 Select하는 Query를 생성한다.
   2) Data Model에서 해당 Column의 Property
      -> Check Box확인 : Read From File Checking
      -> List Value확인 : 옆의 List에서 Image선택
   3) Default Layout생성
      -> Image Column의 Size조정
   4) Run

주의사항: 해당 Image File들은 Working Direcotry나 Current Directory,혹은 Oracle.INI File의 Report25_PATH에 지정된 Directory내에 있어야 한다.-> Building Reports Manual 6-81참조

(V2.5) REPORT2.5에서 ORACLE GRAPHICS를 띄우려면?

Bulletin no : 10323

---

1) Oracle Graphics에서 empsal.ogd라는 Oracle Graphics display를 만든다.

```
query name       : Q_OG_Emp
select statement : SELECT ENAME OG_NAME, SAL OG_SALARY
                           FROM EMP
```

2) Report2.5의 data modeling

query name	group name	select statement
Q_dept	G_dept	select deptno,dname,loc from dept
Q_emp	G_emp	select ename,sal,deptno from emp

Q_dept와 Q_emp는 master/detail로 연결

3) Layout Editor의 Oracle Graphics tool을 사용하여 master repeating frame(R_Dept)안에 Oracle Graphics object를 그려 넣는다.

4) Oracle Graphics object의 property에서 Display Name을 empsal.ogd를 입력한다.(path를 지정하지 않으려면 empsal.ogd를 oracle.ini의 REPORTS25_PATH의 directory에 위치시켜야 한다.)

5) Oracle Graphics object의 Property sheet의 Query tab

Source	Display Query	Report/Display Column
G_emp	Q_OG_Emp	ENAME / OG_NAME SAL / OG_SALARY

Display Query에 Q_OG_Emp로 지정한 것은 Graphics가 database로부터 query를 하지 않고 Oracle Reports에서ENAME 과 SAL를 retrieve하겠다는 의미이다.

* 만약 위의 예제처럼 Reports의 data를 retrieve하지 않고 단순히 parameter만 passing하여 Oracle Graphics에서 database로부터 query를 하려면 -

Graphics의 query에서 where deptno=:parameter1 이라면 Reports의 Oracle Graphics object의 property에서 Display tab내의 Report Column / Display Parameter에 deptno / parameter1 으로 setting하고 query tab부분은 지정하지 않는다.

* 자세한 사항은 Oracle Reports2.5 Reference Manual 8-74 참조.

SOLARIS 2.X SETUP PROBLEM

Bulletin no : 10009

---

1) SPARC system에 SCSI driver를 붙이고자 할때

	SCSI number	device file
CTD driver	4	/dev/rst4
DAT driver	5	/dev/rst5
CD driver	6	/dev/sr0

위와 같이 SCSI number를 맞추고 rebooting을 합니다.
Shutdown이후 Auto booting이 되게 하지말고 "Stop키"와 "A키"를 동시에 눌러 Boot ROM상태로 빠지게 한 후에 SCSI driver가 제대로 연결 되어 있는지를
다음의 command로 확인.

ok probe-scsi
--> ok는 Boot ROM상태의 Prompt

SCSI driver가 제대로 연결 되었다면 각 SCSI driver의 Vendor Name과 SCSI number가 Display되고 제대로 연결 되지 않았으면 그냥 멍청하게 있거나 새로 연결된 driver의 정보가 나타나지 않는데 이때는 SCSI Cable또는 SCSI number를 다시 조사하기 바람.

probe-scsi command의 이후에는 다음 command로 booting
ok boot -r

2) Solaris 2.2에서 Oracle를 install할때는 반드시 Console에서 작업하기 바람.

Console이 아닌 OpenWindows의 "Command tool" 이나 "Shell tool"에서 install 할 경우 Oracle Server를 올리지 못함.
이것은 solaris2.2의 Shared Memory Bug인 듯하며 Solaris2.3에서는 Fix됨. 즉 solaris2.2에서는 Openwindow가 한번 실행되면 Oracle Server를 실행하지 못하기 때문에 rebooting하고, OpenWindows가 수행되기 전에 "Control-C"로 Kill시킨 후에 Oracle를 Install하거나, Install이된 system은 sqldba로 가서 startup시켜야함.

3) SunPro에서 돈주고 사야하는 Compiler를 구입하지 않아도 Oracle Install은 가능함.

사용자가 Pro*C를 쓰지 않는다면 Oracle install과 사용하는데는 별 문제 없음.
^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^

Oracle Install하기 전에 반드시 "/usr/ccs/bin" PATH를 첨가하여 Install하다
"make" command를 못찾아 중간에 끝나는 경우가 생기지 않길바람.

4) Pro*C에서 compile시에 다음과 같은 error message가 나오면

```
Undefined                   first referenced
  symbol                        in file
__cg92_libc                 (command line)
```

LD_LIBRARY_PATH에 십중팔구 "/usr/lib"나 "/lib"가 지정되어 있을 경우이므로 이 PATH를 삭제하기 바람. 이것은 SunPro compiler의 libm.a을 link해야 하는데 "/usr/lib/libm.a"를 link하기 때문에 생기는 문제임.

위의 error는 cc option에 "-xcg92"가 빠져서 나오는 다음의 error message하고는 다른 것임.

```
Undefined                   first referenced
  symbol                        in file
__cg92_used                 $ORACLE_HOME/lib/libsql.a(sqlexp.o)
```

5) 한글 OpenWindows에 한글처리를 htt라는 Input Server를 통하여 한글입력을 받은후에 delimiter(Data의 시작이나 끝을 나타내는 문자로 대표적인 key는 space, enter key)가 오면 그동안의 문자를 X-client에 넘겨주는데 한번에 넘겨주는 문자의 길이를 사용자가 Parameter로 정할 수가 있다.

```
% setenv PREEDIT_LEN 1
```

위에서 보듯이 PREEDIT_LEN의 크기를 1로 정의하면 한글 한글자가 완성되면 delimiter가 입력이 되지않아도 바로 X-client로 전달되며 form에서 한글 data 입력시 다음 라인까지 넘어가는 현상은 발생하게 되지 않음.

PREEDIT_LEN의 default값은 20이다.

UNIX 에서 RAW DEVICE 사용하는 방법

Bulletin no : 10076

RAW DEVICE 는 CHARACTER DEVICE DRIVER 를 사용하는 HW DEVICE 이다.
이러한 RAW DEVICE 를 사용했을때 장점으로는

. RAW DEVICE 는 OS KERNEL 에의해 BUFFERING 이 되지않고 USER BUFFER 와 DEVICE 간에 직접 DATA 가 전송되므로 ORACLE DISK I/O 성능 향상되고 CPU OVERHEAD 가 감소된다.
. OS FILE SYSTEM 의 OVERHEAD 를 피할수있다.
. OS BUFFER SIZE 를 줄일수 있다.

잠재적인 문제점으로서

. SETUP 하기 어렵고 BACKUP 절차가 FILE SYSTEM 보다 복잡하다.
. RAW DEVICE 와 OS FILE 을 혼합하여 사용 할경우 OS FILE 은 ulimit PARAMETER 의 SIZE 보다 작아야 한다. 따라서 ulimit 를 초과하는 TABLE 들은 RAW DEVICE 를 사용하여야 한다.
. OS 는 CYLINDER 0 을보호하지 못하기 때문에 ORACLE 의해 사용된 RAW PARTITION 는 CYLINDER 0 에서 시작하면 안된다.

RAW DEVICE SETTING 방법

1. FREE DISK PARTITION 의 이름을 결정한다.
   FREE PARTITION 은 OS FILE SYSTEM 에의해 사용 된것이 아니어야한다.

2. ORACLE 의 RAW DEVICE SETTING
   $ chown oracle /dev/rdsk/c0s0d1
   $ chmod 600 /dev/rdsk/c0s0d1

3. 새로운 PARTITION 을 추가한다.

```
create database SID
logfile '?/dbs/log1.f' size 100K, '?/dbs/log2.f' size 100K
datafile '/dev/rdsk/c0s0d1' size 10000K reuse;

alter tablespace tablespace_name add datafile
'/dev/rdsk/c0s0d2' size 10000K reuse;
```

RAW DEVICE 의 GUIDLINE

1. REDO LOG FILE 과 DATABASE FILE 을 서로 다른 DRIVE 에 놓는다.
2. 다른 DRIVE 에 자주 사용되는 TABLE 을 분산시킨다.

3. TABLE 과 그 TABLE 의 INDEX 를 분리한다.
4. ORACLE 과 OS I/O 를 분리한다.

RAW DEVICE 의 BACKUP 방법

RAW DEVICE 를 OS FILE 로 COPY 하거나 TAPE 로 직접 COPY 하거나 모두 dd 를 사용하여야 한다. TAPE 로 COPY 하는데 하나의 RAW DEVICE 를 여러개 TAPE 에 COPY 경우는 다음과 같다.

첫번째 TAPE

```
$ dd if=/dev/rdsk/c0s0d1  of=/dev/rst0 bs=20k count=10000
```

두번째 TAPE

```
$ dd if=/dev/rdsk/c0s0d1 skip=10000 of=/dev/rst0 bs=20k count=10000
```

ORACLE 과 UNIX SEMAPHORE 와의 관계

Bulletin no : 10093

---

* 유닉스는 내부 프로세스간의 커뮤니케이션을 위하여 세마포어를 사용한다.
세마포어는 유닉스에 의해서 증감되는 정수(integer)값 이다. 하나의 세마포어에 대해서 하나의 프로세스만이 작업을 할 수 있도록 하여 프로세스 간의 동기화를 유지하는데 이용된다. 다른 프로세스가 사용중인 세마포어를 이용하고자 하는 프로세스는 세마포어의 상태가 증가되거나 0 이 될 때까지 기다리게 된다. (옵션에 따라 다를 수 있다).

* 유닉스에서의 세마포어는 한개씩 이용되는 경우는 거의 없기 때문에 보통 세마포어 집합으로 관리된다. 닉스 커널이 설정될 때 시스템에서 사용 가능한 세마포어의 최대 갯수와 세마포어 집합 당 최대 세마포어 갯수 및 할당 가능한 세마포어 집합의 최대 갯수가 설정된다. 이 값들은 유닉스 커널을 다시 만들고 시스템을 리부팅 하여야만 변경이 가능하다.

<오라클과 세마포어>

오라클은 백그라운드 프로세스들 간의 Concurrency 를 조절하기 위하여 세마포어를 사용한다. 그리고 Fast(Shared Memory) Driver 가 사용되는 경우 유저 프로세스와 쉐도우 프로세스 간의 Two-Task Communication 에도 사용된다.

<사용중인 세마포어>

현재 시스템에 할당된 세마포어를 확인하는 명령은 다음과 같다.

```
# ipcs -sb
```

이 명령은 할당된 모든 세마포어 집합과 ID Number, Owner, 각 집합의 세마포어 수 등을 보여준다. 때때로 비정상적으로 종료된 오라클 프로세스에 의해서 점유된 리소스가 릴리스 되지 않고 남아 있는 경우가 있다. 만약, 오라클이 Shutdown 상태인데도 불구하고 ipcs -sb 를 통해서 보았을 때 오라클이 점유한 세마포어가 사용중으로 나온다면 그것들을 IPCRM명령으로 Free 시켜야 한다.
즉, 오라클이 할당한 세마포어 집합에 대하여

```
#ipcrm -s ID  (ID는 ipcs 에서 출력됨)
```

명령으로 릴리스 시킨다. 아니면 시스템을 리부팅 하여도 된다.

<데이타베이스의 기동>

오라클은 백그라운드 프로세스에 필요한 모든 세마포어를 데이타베이스 기동시에 할당한다. init.ora 화일의 Processes 파라미터는 오라클에서 할당할 최대 세마포어 갯수를 결정한다.

가장 흔하게 발생하는 에러는 데이타베이스 기동시에 나타나는 다음과 같은 에러이다.

ORA-7279 : spcre:semget error, unable to get first semaphore set

시스템은 한 집합내에 가장 많은 세마포어를 가진 것이나 Processes 변수에 의해 설정되는 세마포어 갯수 중 적은 값을 가진 첫번 째 세마포어 집합을 할당하려고 한다. 만약, 시스템에 설정된 세마포어가 부족하거나, 세마포어가 이미 너무 많이 사용중이거나, 사용중이지 않은 세마포어 집합이 세마포어를 너무 많이 갖고 있는가를 체크해 보고 그렇지 않다면 시스템에 충분한 세마포어를 할당하도록 한다.

설정된 세마포어가 없거나 모든 세마포어가 이미 할당된 상태라면 다음과 같은 에러가 발생한다.

ORA-7251 : spcre:semget error, could not allocate any semaphores

첫번째 세마포어 집합이 할당되었지만 두번째 세마포어를 할당받지 못하면 다음과 같은 에러가 발생한다.

ORA-7252 : spcre:semget error, could not allocate semaphores

죽어있는 오라클 프로세스에 의해서 세마포어가 점유되어 있는가를 확인하고 만약 그런 문제가 아니라면 더 많은 세마포어를 할당함으로써 문제를 해결할 수 있다.

Shutdown abort 명령이 내려지면 사용자 Process가 끝나기를 기다리지 않고 오라클 백그라운드 프로세스는 죽게되고 그것에 의해서 점유된 세마포어는 Release 된다. 사용자는 세마포어를 늘리거나 줄여서 데이타베이스에 작업을요구할 때야 비로소 데이타베이스가 Shutdown 되었음을 알게 된다. 이 때 사용자에 의한 세마포어 변경 요구는 실패하게 되며 유저 프로세스와 오라클 Shadow Process도 함께 죽게 된다. 그러면서 다음과 같은 에러가 발생한다.

ORA-7264 : spwat:semop error, unable to decrement semaphore
ORA-7265 : sppst:semop error, unable to increment semaphore

DEC RISC Ultrix, MIPS machine 같은 MIPS RISC 기반의 유닉스 시스템의 경우에는 오라클은 Startup시에 Latching을 위하여 별도의 세마포어를 할당한다. 오라클은 이 때 할당된 세마포어를 사용자가 데이타베이스에 접속할 때 Latch 로 사용한다. 프로세스가 죽게되면 프로세스에 의해서 변경된 세마포어는 원래의 상태로 되돌아 간다. 따라서 접속하는 모든 프로세스는 Undo Structure를 할당할 수 있어야 하는데 Undo Structure가 충분하지 못하면 다음과 같은 에러가 발생한다.

ORA-9702 : sem_acquire: cannot acquire latch semaphore

해결방법은 Undo Structure가 사용가능할 때까지 기다리든지 시스템의 Undo Structure 의 최대값(SEMMNU)를 늘리면 된다.

Fast Driver에 의해서 데이타베이스에 접속되는 경우에는 Shared Memory Buffer에 대 사용허가를 관리하기 위하여 세마포어가 사용된다. 이 때 하나의 접속되는 세션마다 3개의 세마포어로 구성된 세마포어 집합이 할당된다.

시스템의 세마포어가 모두 할당되었다면 다음과 같은 에러가 발생한다.

ORA-2721 : osnseminit: cannot create semaphore set

이 경우에는 이용가능한 세마포어가 생길 때 까지 기다려야 한다. 그리고 비정상적으로 프로세스를 종료하는 경우에는 할당된 세마포어가 Release되지 못하며 어떤 프로세스가 어떤 세마포어를 사용하는가를 구분하기 힘드므로 사용상의 주의가 필요하 .

<세마포어 사용 계획>

세마포어 커널 파라미터의 이름, 조정 방법 및 커널 재생성 방법은 시스템에 따라서 다르다. 필요한 세마포어의 갯수는 다음의 공식으로 구할 수 있다.

```
proc(init.ora에서의 processes 파라미터)
+  test(MIPS RISC 프로세  를 쓰는 경우는 1, 그외에는 0)
+ 3 * f(fast driver connection 갯수)
+ sys(오라클 외의 프로그램에서 필요한 세마포어 수)
------------------------------------------------------------
= tot(필요한 최소한의 세마포어 수)
```

세마포어의 최대 갯수를 지 하는 파라미터(보통 SEMMNS)는 최소한 위에서 계산한 값보다 커야 한다.

일반적으로 하나의 집합마다 가능한 세마포어의 최대 갯수(보통 SEMMSL)는 10~25 정도이다. fast driver를 많이 쓰는 경우라면 세마포어 집합의 갯수가 더 많이 필요하다.

SUN SOLARIS C COMPILE 시 고려 사항

Bulletin no : 10099

---

1. Oracle v7에서 SUN C-Compiler version 2.0 or 2.0.1 사용시
   C-Compiler 위치 : /opt/SUNWspro
   OS Version : Solaris 2.X

   (1) SYSTEM이 SPARCstation 10 , classic/LX , 20, 5
   SYSTEM이 SPACEcenter 1000, 2000일 때
   이 때의 color graphic library는 cg92

   (2) system이 SPARCstation 1, 2, 20, 40, SUN4 series일 때
   이 때의 color graphic library는 cg89

   (3) 만약, cg92를 지원하는 시스템이라면, .cshrc에 다음과 같이 정의.
   setenv LD_LIBRARY_PATH /opt/SUNWspro/SC2.0.1/cg92:$ORACLE_HOME/lib:/usr/lib

   (만약 /opt/SUNWspro directory하에 lib directory가 있다면,
   /opt/SUNWspro/lib:$ORACLE_HOME/lib:/usr/lib로 setting함.)
   set path를 정의한 부분에 /usr/ccs/bin이 나오도록 할 것.

   (4) 만약, cg89를 지원하는 시스템이라면, .cshrc에 다음과 같이 정의.
   setenv LD_LIBRARY_PATH /opt/SUNWspro/sc2.0.1:$ORACLE_HOME/lib:/usr/lib
   set path를 정의한 부분에 /usr/ccs/bin이 맨 앞에 나오도록 할 것.
   (왜냐하면 , sun은 UCB version의 cc도 사용하기 때문)

2. Oracle V7상에서 C-Compiler V3이상을 사용할 때

   (1) cg92를 지원하는 시스템의 경우에는 다음과 같이 함.
   setenv LD_LIBRARY_PATH /opt/SUNWspro/lib:$ORACLE_HOME/lib:/usr/lib

   (2) cg89를 지원하는 시스템의 경우에는 다음과 같이 함.
   setenv LD_LIBRARY_PATH/opt/SUNWspro/SC3.0/cg89:$ORACLE_HOME/lib:/usr/lib

   (3) 두 경우 모두 set path에 /usr/ccs/bin이 먼저 오도록 할 것.

3. 모든 경우에 /usr/ccs/bin directory에 make와 cc가 있는지를 반드시 확인할 것. 만약, cc가 없다면 /opt/SUNWspro/SC2.0.1 혹은 /opt/SUNWspro/SC3.0하에 있는 cc 와 symbolic link를 시켜줄 것.
   ex) cd /usr/ccs/bin ln -s /opt/SUNWspro/SC2.0.1/cc cc

4. pro*c의 문제가 생기면, 먼저 일반 c program을 먼저 test하는 것이 좋을 것임.

5. SUN C-Compiler version 3.0은 Oracle 7.0.16.4에서도 지원됨.

ORACLE DATABASE 의 AUTOMATIC STARTUP/SHUTDOWN

Bulletin no : 10145

Introduction

This bulletin addresses the topic of automatic startup and shutdown of Oracle instances while a UNIX system is being brought up or down respectively.

The bulletin addresses specific schemes on the more popular UNIX platforms and lists the important scripts and directories that come into play in the process.

For more detailed explanations and alternatives, the reader should consult the System Administrator's Guide for the appropriate UNIX port.

Files of interest in your Oracle Installation

```
  $ORACLE_HOME/bin/dbstart -- startup database instance(s)
  $ORACLE_HOME/bin/dbshut  -- shutdown database instance(s)
  /etc/oratab    or        -- specifies instance information
  /var/opt/oracle/oratab
```

Details:

This section briefly describes the Oracle startup and shutdown scripts invoked at system startup and shutdown

The dbstart script ensures a clean startup of database instance(s) even after a system failure. dbshut ensures a clean shutdown for database instance(s).

The /etc/oratab file contains a field that specifies whether a particular database instance should be brought up/down at system startup/shutdown time. By specifying 'Y' in this field, the dbstart and dbshut scripts bring this database instance up/down.

System V v/s BSD UNIX

This section describes the generic approach taken by System V and BSD based UNIX operating systems at system startup and shutdown time.

For more details, the individual section for a particular operating system should be referred to.

System V based systems have their initialization scripts in the directories named as /etc/rc.d where is the run-level of the script. Usually, run-level 0 signifies power shutdown mode while run-level of 2 signifies multi-user mode.

The initialization scripts in these directories are named using the following method:

[K¦S][two-digit number][descriptive filename]

"S75cron" and "K30tcp" are examples of scripts that are named using the above convention.

The scripts whose name starts with an 'S' are called at startup, while the sscripts whose name starts with a 'K' are called at shutdown time.

Also, note that the digits in the name have some importance associated with them.
A script with a higher number in its name is executed later in the series of scripts being executed.

Thus, usually, oracle scripts are named something like 'S99oracle', 'K01oracle' etc.
This makes a lot of sense since we want the Oracle startup to happen *after* all the system stuff has been started. On the other hand, we want to shutdown Oracle *before* everything else on the system goes down.

BSD based systems, on the other hand, use the files /etc/rc*, such as /etc/rc, /etc/rc.local etc. at system startup time.

System shutdown in BSD based systems is very simple as well. Usually, no system or system defined scripts are invoked before the command /etc/shutdown is invoked.
Some implementations invoke /etc/rc.shutdown at shutdown time.

Platform Specific Details

Following are some details for implementing automatic startup and shutdown of Oracle instances on six major flavors of UNIX.
Again, for more details, please refer to your System Administrator's Guide or the on-line man pages.

(1) SunOS 4.1.3
    ===========

Important Files

   /etc/rc.local - The startup script for the system

Details

On a SunOS system, the startup and shutdown procedures are very simple. At startup time, the script /etc/rc.local is executed. Thus, an entry such as:
  su -c <$ORACLE_HOME>/bin/dbstart

would be needed to startup the Oracle instance(s).

There is no script that is executed at shutdown time in SunOS. Thus, the only way to invoke 'dbshut' at system shutdown time is to either do it manually or to write a shell script such as:

    #!/bin/csh
    su -c <$ORACLE_HOME>/bin/dbshut
    /etc/shutdown $*

which invokes Oracle's dbshut script before invoking the /etc/shutdown executable. Ofcourse, the above script MUST be owned by root and be executable only by root as well.

NOTE: If the system is shutdown without shutting down the databases, restarting
         the instance again after powering back will fail.
         In this case you must delete the SGA files in $ORACLE_HOME/dbs and restart
         the instance.

NOTE: The two 'su' statements in the above section have $ORACLE_HOME inside angled brackets
         < >. This is to point out that the expanded form of $ORCALE_HOME should be written
         in those lines since the value of $ORACLE_HOME might not be set for the superuser at
         that point.

(2) OSF/1
    =====

Important Files

    /etc/inittab -- Controls the initialization process
    /sbin/rc3   -- Script to start/stop processes at run level 3
    /sbin/rc0   -- Script to start/stop processes at run level 0

/sbin/rc3.d/S99oracle -- link to the script /sbin/init.d/oracle
/sbin/rc0.d/K01oracle -- same as above

Details

The /etc/inittab contains instructions for the init process. It tells it when to create and run initialization processes. One of the entries in the inittab file is:

s3:3:wait:/sbin/rc3 < /dev/console > /dev/console 2>&1

which invokes the script /sbin/rc3 and waits for it to finish. The script is invoked when the initialization is at run level 3 (multiuser state).

Now, /sbin/rc3 invokes scripts that are in the /sbin/rc3.d directory.
If the name of the script starts with a 'K', it starts the script with a 'stop' parameter, thereby stopping a process identified by the script.
On the other hand, if the script name begins with a 'S', /sbin/rc3 provides it with a runtime parameter of 'start', thereby starting processes identified by that particular script.

Thus, the /sbin/rc3.d/S99oracle is an oracle script that will be kicked off by /sbin/rc3 and get a runtime parameter of 'start'. It is this script that in turn invokes the SQL*Net V1 and V2 listeners, if installed and starts up the script $ORACLE_HOME/bin/dbstart.

The shutdown process works in a very similar fashion. The entry in /etc/inittab, for this purpose is:

s0:0:wait:/sbin/rc0 off < /dev/console > /dev/console 2>&1

This kicks off the script /sbin/rc0. /sbin/rc0, in turn, calls all scripts in /sbin/rc0.d directory providing with a runtime parameter of 'start' if their name begins with a 'S', or with a runtime parameter of 'stop', of their name begins with a 'K'.

Thus, the script /sbin/rc0.d/K01oracle gets invoked with a parameter of 'stop'. This script stops the the SQL*Net V1 and V2 listeners and invokes the script $ORACLE_HOME/bin/dbshut.

Note that both /sbin/rc3.d/S99oracle and /sbin/rc0.d/K01oracle are both symbolic links to the script /sbin/init.d/oracle. They get invoked with a 'start' and 'stop' parameter respectively which inturn controls the behaviour of the script.

NOTE: The links to /sbin/init.d/oracle might not be correctly established.
There may be more links than these 2, forcing the startup or shutdown
to happen more than once.

(3) Solaris 2.3

==========

Important Files

```
   /etc/inittab          -- Controls the initialization process
   /etc/rc2              -- Script to start/stop processes at run level 2
   /etc/rc0              -- Script to start/stop processes at run level 0

   /etc/rc2.d/S99dbstart -- link to the script /etc/init.d/dbstart
   /etc/rc0.d/K01dbshut  -- link to the script /etc/init.d/dbshut

   /var/opt/oracle/oratab-- Oracle oratab file
```

Details

The process of startup and shutdown is very similar to that in OSF/1. The only difference being that all the appropriate scripts and directories can be accessed from /etc instead of /sbin. Actually, the directories, such as rc0.d, rc2.d etc. are physically located in /etc. The scripts, such as rc0, rc2 etc. are symbolic links to corresponding scripts in /sbin.
Also, note that the scripts /etc/init.d/dbstart and /etc/init.d/dbshut are very simple one scripts containing the following commands respectively:

```
   su  -c <$ORACLE_HOME>/bin/dbstart
   and
   su  -c <$ORACLE_HOME>/bin/dbshut
```

In the commands above, <$ORACLE_HOME> should be replaced by the actual physical path that $ORACLE_HOME points to, and should be replaced by the actual userid who owns that Oracle installation.

In Solaris, unlike OSF/1, the "start" and "stop" parameters that the system passes to the scripts are ignored by the /etc/init.d/dbstart and /etc/init.d/dbshut.

NOTE: Please make sure that both dbstart and dbshut in /etc/init.d are executable
       by the oracle userid.

For more details of the initialization process, please refer to /etc/init.d/README.

(4) AIX
    ===

Important Files

    /etc/inittab   -- System initialization file
    /etc/rc   -- System initialization script
    /etc/mkitab   -- Utility to make an entry in /etc/inittab
    /etc/shutdown -- System shutdown script

Details

  There are many ways to incorporate startup of Oracle instances in the general scheme of system startup. On the other hand, system shutdown in AIX is a very straight forward process.

The user can choose one of the following methods for startup:

1. Use /etc/mkitab to make an entry in /etc/inittab :

   The command:

   /etc/mkitab ":2:wait:/bin/su   -c
     <$ORACLE_HOME>/bin/dbstart > /dev/console 2>&1"

   can be used to introduce dbstart in the /etc/inittab file. Thus, on system startup, at run level 2, $ORACLE_HOME/bin/dbstart will be invoked by the system as the Oracle owner.

2. This method is essentially the same as (1) above, but with one level of indirection.
   Use /mkitab as follows:

   /etc/mkitab "rcoracle:2:wait:/etc/rc.oracle > /dev/console 2>&1"

   Also create a script /etc/rc.oracle, owned and executable only by root.
   The contents of this script should be:

    #
    # Startup the oracle database
    #
    su </bin/dbstart
    EOF

3. Make the following entry at the end of the startup script /etc/rc:

    /bin/su -c <$ORACLE_HOME>/bin/dbstart

The system shutdown process in AIX just involves the script /etc/shutdown.
Thus, besides the option of manual shutdown, there is only option left to the user to automate shutdown of Oracle instances prior to system shutdown.

The user could write a shutdown script of the own that first shuts down the Oracle instances and then invokes the system shutdown script. The sample script could be as follows:

```
#!/bin/sh

/bin/su  -c /bin/dbshut
/etc/shutdown $*
```

Please note that the above script should be owned by root and be executable only by root as well. Changing the system shutdown script is NOT recommended by Oracle.

NOTE: In all the above cases, should be replaced by the userid of the owner of the Oracle installation.
<$ORACLE_HOME> should be replaced by the actual path pointed to by $ORACLE_HOME.

(5) HP/UX
=====

Important Files

```
/etc/rc         -- System startup file
/etc/shutdown   -- System shutdown executable
/etc/shutdown.d -- System shutdown directory
```

Details

Unlike Solaris and OSF/1, the HP/UX system startup and shutdown are relatively straight forward procedures.

At system startup, the script /etc/rc is executed. Therefore, to provide automatic startup for Oracle, simply put the following line just before the logical end of the script:

```
su  -c <$ORACLE_HOME>/bin/dbstart
```

System shutdown is brought about by running the binary /etc/shutdown.
This binary first executes all scripts and executables in the directory /etc/shutdown.d and then goes on its merry way to bring the system down normally.

Thus, to bring the Oracle instances down before the system is halted, you can do one of the following:

1. Copy the script $ORACLE_HOME/bin/dbshut to /etc/shutdown.d.
2. Make /etc/shutdown.d/dbshut a symbolic link to
   $ORACLE_HOME/bin/dbshut.
3. Write a small script such as:

```
#!/bin/sh

su  -c <$ORACLE_HOME>/bin/dbshut
```

In this case, make sure that the script is owned and executable only by the superuser.

If you are running Oracle instances from both versions 6 and 7, ou should use the dbshut script that comes with Oracle version 7.

NOTE: In the above section, <$ORACLE_HOME> should be replaced by the actual path pointed to by $ORACLE_HOME, and should be replaced by the userid who owns the Oracle installation.

SCO UNIX
========

Important Files

```
/etc/inittab          -- Controls the initialization process
/etc/rc2              -- Script to start/stop processes at run level 2
/etc/rc0              -- Script to start/stop processes at run level 0

/etc/rc2.d/S99dbstart -- link to the script /etc/init.d/dbstart
/etc/rc0.d/K01dbshut  -- link to the script /etc/init.d/dbshut

/etc/oratab           -- Oracle oratab file
```

Details

Please refer to the section for Solaris 2.3. The implementation details are very similar to those.

ORACLE 에서 CRON 사용

Bulletin no : 10146

The objective of this bulletin is to acquaint the reader with the use of cron and to provide some examples on how it is used with ORACLE. We will use the Unix att universe for the purpose of discussion. Differences between att and bsd are documented at the end of this report.

For simplicity, let us assume that the database is owned by the account 'oracle' and that the demo account is called 'scott'. Let us also assume that $ORACLE_HOME is located at /usr/oracle and the SID is S

What is cron?. Cron is a clock daemon utility provided by unix. This utility, which is located in /etc/cron executes commands specified in the file /usr/lib/crontab at times specified in crontab by the user. cron checks this file EVERY MINUTE.

Not all users are allowed to use cron. In order to use it, your userid must be listed in a file called /usr/lib/cron/cron.allow which is owned by root. Contact your system administrator to get your userid in this file. From now on, let us assume that the file cron.allow only contains userids root and oracle.

Every user listed in cron.allow will have his/her personal crontab file. Each user can only have one crontab file. This crontab will list all the commands that the user wants to execute and what times thereof.

How to set up your own crontab file

All crontab file contain lines in the following format:

* * * * *

where the *'s (often called the five time fields) reference digits that collectively make up the time, and reference the command to be executed. The five time fields in successive order are:

Field	Range
minute	0-59
hour	0-23
day of the month	1-31
month of the year	1-12
day of the week	0-6 (0 is Sunday)

You may directly use * for any one of these values which means that all possible values are accepted. For example, to execute an echo command everyday at 7:29 in the evening you can set up the following line

```
29 19 * * * echo its dinner time.
```

You can enter more requests on consecutive lines. Once you have entered all these into a file (say, mycron) using your favorite editor, you can now submit this file as YOUR CRONTAB file to the system. To do this type:

```
crontab mycron
```

You can always add more requests in your file mycron and then resubmit it using the above command. The system will only retain your most recent submission.

To view your current submitted file type:

```
crontab -l
```

To delete your current submitted file type:

```
crontab -r
```

Your Results

Now that you have submitted your crontab file, the echo command will execute everyday at 7:30 in the evening. The results are sent to you via unix mail.

Uses of Cron

As you can see, cron is very useful. System administrators use cron often to perform administrative chores automatically. Typically such activities are performed at night when the system load is low. From a database standpoint, we can use cron to perform backups of the database every night or we can run sql reports that will spool output files every night to keep track of system usage.

Scenario

Let us set up a very simple example - let us say that we wish to know the contents of the emp and dept tables in the scott account everyday at 11:30 at night and save the results in a file called scottout. In order to do this, we will need 3 files:

```
sql script containing the queries       - call this scottcron.sql
a shell script to execute scottcron.sql - call this scottshell
a crontab file to run the shell script  - call this mycron
  (this last file has to be submitted to cron)
```

Let us also say that these files and the results will be located in the directory /usr/oracle/personal.

Step 1 - Writing the sql script

```
spool /usr/oracle/personal/scottout
select * from emp;
select * from dept;
exit;
```

Now we need to set up a shell script that will enter sqlplus as scott/tiger and execute scottcron.sql. We also need to write the crontab file. Let us approach this from two angles:

Approach A - Executing as ROOT

In this approach we will let the ROOT execute the commands and save it in /usr/oracle/personal. We can set up the shell file scottshell as follows:

```
sqlplus scott/tiger @/usr/oracle/personal/scottcron
```

and the crontab file as

```
30 23 * * * su - oracle -c "/usr/oracle/personal/scottshell" >
                          "/usr/oracle/personall/scotterror"
```

Essentially, root is the super user and can thus impersonate oracle without a password. su oracle will enable root to become oracle. The -c means 'execute the following command as oracle'. The extra '-' between su and oracle is crucial - because of this, the script is being executed as if the user oracle had logged in himself. In other words all environment variables like $ORACLE_HOME,$ORACLE_SID and $PATH are set correctly.

This is the approach we use at Oracle Support when maintaining our own online support system. Our crontab file looks like the following

```
<5 time fields> su - oracle -c "shell1"
<5 time fields> su - oracle -c "shell2"
<5 time fields> su - oracle -c "shell3"
<5 time fields> su - oracle -c "shell4"
<5 time fields> su - oracle -c "shell5"
```

Approach B - Executing as oracle

In this approach, oracle will submit the crontab file instead of root. The shell file scottshell will now look like:

```
ORACLE_HOME=/usr/oracle; export ORACLE_HOME;
```

ORACLE_SID=S; export ORACLE_SID;
/usr/oracle/bin/sqlplus scott/tiger @/usr/oracle/personal/scottcron and the crontab file will look like:

```
30 23 * * * "/usr/oracle/personal/scottshell" >
          "/usr/oracle/personal1/scotterror"
```

A common mistake is to forget to set ORACLE_HOME and ORACLE_SID - this usually yields to a ORA 2700 error. This is not a bug - when cron runs the script it uses the environment similar to the one used by root when it does an 'su' to oracle WITHOUT executing the login scripts of oracle. Thus the ORACLE variables are not set. As a matter of fact, you can also set the PATH explicitly to avoid using full path names.

We can do an export or run a report in the same way.

Differences with BSD Environment.

In the ATT environment, we could never edit the crontab file directly - we had to keep resubmitting our personal cronfiles. In BSD versions, there is only one configuration file called /usr/lib/crontab. Version 4.3 also has a crontab.local. You may either use one or the other, not both.

Also on some systems, the crontab files are set up in a seven field format. The first 5 fields are the time fields, the sixth is the userid and seventh is the command to be executed.

If you are not using cron, be sure to remove your crontab file - otherwise the system will keep executing it forever!!!.

UNIX 에서 ORACLE PERMISSION 문제 - ORA-1034, ORA-7320,7429

Bulletin no : 10150

문제 설명

========

ORACLE DBA 사용자만 데이타베이스를 ACESS할수 있고 다른 사용자는 SQL*PLUS 등을 통하여 CONNECT 를 하려고 할때 다음 에러가 발생 할경우가 있다.

```
ORA-01034, "ORACLE not available"

ORA-07320, "smsget: shmat error when trying to attach sga."
// *Cause:  Unable to attach segment.
// *Action: Check errno, sercose[0] returns segment id. Verify segment
//          exists and that permissions are correct.

UNIX System V Error 13    /* Permission denied */
```

or:

```
ORA-01034, "ORACLE not available"

ORA-07429, "smsgsg: shmget() failed to get segment."
// *Cause:  a shared memory segment used for all part of the SGA could
//          not be retrieved.
// *Action: Use the system error number in the error message to
            determine
//          why the segment could not be retrieved. If it does not
            exists,
//          shutdown the database using the "abort" option, and then
//          restart it. If the get failed because the permissions are
//          incorrect, make sure that the ownership of the oracle
//          executable is the same as that on the shared memory
            segment.
```

해결 방법

==========

Check for these permission and ownership problems:

o Check the "oracle" and "orasrv" executables. (You may not have an "orasrv" executable; you may have "tnslsnr" instead.
"oracle" and "orasrv" should have the following permissions (be setuid):

```
-rwsr-s--x   1 usupport dba   7330847 Oct 20 11:01 oracle*
-rwsr-x--x   1 root     dba     62009 Dec 31 1993  orasrv*
```

If they are not:

```
% chmod 6751 oracle
% chmod 4751 orasrv
```

o Verify that the filesystem is not mounted no set uid. Check /etc/fstab for "nosuid". For example:

```
/dev/dsk/c3d0s3   /dev/rdsk/c3d0s3      /mnt4 ufs    - yes rw,nosuid
```

o Check the "umask" of the oracle user. If Oracle is installed with any "umask" other than "022", such as "077", the directory and file permissions will be set incorrectly. To see the current umask, type this at the oracle user's UNIX prompt:

```
% umask
```

You should see:

```
022 or 22
```

o Check the ownership of the executables in $ORACLE_HOME/bin.
All files in the oracle installation must be owned by the oracle user except the following, which when they are present must be owned by root:

```
-rw-r--r--   1 root     dba      7332 Jul 12 09:47 gateway
-rwsr-xr-x   1 root     dba   3348364 Jun  6 10:44 oomgr*
-rwsr-xr-x   1 root     dba       246 Jun  6 10:44 oomgrf*
-rwsr-s--x   1 root     dba     62009 Dec 31 1993  orasrv*
-rwsr-xr-x   1 root     dba      6478 Jun 28 08:59 osh*
```

Use the UNIX "chown" command to correct minor ownership problems or reinstall. For example:

```
% chown root osh
```

changes the owner of the executable "osh" to the "root" user:

o Verify that all oracle files have as their group the oracle dba group.
Use "chgrp" to correct group ownership problems:

```
% chgrp dba *
```

changes the group of all files in the current directory to that of the oracle dba. However, if permissions are not correct in one directory it is likely to be an installation wide problem.

o Check that all directories in the path of ORACLE_HOME have at least a 755 mask.
For example, if ORACLE_HOME = /usr/oracle/714, then the directories /usr /usr/oracle and /usr/oracle/714 should all have a mask of 755.
This means their permissions should be like:

```
drwxr-xr-x   35 root     sys          1024 Oct  8 18:27 usr
```

If they are not:

```
% chmod 755
```

UNIX 에서 SHARED MEMORY REQUIREMENT

Bulletin no : 10151

Problem Explanation:

Oracle uses shared memory and semaphores to communicate between processes and the SGA (System Global Area). There are certain requirements for shared memory and the semaphores. When the Oracle instance comes up, it allocates a certain portion of the main memory to create the SGA. If the shared memory or the semaphores are not set properly, then it gives an error related to shared memory or semaphores.

The following are the recommended values for shared memory and semaphores for running a SMALL size Oracle database. These values are set at the Unix kernel level.

Solution Description:

The shared memory feature of the UNIX operating system is required by Oracle.

The System Global Area (SGA) resides in shared memory; therefore, shared memory must be available to each Oracle process to address the entire SGA.

Definitions of Shared Memory and Semaphore Parameters

SHMMAX = The maximum size(in bytes) of a single shared memory segment.
SHMMIN = The minimum size(in bytes) of a single shared memory segment.
SHMMNI = The number of shared memory identifiers.
SHMSEG = The maximum number of shared memory segments that can be attached by a process.
SEMMNS = The number of semaphores in the system.
SEMMNI = The number of semaphore set identifiers in the system; determines the number of semaphore sets that can be created at any one time. SEMMSL = The maximum number of sempahores that can be in one semaphore set.

It should be same size as maximum number of Oracle processes (The PROCESSES parameter in the init.ora file).

Recommended Semaphore and Shared Memory Parameters

Operating System	Shared Memory Parameters	Semaphore
Sun OS	SHMSIZE = 32768	SEMMNS = 200
	SHMMNI = 50	SEMMNI = 50
Solaris	SHMMAX = 8388608	SEMMNS = 200
	SHMSEG = 20	SEMMSL = 50
	SHMMNI=100	SEMMNI = 70
HP/UX	SHMMAX = 0x4000000 (64 Mb)	SEMMNS = 128
	SHMSEG = 12	SEMMNI = 10
Digital Unix	SHMMAX = 4194304	SEMMNS = 60
(DEC Alpha OSF/1)	SHMSEG = 32	SEMMSL = 25
Ultrix	Use system default	SEMMNS = 60
		SEMMSL = 5
AT&T Unix	SHMMAX = RAM-Dependant	SEMMNS = 200
8 or 16 Mb RAM	SHMMAX = 5 Mb	for all RAM
32 Mb RAM	SHMMAX = 8 Mb	values
64 Mb RAM	SHMMAX = 16 Mb	
128 Mb RAM	SHMMAX = 32 Mb	
256 Mb RAM	SHMMAX = 64 Mb	
512 Mb RAM	SHMMAX = 128 Mb	
1024 Mb RAM	SHMMAX = 256 Mb	
2048 Mb RAM	SHMMAX = 512 Mb	
	SHMSEG = 6 for all RAM values	
	SHMMIN = 1 for all RAM values	
Dynix/PTX	SHMMAX = 11010048	SEMMNS = 200
	SHMSEG = 20	SEMMSL = 85
Other parameter:	NOFILES = 128	
DG/UX	SHMMAX = 4194304	SEMMNS = 200
	SHMSEG = 15	

Setting the Shared Memory and Semaphore Parameters

The values of the shared memory and semaphore parameters must be set in the kernel configuration file of your operating system (OS). The location of that file is port-specific.

To check your current shared memory and semaphore configuration you can use the command:

```
% sysdef | more
```

Example on HP-UX (relevant sections only):

```
-----------------------------------------
```

Semaphore Related Parameters

```
maximum value for semaphores(semaem)= 16384
Semaphore  map(semmap)= 4098
number of semaphore identifiers(semmni) = 4096
total number of semaphores in the system(semmns) = 8192
number of semaphore undo structures(semmnu) = 1536
semaphore undo entries per process(semume) = 512
semaphore maximum value(semvmx) = 32767
```

Shared Memory Related Parameters

```
maximum shared memory segment size in bytes(shmmax) = 536870912
minimum shared memory segment size in bytes(shmmin) = 1
maximum shared memory segments in system (shmmni) = 512
maximum shared memory segments per process(shmseg) = 512
```

NOTE: The SHMMAX is quite large on this system as there are 8 instances running on this system.

To make changes in shared memory or semaphore parameters:

1. Shut down any running Oracle instances
2. Locate the kernel configuration file for your OS
3. Make the necessary changes using the system utilities or your favorite editor. System utilities include:

OS	Utility
HP-UX	SAMS
SCO	SYSADMSH
AIX	SMIT
Solaris	ADMINTOOL

4. Reconfigure the kernel.
5. Reboot your machine.
6. Restart your Oracle instances

Example, using Solaris 2.3/2.4 parameters and commands:

1. Log into SQLDBA and type:
   SQLDBA> shutdown
   SQLDBA> exit

2. Log in as the superuser (root) and:
   # cd /etc/system

3. Add the following lines to the /etc/system file:

   set shmsys:shminfo_shmmax=8388608
   set shmsys:shminfo_shmmin=1
   set shmsys:shminfo_shmmni=100
   set shmsys:shminfo_shmseg=20
   set semsys:seminfo_semmns=200
   set semsys:seminfo_semmni=70

4. Reconfigure the kernel:
   # touch /reconfigure

5. Reboot the machine:
   #init 6

6. Log into SQLDBA and type:
   SQLDBA> startup
   SQLDBA> exit

Oracle, Shared Memory, and SGA Size

There are Oracle init.ora parameters which can be modified to influence the size of the SGA. These settings of these parameters, in conjunction with OS shared memory and semaphore parameters, can influence both system and Oracle performance.

UNIX 에서 SHARED MEMORY/SEMAPHORE 의 이해

Bulletin no : 10152

Understanding Oracle and Shared Memory/Semaphores

Shared memory and semaphores are two important resources for an Oracle instance on Unix. An instance cannot start if it is unable to allocate what it needs. This paper primarily discusses the process Oracle goes through to allocate shared memory and semaphores at instance startup. Other important points unrelated to startup as well as some troubleshooting information will be touched.

General

Shared memory is exactly that - a memory region that can shared between different processes. Oracle uses shared memory for implementing the SGA, which needs to be visible to all database sessions. Shared memory is also used in the implementation of the SQL*Net V1 Fast driver as a means of communicating between the application and shadow process. On the RS/6000, each shadow process stores its PGA in a shared memory segment (however, only the shadow attaches this segment). In the latter two cases, Oracle allocates the shared memory dynamically as opposed to the allocation of the SGA, which occurs at instance startup. This allocation will not be discussed in this paper.

Semaphores can be thought of as flags (hence their name, semaphores).
They are either on or off. A process can turn on the flag or turn it off.
If the flag is already on, processes who try to turn on the flag will sleep until the flag is off. Upon awakening, the process will reattempt to turn the flag on, possibly suceeding or possibly sleeping again. Such behaviour allows semaphores to be used in implementing a post-wait driver - a system where processes can wait for events (i.e. wait on turning on a semphore) and post events (i.e. turning of a semaphore). This mechanism is used by Oracle to maintain concurrency control over the SGA, since it is writeable by all processes attached.
Also, for the same reasons, useof the Fast Driver requires additional semaphores. However, these semaphores will be allocated dynamically instead of at instance startup. This allocation will not be discussed in this paper.

Instance startup

On instance startup, the first things that are done are: read the init.ora, start the background processes, and allocate the shared memory and semphores required. The size of the SGA will be calculated from various init.ora parameters; this will be the amount of shared memory required. The SGA is broken into 4 sections - the fixed portion, which is constant in size, the variable portion, which varies in size depending on init.ora parameters, the redo block buffer, which has its size controlled by log_buffers, and the db block buffer, which has its size controlled by db_block_buffers.

The size of the SGA is the sum of the sizes of the 4 portions.

There is unfortunately no simple formula for determining the size of the variable portion. Generally, the shared pool dominates all other parts of the variable portion, so as a rule of thumb, one can estimate the size as the value of shared_pool_size (in v6, one can ignore the size of the variable portion). The number of semphores required is much simpler to determine. Oracle will need exactly as many semaphores as the value of the processes init.ora parameter.

Note that the recommended kernel parameter values in the ICG are enough to support the default database (4M SGA, 50 processes), but may be insufficient to run a larger instance. With the above estimations and the information which follows, a DBA should be able to build a kernel with appropriate settings to support the instance.

Shared memory allocation

Oracle has 3 different possible models for the SGA - one-segment, contiguous multi-segment, and non-contiguous multi-segment.

When attempting to allocate and attach shared memory for the SGA, it will attempt each one, in the above order, until one succeeds or raises an ORA error. On other, non-fatal, errors, Oracle simply cleans up and tries again using the next memory model. The entire SGA must fit into shared memory, so the total amount of shared memory allocated under any model will be equal to the size of the SGA. This calculated value will be referred to below as SGASIZE.

The one-segment model is the simplest and first model tried. In this model, the SGA resides in only one shared memory segment. Oracle attempts to allocate and attach one shared memory segement of size equal to total size of the SGA. However, if the SGASIZE is larger than the configured SHMMAX, this will obviously fail (with EINVAL). In this case, the SGA will need to be placed in multiple shared memory segments, and Oracle proceeds to the next memory model for the SGA. If an error other than EINVAL occurs when allocating the shared memory with shmget(), Oracle will raise an ORA-7306.

If the segment was gotten (i.e. if SHMMAX > SGASIZE), Oracle attempts to attach it at the start address defined in ksms.o. An error on the attach will raise an ORA-7307. Note: k는.

o is only used in SVR4-based Operating Systems such as Solaris and is not used in BSD- based Operating Systems such as Sun4 and Ultrix.

With multiple segments, there are two possibilities. The segments can be attached contiguously, so that it appears to be one large shared memory segment, or non-contiguously,

with gaps between the segments. The former wastes less space that could be used for the stack or heap, but depending on alignment requirements for shared memory (defined by SHMLBA in the kernel), it may not be possible.

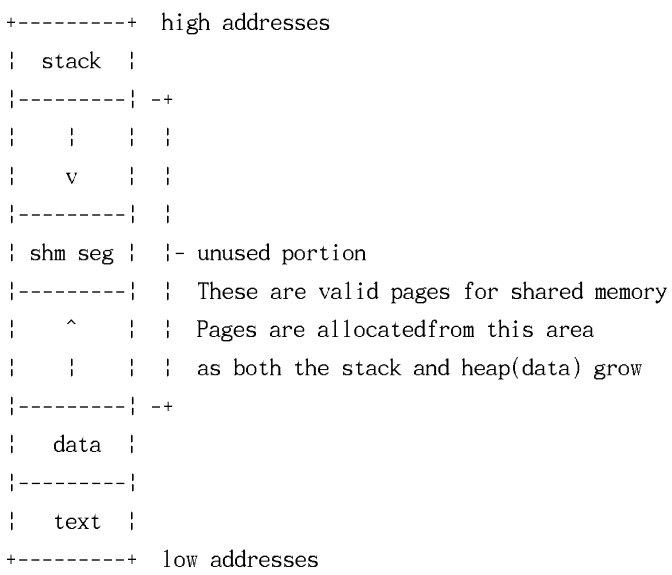
At this point, Oracle needs to determine SHMMAXso it can determine how many segments will be required. This is done via a binary search algorithm over the range [1...SGASIZE] (since Oracle is trying this model and not the one segment model it must be that SHMMAX.dbf file is used to get the necessary information. In version 7, the SGA itself contains the information about the shared memory and semaphores (how the bootstrap works will be explained later). In either case, the information stored is the same - the key, id, size, and attach address of each shared memory segment and the key, id, and size of each semaphore set. Note that we need not do anything special to initialize the semaphores. We can use them with the data structure we read in on connecting.

The version 6 approach is rather simple. It first tries to open the sgadef.dbf file. If it cannot, an ORA-7318 is raised. Once opened, the data written earlier on startup is read. If an error occurs for some reason on the read, an ORA-7319 occurs. Once all the data is read in, Oracle attaches each segment in turn. First, it generates what it believes the key for the segment should be. It then gets that segment, returning ORA-7429 if it fails. The key used and the key stored are then compared. They should be equal, but if not, an ORA-7430 occurs. Once the key is verified, the segment is attached. A failure to attach the segment raises an ORA-7320. If the segment is attached, but not at the address we requested, an ORA-7321 occurs. This process is repeated for all segments until the entire SGA is attached.

Version 7 differs only in the first part, when the shared memory and semaphore data is read. Once that data is read in, Oracle proceeds in the same manner. To fetch this data, Oracle generates what it thinks should be the key for the first segment of the SGA and attaches it, as if it were the only segment. Once it is attached, the data is copied from the SGA. With this data, Oracle attaches any remaining segments for the SGA. There is one possible problem. If somehow two instances have a key collision (i.e. they both generate the same key for their first segment), it is possible to only have one of the two instances up at a time! Connection attempts to either one will connect a user to whichever instance is up. This is rare, but can happen. Development is currently working on a better key generation algorithm.

Attaching shared memory

As seen in previous sections, shared memory must be gotten (this may mean allocating the shared memory, but not necessarily) and then attached, to be used. Attaching shared memory brings the shared memory into the process' memory space. There are some important things about attach addresses. For one thing, they may need to be alligned on some boundary (generally defined by SHMLBA). More importantly, shared memory must mapped to pages in the process' memory space which are unaccounted for. Every process already has a text, a data, and a stack segment laid out as follows (in general):



So valid attach addresses lie in the unused region between the stack and the data segments (a shared memory segment is drawn in the diagram to aid in visualization - not every process has shared memory attached!). Of course, the validity also depends on the size of the segment, since it cannot overlap another segment. Note that both the stack and data segments can grow during the life of a process. Because segments must be contiguous and overlapping is not allowed, this is of some importance. Attaching shared memory creates a limit on how much the stack or data segment can grow. Limiting the stack is typically not a problem, except when running deeply recursive code. Neither is limiting the data segment, but this does restrict the amount memory that can be dynamically allocated by a program. It is possible (but seldom) that some applications running against the database may hit this limit in the shadow (since the shadow has the SGA attached). This is the cause of ORA-7324 and ORA-7325 errors. How to deal with these is discussed in the troubleshooting section.

The SGA is attached, depending on the allocation model used, more or less contiguously (there may be gaps, but those can be treated as if they were part of the shared memory). So where the beginning of the SGA can be attached depends on the SGA's size. The default address which is chosen by Oracle is generally sufficient for most SGAs. However, it may be necessary to relocate the SGA for very large sizes. It may also need to be changed if ORA-7324 or ORA-7325 errors are occuring. The beginning attach address is defined in the file ksms.s. Changing the attach address requires recompilation of the Oracle kernel and should not be done without first consulting Oracle personnel. Unfortunately, there is no good way to determine what a good attach address will be. When changing the address to allow a larger SGA, a good rule of thumb is taking the default attach address in ksms.s and subtracting the size of the SGA. The validity of an attach address can be tested with the Oracle provided tstshm executable. Using tstshm -t -b will determine if the address is usable or not.

Trouble shooting

Errors which might have multiple causes are discussed in this sections. Errors not mentioned here generally have only one cause which has a typically obvious solution.

ORA-7306, ORA-7336, ORA-7329

Oracle received a system error on a shmget() call. The system error should be reported. There are a few possibilities:

1) There is insufficient shared memory available. This is indicated by the operating system error ENOSPC. Most likely, SHMMNI is too small. Alternatively, there may shared memory already allocated; if it is not attached, perhaps it can be freed. Maybe shared memory isn't configured in the kernel.

2) There is insufficient memory available. Remember, shared memory needs pages of virtual memory. The system error ENOMEM indicates there is insufficient virtual memory. Swap needs to be increased, either by adding more or by freeing currently used swap (i.e. free other shared memory, kill other processes)

3) The size of the shared memory segment requested is invalid. In this case, EINVAL is returned by the system. This should be very rare -however, it is possible. This can occur if SHMMAX is not a mulitple of page size and Oracle is trying a multi-segment model. Remember that Oracle rounds its calculation of SHMMAX to a page boundary, so it may have rounded it up past the real SHMMAX! (Whether this is a bug is debatable).

4) The shared memory segment does not exist. This would be indicated by the system error ENOENT. This would never happen on startup; it only would happen on connects. The shared memory most likely has been removed unexpectedly by someone or the instance is down.

ORA-7307, ORA-7337, ORA-7320

Oracle received a system error on a shmat() call. The system should be reported. There a a few possibilities:

1) The attach address is bad. If this is the cause, EINVAL is returned by the system. Refer to the section on the attach address to see why the attach address might be bad. This may happen after enlarging the SGA.

2) The permissions on the segment do not allow the process to attach it. The operating system error will be EACCES. Generally the cause of this is either the setuid bit is not turned on for the oracle executable, or root started the database (and happens to own the shared memory). Normally, this would be seen only on connects.

3) The process cannot attach any more shared memory segments. This would be accompanieed by the system error EMFILE. SHMSEG is too small. Note that as long as SHMSEG is greater than SS_SEG_MAX, you should never see this happen.

ORA-7329, ORA-7334

Oracle has determined the SGA needs too many shared memory segments. Since you can't change the limit on the number of segments, you should instead increase SHMMAX so that fewer segments are required.

ORA-7339

Oracle has determined it needs too many semaphore sets. Since you can't change the limit on the number of semaphore sets, you should increase SEMMSL so fewer sets are required.

ORA-7250, ORA-7279, ORA-7252

Oracle received a system error on a semget() call. The system error should be reported. There should be only one system error ever returned with this, ENOSPC. This can mean one of two things. Either the system limit on sempahore sets has been reached or the system limit on the total number of semaphores has been reached. Raise SEMMNI or SEMMNS, as is appropriate, or perhaps there are some semaphore sets which can be released. In the case of ORA-7250, ORANSEMS may be set too high (>SEMMSL). If it is, raise SEMMSL or decrease ORANSEMS.

ORA-7251

Oracle failed to allocate even a semaphore set of only one semaphore. It is likely that semaphores are not configured in the kernel.

ORA-7318

Oracle could not open the sgadef file. The system error number will be returned. There are a few possible causes:

1) The file doesn't exist. In this case, the system error ENOENT is returned. Maybe ORACLE_SID or ORACLE_HOME is set wrong so that Oracle is looking in the wrong place. Possibly the file does not exist (in this case, a restart is necessary to allow connections again).

2) The file can't be accessed for reading. The operating system error returned with this is EACCES. The permissions on the file (or maybe directories) don't allow an open for reading of the sgadef file. It might not be owned by the oracle owner. The setuid bit might not be turned on for the oracle executable.

ORA-7319

Oracle did not find all the data it expected when reading the sgadef.dbf file. Most likely the file has been truncated. The only recovery is to restart the instance.

ORA-7430

Oracle expected a key to be used for the segment which does not match the key stored in the shared memory and semaphore data structure. This probably indicates a corruption of the sgadef file (in version 6) or the data in the first segment of the SGA (in version 7). A restart of the instance is probably necessary to recover in that case.It may also be a key collision problem and Oracle is attached to the wrong instance.

ORA-7321

Oracle was able to attach the segment, but not at the address it requested. In most cases, this would be caused by corrupted data in the sgadeffile (in version 6) or the first segment of the SGA (in version 7). A restart of the database may be necessary to recover.

ORA-7324, ORA-7325

Oracle was unable to allocate memory. Most likely, the heap (data segment) has grown into the bottom of the SGA. Relocating the SGA to a higher attach address may help, but there may be other causes. Memory leaks can cause this error. The init.ora parameter sort_area_size may be too large, decreasing it may resolve the error. The init.ora parameter context_incr may also be too large, decreasing it may resolve this

ORA-7264, ORA-7265

Oracle was unable to decrement/increment a semaphore. This generally is accompanied by the system error EINVAL and a number which is the identifier of the semaphore set. This is almost always because the semaphore set was removed, but the shadow process was not aware of it (generally due to a shutdown abort or instance crash). This error is usually ignorable.

SystemParameters

SHMMAX - kernel parameter controlling maximum size of one shared memory segment
SHMMHI - kernel parameter controlling maximum number of shared memory segments in the system
SHMSEG - kernel parameter controlling maximum number of shared memory segments a process can attach
SEMMNS - kernel parameter controlling maximum number of semphores in the system
SEMMNI - kernel parameter controlling maximum number of semaphore sets. Semphores in Unix are allocated in sets of 1 to SEMMSL.
SEMMSL - kernel parameter controlling maximum number of semaphores in a semphore set.
SHMLBA - kernel parameter controlling alignment of shared memory segments; all segments must be attached at multiples of this value.
Typically, non-tunable.

System errors

ENOENT - No such file or directory, system error 2
ENOMEM - Not enough core, system error 12
EACCES - Permission denied, system error number 13
EINVAL - Invalid argument, system error number 22
EMFILE - Too many open files, system error number 24
ENOSPC - No space left on device, system error number 28

Oracle parameters

SS_SEG_MAX - Oracle parameter specified at compile time (therefore, unmodifiable without an Oracle patch) which defines maximum number of segements the SGA can reside in. Normally set to 20.

SS_SEM_MAX - Oracle parameter specified at compile time (therefore, unmodifiable without an Oracle patch) which defined maximum number of semaphore sets oracle will allocate. Normally set to 10.

RAID 와 ORACLE의 관계 소개

Bulletin no : 10157

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1. RAID - General Description

RAID (Redundant Arrays of Inexpensive Disks) is frequently used to provide data redundancy in case of disk failure and involves, in part, striping of data across several physical drives. RAID can be implemented in any one of several levels, ranging from 0 to 6. Those levels are characterized by the following:

RAID 0: This is a 'normal' file system, in which there will be data loss with any disk failure.
This level provides good performance and speed of access but no data redundancy. If something on the disk becomes corrupted, there is no way to restore or recalculate that data.

RAID 1: Full data redundancy is provided by complete mirroring of all files.
This method requires twice as much disk space as there is data.
No error correction codes (ECC's) are used, and there is hence no parity information involved.

RAID 2: Redundancy is provided through ECC's (also sometimes called parity), which are calculated for each 'chunk' of data. These ECC's can be used to recreate the data in question and so this method requires less disk space that RAID level 1. However, the need to calculate ECC's will generally make writes slower.

RAID 3: ECC's are again calculated to provide data redundancy, but all the parity bytes are stored on one physical disk with bit level parity.

RAID 4: The same as RAID level 3 but with block level parity.

RAID 5: Redundancy is provided by ECC's as in RAID 3 and 4, but in this ****** case, parity information is stored along with the data and is hence stripped across several physical disks. RAID 5 should not be used for write-intensive files, since the continuous calculation of ECC's makes writing data to disk slower.
RAID 3 should be used instead. RAID 5 works best for read-only data files.

RAID 6: ECC's are calculated in a more complex way, and redundancy is provided with an advanced 2-dimensional parity.

Raid levels 1 and 5 are most commonly used. However, the choice of which specific RAID level to choose and how to implement that choice can be a complex question which must balance

performance, cost, and data availability issues.

2. Pro's and Cons of Implementing RAID technology

There are benefits and disadvantages to using RAID, and those depend on the RAID level under consideration and the specific system in question.

In general, RAID level 1 is most useful for systems where complete redundancy of data is a must and disk space is not an issue. For large datafiles or systems with less disk space, this RAID level may not be feasible. Writes under this level of RAID are no faster and no slower than 'usual'. For all other levels of RAID, writes will tend to be slower and reads will be faster than under 'normal' file systems. Writes will be slower the more frequently ECC's are calculated and the more complex those ECC's are.

Depending on the ratio of reads to writes in your system, I/O speed may have a net increase or a net decrease. RAID can improve performance by distributing I/O, however, since the RAID controller spreads data over several physical drives and therefore no single drive is overburdened.

The striping of data across physical drives has several consequences besides balancing I/O. One additional advantage is that logical files may be created which are larger that the maximum size usually supported by an operating system. There are disadvantages, as well, however. Striping means that it is no longer possible to locate a single datafile on a specific physical drive.

This may cause the loss of some application tuning capabilities. Also, in Oracle's case, it can cause database recovery to be more time -consuming. If a single physical disk in a RAID array needs recovery, all the disks which are part of that logical RAID device must be involved in the recovery.

One additional note is that the storage of ECC's may require up to 20% more disk space than would storage of data alone, so there is some disk overhead involved with usage of RAID.

3. RAID and Oracle

The usage of RAID is transparent to Oracle. All the features specific to RAID configuration are handled by the operating system and go on behind- the-scenes as far as Oracle is concerned. Different Oracle file-types are suited differently for RAID devices. Datafiles and archive logs can be placed on RAID devices, since they are accessed randomly. Redo logs should be not be put on RAID devices, since they are accessed sequentially and performance is enhanced in their case by having the disk drive head near the last write location. However, mirroring of redo log files is strongly recommended by Oracle.

References:

1. Oracle for Unix Performance Tuning Tips pp. 26-27 (A22535-1)
2. RAID: High-Performance, Reliable Secondary Storage (Chen, Peter etal, 1994)
3. Personal Communications with Oracle Analysts COMET and GASFOOR.

UNIX 에서 사용되는 ORACLE 환경 변수

Bulletin no : 10161

The document describes each Oracle environment variable with its name, it use, general sample values, and specific examples in the following manner:

ENVIRONMENT VARIABLE NAME
    Use: What is this environment variable?
    General: What are some sample values?
    Example: Specific example

In addition, UNIX environment variables that affect Oracle are briefly described.

General Notes

1. $O_H == $ORACLE_HOME
2. $O_S == $ORACLE_SID
3. An environment variable followed by a "*" applies to Oracle7.
4. An environment variable followed by a "#" is used during database creation for Oracle Version 6 only.
5. For more information, see your IUG, ICG, or the "ORACLE7 Server for UNIX Administrator's Reference Guide"
6. Note that not all UNIX platforms use all of these environment variables.

Oracle Environment Variables

APIPATH*
    Use: version 7.0.12 only: directory containing Tool Kit II *.res files
    General:
    Example: $O_H/orainst:$O_H/tk2/admin

BOOK_LOCALPREFERENCE*
    Use: Oracle*Book env var
    General: $ORACLE_HOME/book/admin
    Example: $ORACLE_HOME/book/admin

BOOK_RESOURCE*

Use: directory for Oracle*Book resource files

General: $O_H/book/admin/resource/US, $O_H/book/admin/resource/JA

Example: $ORACLE_HOME/book/admin/resource/US

BOOK_GLOBALPREFERENCE*

Use: Oracle*Book env var

General: $ORACLE_HOME/book/admin

Example: $ORACLE_HOME/book/admin

BOOK_HELP*

Use: directory for Oracle*Book help files

General: $ORACLE_HOME/book/admin/help/US

Example: $ORACLE_HOME/book/admin/help/US

CASE_HP_CMD

Use: Command to plot a file from CASE*Designer

General: Actual print command or name of a script

Example: lpr -Pplot

CASE_PS_CMD

Use: Command to print postscript file from CASE*Designer

General: Actual print command or name of a script

Example: lpr -P

CASE_RESOURCE

Use: Tool Kit I terminal type for CASE

General: file:device (file.r from $O_H/dict50/admin/etc) and (device from Oraterm)

Example: case_hpx:xterm

CASE_SDPRINT

Use: to print CASE*Designer screen prints to line or PostScript printers

General: printer name

Example: lw

CASE_XTERM

Use: Tool Kit I GUI window type for CASE

General: xterm, hpterm, aixterm, dxterm, etc.

Example: xterm

CGEN_HOME

Use: home directory of CASE Generator

General:

Example: $O_H/cgen20

DEBUG_SLFIND*

Use: Debug for TK2 tools

General: 0,1,?

Example: 1

DBS_FILE#

Use: initial database file name

General: $O_H/dbs/dbs$O_S.dbf will hard code db file (you will not be able to move $O_H); setting to dbs$O_S.dbf will allow you to move $O_H

Example: dbsoracle.dbf

DBS_SIZE#

Use: initial database file size

General: numberK, numberM

Example: 5000K or 5M minimum, up to partition size

FORMS_DEVICE*

Use: devicename for TK2 tools (CDE tools)

General: vt220, vt100, hp, sun(see $ORACLE_HOME/tk2/admin/terminal)

Example: hp

FORMS30PATH

Use: directory containing Forms 3.0 (TK I) resource files

General:

Example: $ORACLE_HOME/forms30/admin/resource

LOG

Use: file to record the install procedure

General: $O_H/install/install.log

Example: $O_H/install/install.log

LOG_FILE1#

Use: first redo log file name

General: $O_H/dbs/log1$O_S.dbf will hard code log file (you will not be able to move $O_H); setting to log1$O_S.dbf will allow you to move $O_H

Example: log1oracle.dbf

LOG_FILE2#

Use: first redo log file name

General: $O_H/dbs/log2$O_S.dbf will hard code log file (you will not be able to move $O_H); setting to log2$O_S.dbf will allow you to move $O_H

Example: log2oracle.dbf

LOG_SIZE#

Use: initial log file size

General: numberK, numberM

Example: no minimum, 500K default, no maximum (tune to your database needs)

MENU5PATH

Use: directory containing Menu 5.0 (TK I) resource files

General:

Example: $ORACLE_HOME/menu5/admin/resource

MM_RESOURCE*

Use: ?

General: $ORACLE_HOME/mm/admin/resource/US

Example: $ORACLE_HOME/mm/admin/resource/US

NLS_LANG

Use: National Language Support language, territory, and character set

General: language_territory.characterset

Example: american_american.us7ascii

NO_MAKE

Use: set to true if your system doesn't have C dev tools (cc, make, ld, etc.)

General:

Example: false

ORACLE_BASE*

Use: new directory structure for Oracle 7 to follow Oracle Flexible Architecture

General: see ICG "Preparing to install Oracle Products"

Example: can be any directory

ORACLE_HELP

Use: directory containing help file?

General: $ORACLE_HOME/help/admin/resource

Example: $ORACLE_HOME/help/admin/resource

ORACLE_HOME

Use: dir containing top level Oracle directories

General: any directory with enough space

Example: /usr/oracle

ORACLE_ICON*

Use: dir containing icons for CDE tools

General: $ORACLE_HOME/guicommon/tk2/admin/icon

Example: $ORACLE_HOME/guicommon/tk2/admin/icon

ORACLE_LPARGS

Use: which lp arguments to use for Easy*SQL,SQL*Calc,SQL*Forms, SQL*Menu, SQL*Report, or SQL*ReportWriter

General:

Example: -c -s

ORACLE_LPPROG

Use: which lp command to use for Easy*SQL,SQL*Calc, SQL*Forms, SQL*Menu, SQL*Report, or SQL*ReportWriter

General: lp, lpr, print

Example: lp

ORACLE_LPSTAT

Use: which lp status command to use

General: lpstat, lpq

Example: lpstat

ORACLE_OWNER

Use: UNIX Operating System usercode who owns the Oracle files

General: can be any user, default is oracle

Example: oracle

ORACLE_PAGER

Use: which UNIX pager to use (more, less, pg) in Oracle Products like SQL*Menu

General:

Example: more

ORACLE_PATH

Use: directory for reading and writing to and from SQL*Forms, SQL*Menu 5.0, SQL*Plus

General: can be any directory (if not set, read and write from current directory)

Example: $HOME/oracle

ORACLE_SERVER#

Use: For client only installs

General: T if client only, F if there will be a local database

Example: T

ORACLE_SID

Use: Oracle System Identifier

General: must begin with a letter, followed by a number or character; limited by some OS's to 4 chars

Example: v712

ORACLE_TERM*

Use: Tool Kit II env var pointing to the tk2c${ORACLE_TERM}.res file under $ORACLE_HOME/rdbms/admin/terminal (tk2c stands for character toolkit2)

General: any file like tk2c${ORACLE_TERM}.res

Example: vt100

ORACLE_TERMINAL

Use: directory where the Tool Kit II .res files reside

General: $O_H/tk2/admin/terminal

Example: $O_H/tk2/admin/terminal

ORACLE_TRACE

Use: allow verification/trace to be turned on and off during install (echoes everything done by sh)

General: T does set -x; anything else is off

Example: T

ORACLE_VERIFY

Use: if true, runs the .verify scripts during install

General: T; anything else is off

Example: T

ORACLE_DOC*

Use: points to directory containing on-line doc starting v7.0.16

General: to any directory containing online doc (file.obd)

Example: $ORACLE_HOME/doc

ORAENV_ASK

Use: if set to anything, prompts for $O_S or $O_H when (c)oraenv is invoked

General: NO or anything

Example: NO

ORAKITPATH

Use: set to directory containing the v1 orakit resource file

General: $ORACLE_HOME//admin/resource

Example: $ORACLE_HOME/forms30/admin/resource

ORAMAIL_EDITOR
Use: which editor to use for Oracle*Mail
General: vi, emacs, ed
Example: vi

ORANSEMS#
Use: number of semaphores that make up a set
General: 1-UNIX maximum
Example: 60

ORAPIPES*
Use: sets the default pipe driver to v1 or v2
General: v1, v2
Example: v2

ORATERMPATH
Use: directory containing Oraterm (TK I) resource files
General:
Example: $ORACLE_HOME/oraterm/admin/resource

PATCH_HOME
Use: directory containing patches and patch.list
General: any directory,$O_H/patch,$O_H/install/patch
Example: $ORACLE_HOME/install/patch

RT_STATUS
Use: directory for SQL*TextRetrieval V2.0 to find forms
General: can be any directory (if not set, read and write from current directory)
Example: $HOME/oracle

SDD_HOME
Use: directory containing top level CASE Dictionary directories
General: can be any directory containing top level CASE directories
Example: $O_H/dict50/admin

SDD_PRINT
Use: for CASE: set to UNIX print command to use
General: lp, lpr
Example: lp

SDD_WPRINT
Use: for CASE: set to UNIX print command to use for wide
General: lp, lpr
Example: lp

SQLPATH

Use: now superceded by $ORACLE_PATH

General: can be any directory (if not set, read and write from current directory)

Example: $HOME/oracle

SQLTR_STATUS

Use: directory for SQL*TextRetrieval V1.1 to find forms

General: can be any directory (if not set, read and write from current directory)

Example: $HOME/oracle

SRW_TMP

Use: directory to use for SQL*ReportWriter temporary files

General: any directory with >10M of disk space

Example: /usr/tmp

TK2DEV*

Use: version 7.0.12 only: Tool Kit II terminal resource file

General: vt100 portion from tk2_vt100.res

Example: vt100

TNS_ADMIN*

Use: directory containing SQL*Net v2 files

General: any dir (see manual for order followed--looks at /etc, $HOME, and $TNS_ADMIN

Example: $O_H/network/admin

TWO_TASK

Use: hoststring to use for SQL*Net v1

General: p:, f:, a:host:$O_S, d:host:$O_S, star:host:$O_S, t:host:$O_S, tt:host:$O_S, x:host:$O_S

Example: t:host:oracle

UNIX Environment Variables Affecting Oracle

==========

ADA_PATH set to directory containing Ada compiler

ALSYCOMP_DIR set to directory containing Ada compiler

DISPLAY

Use: Tells X-based tools which Display to use

General: `hostname`:0.0

Example: fubar:0.0

HOME $HOME

LANG en_US, De_DE

LANGUAGE american_america

LDOPTS ld options for compiling

LDPATH directory containing shared object libraries

LD_LIBRARY_PATH
directory containing shared object libraries

LOG when set to a filename, logs information from install session

LOGNAME `logname` or `whoami`

MALLOCTYPE 3.1, 3.2 for telling AIX which malloc type to use

NLSPATH OS directory containing language message files

NONAMESERVER
Use: For tcp/ip networks without a named server
General: 1 for no named server, 0 for named server
Example: 1

PATH $PATH

PRINTER

SHELL /bin/csh or /bin/sh

TERM any valid term: ansi, hft, hp, mac2, sun, xterm, vtxxx

TMPDIR some UNIX boxes allow /tmp to be renamed to $TMPDIR (any directory)

TZ sets local time zone

XENVIRONMENT set to file Orakit, or other X11 resource files

XAPPLRESDIR set to dir containing resource specifications file Orakit, or other X11 resource files

CORE DUMP TRACING 방법

Bulletin no : 10188

ORACLE product 사용중 발생하는 core dump를 tracing하는 방법은 아래와 같다.

1. 준비사항

(1) Product의 Full version number,RDBMS,PL/SQL(if used), 그리고 관련된 Products
예를들면,Forms와 Menu가 link되어 있다면, menu의 version도 필요

(2) core dump를 야기시킨 정확한 command
예를들면,sqlforms30...

2. Tracing 순서

(1) core dump file의 위치확인 (일반적으로,current working directory)
(2) log on as oracle 후 cd $ORACLE_HOME/bin
(3) $ file program (where,program = core를 야 시킨 command,예를들면,
sqlforms30,sqlplus,runrep...)
이 결과를 있는그대로 적어둔다.
여기에서,결과가 'dynamic' or 'dynamically linked'등로 나오면,반드시 당 executable을 relink 해야 함. 왜냐하면,Oracle은 현재 dynamic linking을 support하지 않으므로. 또한,relinking후 core 를 다시 생성하여야 한다. version도 다시 확인할것.

(4) $ chmod +r program
(5) log out 후 core dump를 야기했던 user로 log on하고 해당 unix version 에 따라서,
아래에서 선택.

Command	Note	Exit command
dbx		quit
xdb	(HP-UX only)	quit
sdb		q
adb		Ctrl-d

core dump가 발생했던 directory 로 이동후
dbx $ORACLE_HOME/bin/program core 2>&1 ¦tee /tmp/mystack
(In Bourne or Korn shell)
dbx $ORACLE_HOME/bin/program core ¦tee /tmp/mystack
(In C shell)

(6) 상기의 결과가 unix prompt를 바로 return하면,permission이나 file type에 문제가 있는 것임.간혹,unix에 라서는 core file의 limit를 두는 경우가 있는데 만일 이것이 너무 작으면,core file이 쓸모가 없을 수 있음.

some information을 얻은 후 prompt가 나오면,아래의 command실행.

Command	Enter
dbx	where
xdb	t
sdb	t
adb	$c

(7) 상기의 처리과정에서, /tmp/mystack이라 stack trace file이 생성되 ,(5)항에서 적당한 exit command를 참고하여 exit한다.

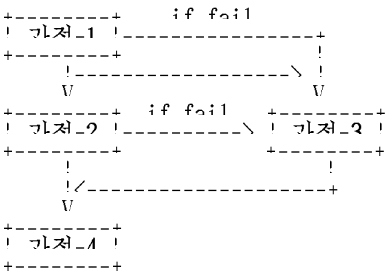
(8) stack trace file을 print하여 해당개발부서(본사 or RND dept.)에 reporting한다.

UNIX SVR4 에서 O/S의 REBUILD

Bulletin no : 10191

요즘 표준화 방향으로 자리가 잡힌 System V Release 4(SVR4) Unix System은 AT&T 3.2, SunOS4, BSD 3.4, XENIX 등을 근간으로 하여 system이 구성되었고, 이 SVR4를 근간으로 여러 H/W vendor이 자신의 H/W에 맞게 각각의 Unix System을 만들었다. 이 과정에서 각 H/W vendor는 자신들이 공급하던 이전 System에 호환성이 있게 system을 만들기도하고, 또는 SVR4를 거의 수정없이 자신의 H/W에 porting하기도 하였다.

다음의 내용은 Unix Lab.에서 개발한 Original SVR4에서 System Parameter를 수정 하고 이를 system에 적용시는 과정을 설명한다.



~~~~~~~~~~~~~~~~~~~~~~(과정-1)~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~

```
# sysdef  -----> 실패, 또는 정보가 부족할 경우 (과정-3)을 참조.
.......................................
*
* IPC Shared Memory
*
524288  max shared memory segment size (SHMMAX)
     1  min shared memory segment size (SHMMIN)
    10  shared memory identifiers (SHMMNI)
    10  max attached shm segments per process (SHMSEG)
.......................................
```

~~~~~~~~~~~~~~~~~~~~~~(과정-2)~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~

(과정-1)에서 나온 데이타를 기반으로 system parameter을 고치면

```
# cd /etc/conf/bin
# ls
idbuild    idconfig   idmaster   idmkinit   idmkunix   idspace    idval
idcheck    idinstall  idmkenv    idmknod    idreboot   idtune
# ./idtune SHMMNI 20 ----> 실패, 또는 정보가 부족할 경우 (과정-3)을 참조
```

~~~~~~~~~~~~~~~~~~~~(과정-3)~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~

만약 sysdef 명령으로 system parameter를 알수 없는 경우, 또는 idtune 명령 대신 수작업으로 system parameter를 변경시키려 한다면,
/etc/conf/cf.d/mtune과 /etc/conf/cf.d/stune의 file 내용을 참조하면 된다.

1) /etc/conf/cf.d/mtune -->system이 가진원래의 값

```
        paramet_name    current_value   min_value       max_value
.................
        SHMMNI          10              10              1000
.................
```

2) /etc/conf/cf.d/stune --> 수정된 내용만 포함.

```
.................
        SHMMNI  20
.................
```

3) /etc/conf/cf.d/mtune으로 정보를 참조하고, /ect/conf/cf.d/stune에 변경될 내용을 2)와 같이 만들면됨.

~~~~~~~~~~~~~~~~~~~~(과정-4)~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~

O/S를 재구성하고 이를 적용시킨다.

```
# ./idbuild
# cd /
# Shutdown -y -g0 -i6
```

위의 과정을 응용하면 ORACLE에서 필요한 system parameter를 System(SVR4계열)에적용할수 있다. 위의 shutdown 과정에서 /stand/unix는 /stand/unix.old 로 save되고 /stand/unix는 새로운 O/S로 대치된다. 그러므로 shutdown전에 /stand/unix를 다른 이름으로 복사하면 현재 상태의 O/S를 보전할수 있다.
예를들어 /stand/unix를 /stand/unix.1st로 복사하여 현재 상태의 O/S를 보존하고 shutdown을 진행하면 다음과 같은 결과를 얻을수 있다.

```
/stand/unix                         ----> 새로운 환경의 O/S
/stand/unix.old, /stand/unix.1st ----> 이전 환경의 O/S
```

~~~~~~~~~~~~~~~~~~~~(첨부)~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~

System vendor는 system이 올라올때(booting), H/W 또는 O/S 마다 다르지만, booting되는 O/S를 사용자가 지정할수 있는 능력을 제공한다. 예를 들면 bootstrap이란 program에 이런 능력을 제공하기도 하고, 또는 firmware mode를 제공함으로써 이런 능력을 제공한다(firmware에서 제공되는 명령을 사용하여 booting O/S를 지정함). 이중 firmware로서 제공되는 것은 각 H/W 특성에 적합하게 전부 다른 형식, 다른 명령어를 갖는다.
~~~~~~~~~~~~~~~~~~~~

아래에는 bootstrap program에 의해 booting O/S를 지정하는 SVR4 i386 version (H/W:i386 or i486 PC, O/S:Unix SVR4)를 예로 간단하게 설명하겠다.

system booting시 다음과 같은 메세지 출력....

Booting Unix System.....

위의 메세지 출력시 Enter Key를 친다. 다음과 같은 메세지 출력....

Booting OS:

위의 메세지를 출력하고 System은 중지한다.
이때 다음과 같은 입력을 할수 있다.

```
Booting OS:(Enter key)                  ----> 현재의 진행을 계속
또는
Booting OS:/stand/unix(Enter key)       ----> /stand/unix를 O/S로 사용함.
또는
Booting OS:/stand/unix.old(Enter key)   ----> /stand/unix.old를 O/S로 사용함.
또는
Booting OS:/stand/unix.1st(Enter key)   ----> /stand/unix.1st를 O/S로 사용함.
```

SYSTEM RESOURCE FOR TUXEDO

Bulletin no : 10198

Semaphore for Tuxedo System/T

==========

; process는 semaphore를 얻기위해 Tuxedo System Bulletin Board에 접근한다.

* semmns (maximun number of semaphore in use in the system)
  - 대부분의 Unix system V 구조에서 default = 60
  - Tuxedo configuration parameter (MAXACCESSERS) 가 semmns로 간주되어 사용되며 일반적으로 250은 되어야 한다.
* semmni (maximum number of active semaphore sets)
  - default = 10
* semmsl (maximum number of semaphores per semaphore set)
  - default = 25
  - equals semmns
* semmap (size of control map used to manage semaphore sets)
  - default = 10
  - equals semmns
* semmnu (number of undo structures in the system)
  - 동시에 Bulletin board 에 접근 시도하는 Unix system process의 수를 통제
  - Bulletin Board 에 접근시도하는 수가 이 값보다 크면 Unix System Error (ENOSPC)발생
  - default = 10
  - equals semmns

* semume (maximum number of undo entries per undo structure)
  - default = 10
  - equals semmns

Message Queue for Tuxedo system

==========

; Tuxedo system은 Client/Server communication을 위해 message & message queue를 사용
; 각각 독립적인 server는 message를 받기위해 message queue(MSSQ set)와 reply queue(REPLYQ parameter)가 있다.
; 각각의 Client process는 자신의 queue를 가진다.
; message parameter에 의해 정의된 한계값이 초과되면 "Blocking"발생
; 모든 process가 send message를 시도하면서 block되었다면 application은 deadlock에 빠진 것임.

* msgmni (number of unique message queue identifiers)
  - default = 50 , 그러나 너무 적은 값이다
  - process수 (queue수) + one for DDBL + one for BBL + two for the BRIDGE process
  - 만약 distributed transaction processing을 한다면 각 server qroup을 위해 여분의 queue를 더한다.

* msgmap (size of control map to manage message segments)
  - msgmni의 두배 이상이어야 한다.
* msgmax (maximum messasge size)
  - Byte단위
  - Tuxedo system application을 위한 일반적인 값은 16KB ~ 32KB범위
* msgmnb (maximum message queue length)
  - same value (or higher) as msgmax
  - msgmnb의 75%보다 큰 message는 file transfer로 보내어진다.
* msgssz (size of message segment)
  - text를 위해 충분히 큰 연속적인 segment, Tuxedo system application = 32
* msgtql (number of outstanding messages)
  - default = 40 , 그러나 busy system을 위해서는 200보다 커야한다.
* msgseg (number of message segments in the system)
  - msgssz의 multiple
  - 대부분의 platform에서는 <= 128KB(그러나 Sun/solaris2.4에서는 <= 32KB)

실전 테스트 (from 대한보증보험 BMT)

========

문 제)

---

* Transaction 발생 sytem A에서 Oracle(including *XA)을 탑재한 Server B로 Tuxedo application (oracle DB에 접속하는 100개의 process를 발생)이 수행 도중 대부분의 process가 kill되는 현상
  - semaphore에 관련된 듯한 message
  - Error : ora-00050
  (Could not obtain the operating system resources necessary for oracle enque)
  - Error : ora-00060 (deadlock detected while waiting for resource)

해결방안)

---

* O/S parameter 수정
  - sun/solaris2.4 에서 /etc/system
    ===>>

```
set shmsys:shminfo_shmmax = 314572800  (=300M,  > oracle SGA size )
          "        shmmin = 1
          "        shmmni = 100
          "        shmseg = 1000
set semsys:seminfo_semmns = 1024    ( Sequent에서 maximum = 4096 )
          "        semmni = 1024
          "        semmsl = 1024
          "        semmap = 1024
          "        semmnu = 1024
          "        semume = 1024
set msgsys:msginfo_msgmni = 700
          "        msgmap = 1400       ( >= msgmni * 2 )
```

" msgmax = 65535 (64KB)

===>>

* Oracle initSID.ora 수정 (-> PROCESSES = 500)

< Auto setting >

```
+-------------------------------------------+
; SESSIONS = 1.1 * POOCESSES                 ;
; ENQUEUE_RESOURCES = ((SESSIONS - 10)*2)+55 ;
; TRANSACTIONS = 1.1 * SESSIONS              ;
; DML_LOCKS = 4 * TRANSACTIONS               ;
; DISTRIBUTED_TRANSACTIONS < TRANSACTIONS    ;
+-------------------------------------------+
```

; Queue & Lock에 관련된 parameter들이 위와 같이 자동 정의되므로 일반적으로는 PROCESSES 값만 조정하면 된다.

; OPS 환경에서는 GC_?? parameter와 함께 적절한 PROCESSES 값 setting.

SCRIPT FOR HOT BACKUP

Bulletin no : 10199

```
# ------------------------------------------------------------*
#  Hot Backup for oracle7 dbms tablespace                      *
#  OTPT_TBL                                                    *
#  For each tablespace                                         *
#       1. begin backup for data file                          *
#       2. dd backup                                           *
#       3. end backup                                          *
# ------------------------------------------------------------*

sqldba lmode=y <
```